

Original conductor roots, native singular directions, and complete observation receivers

Split-Zero research programme

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The new conductor calculation controls the fixed low-degree contribution with an explicit error below kq . The root calculation bounds the complete polynomial produced by that conductor relation. The native singular-value calculation evaluates the unique smallest direction and its late heat transition; the original observation receives an explicit polynomial line. The full conductor graph-volume correction is now bounded by $O_A(q \log q) = o(kq)$. Its exact complementary invariant covariances, the fixed-kernel allocation, and the individual complex projected-current signs remain under calculation.

The complete arguments follow. The new graph map always carries its transformed metric. The small-singular asymptotic uses the retained written monic-profile theorem, whose exact H1–H9 derivation is included. The numerical fixtures verify finite identities and constants; they do not exhibit an off-critical zeta zero.

Contents

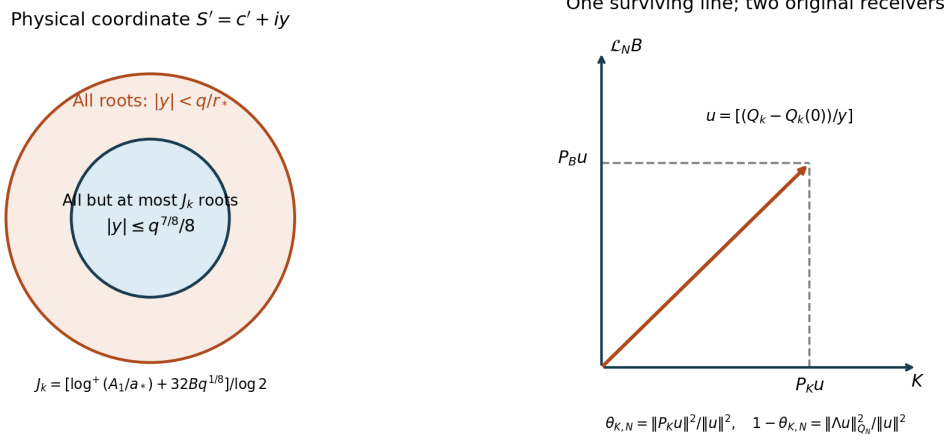
I	Every original eigenclass meets the action boundary	6
1	Original source and the complete boundary	6
2	The whole root grid strengthens the boundary bound	7
3	An evaluated residual for every original eigenclass	8
4	The receiver in the fixed conductor kernel	8
II	The roots of the complete conductor polynomial	11
5	The original polynomial and its reciprocal expression	11
6	A finite count of distant roots	12
7	Every root has a finite linear bound in q	13
8	The exact symbol governing the roots at q -scale	14
9	Return to the full conductor quotient	14
III	The late heat direction through the original observation	16

10	One specified polynomial class controls the late heat	16
11	The original kernel and its complementary observation	17
12	The exact relation to the observed action's own heat	17
13	The surviving class is detected by the actual conductor	18
IV	The full invariant kernel and conductor graph	20
14	The actual invariant kernel: degree filtration, quantitative minors, and its complete conductor graph	20
14.1	1. Original data and the exact residual constraint	20
14.2	2. Exact degree filtration and high-degree projection minors	21
14.3	3. The actual low-degree angle from the full invariant image	23
14.4	4. Why top-degree transversality cannot be assumed from the five-orbit condition .	25
14.5	5. Uniform Gamma condition bounds in the original value coordinates	25
14.6	6. Removing the fixed low graph from the actual kernel determinant	26
14.7	7. The complete relation graph and its source transport	27
14.8	8. What has been evaluated and the remaining connecting map	29
14.9	9. An exact graph over the actual outer polynomial	29
V	The complete conductor graph volume and its invariant receiver	31
15	The complete conductor graph: a quantitative full-volume comparison	31
15.1	1. Original objects and finite domain	31
15.2	2. Every Gamma linear factor has a polynomial lower bound	32
15.3	3. All conductor roots, including those on the q-scale	32
15.4	4. Exact physical centers and complete relation norms	34
15.5	5. The full g-dimensional graph volume is subleading	34
15.6	6. The exact unresolved receiver on the original kernel plane	36
15.7	7. The remaining allocation is an explicit original invariant receiver	36
15.8	Proof-source coverage and citations	37
VI	Complete finite source for the eigenclass phase criterion	38
16	The finite eigenclass phase criterion	38
16.1	7. The supplied mixed phase has an exact eigenclass criterion	38
16.1.1	A projection-independent sign is equivalent to a reducing eigenline	38
VII	The unique small singular direction	40
17	The unique small singular direction of the native arithmetic quotient	40
17.1	The theorem	40
17.2	1. The inverse itself has one unbounded direction	41
17.3	2. Parity gives a determinant identity at zero	42
17.4	3. The logarithmic reference limit is justified at zero	43
17.4.1	3.1 A hard gap for the reference relation measure	43
17.4.2	3.2 Moving the negative characteristic-value comparison to zero	44
17.5	4. Deriving the unique smallest singular value	44

17.5.1	The adjacent even cutoff is not omitted	45
17.6	5. Evaluating the remaining endpoint constant	46
17.7	6. An explicit late-time heat return	47
17.8	Verification and scope	47
17.9	Sources	48
VIII	General-family mixing, flag quotients and rank changes	49
18	General-family continuation: uniform mixing bounds and the original kernel determinant	49
18.1	1. Retain the original mixed source and its complete minimum	49
19	2. The entire kernel determinant has a positive spectral-shift representation	51
19.1	2.1 The large-activation exponent and constant are evaluated	51
19.2	2.2 The slope counts actual unobserved normal values	53
20	3. The section's total motion is bounded independently of all covariance scales	53
20.1	3.1 Resolve all positive covariance modes without assuming they commute with the kernel blocks	53
20.2	3.2 The observed-dimension constant is sharp	54
20.3	3.3 The finite motion does not erase determinant growth	55
20.4	3.4 Return the estimate through the arithmetic action	55
21	4. Every fixed flag quotient has the same rank-controlled determinant calculus	56
22	5. The new GitHub kernel formula now has an explicit $o(kq)$-accurate extremal expression	57
22.1	5.1 Use source-index orthogonality without deleting root interference	57
22.2	5.2 The original Gamma comparison gives the required scale of error	58
23	6. Several noncommuting source additions have exact growth exponents	59
23.1	6.1 General weighted paths	59
23.2	6.2 An exact noncommuting example	60
23.3	6.3 Complex parameter control is also explicit	60
24	7. Rank-changing sources require a finite correction, not a continuity assertion	61
24.1	7.1 Complete analysis along an analytic parameter curve	61
24.2	7.2 Return the missing directions through the actual minimum	62
24.3	7.3 A moving observation has an additional derivative term	62
25	8. What is now established for the programme	63
IX	Dual pencils and singular certificates	64
26	I. A division-free spectral pencil, all singular directions, and the moving-observation boundary	64
26.1	I.1 Original coordinates and the exact dual pencil	64
26.2	I.2 Several noncommuting source additions still have positive coefficients	64
26.3	I.3 Recover each squared singular value, including the smallest	65
26.4	I.4 Heat, inverse exterior powers, and simultaneous scales	66
26.5	I.5 The moving observation needs a different pencil	66

X	Calibrated heat and the observation	68
27	II. The resolved, calibrated heat-transition layer and the actual observed direction	68
27.1	II.1 What is inherited and what is being calculated	68
27.2	II.2 Finite identities and constants	68
27.3	II.3 Native consequences and what remains unknown at the profile threshold . . .	69
27.4	II.4 The observed heat deficit measures a product of actual projection fractions . .	69
27.5	II.5 An exact finite example for validation	70
XI	Finite unit certificates	71
28	III. Finite unit certificates without a Neumann smallness hypothesis	71
28.1	III.1 The only algebraic inverse gate is the leading coefficient	71
28.2	III.2 The retained jet metric improves rather than destroys the estimate	71
28.3	III.3 Two-sided physical-metric bounds with no smallness restriction	72
28.4	III.4 Tensoring, quotient minima, and the original labels	73
XII	ES heat jets and full moving blocks	74
29	IV. The ES trace-rigidity theorem as two exact heat jets, with the full mixed return	74
29.1	IV.1 What the newer ES source has proved	74
29.2	IV.2 Two centres recover both arithmetic traces	74
29.3	IV.3 Finite samples give rigorous jet intervals	75
29.4	IV.4 The complete mixed minimum retains the missing second-trace term	75
29.5	IV.5 The raw gate has an exact finite spacing certificate	76
XIII	The retained full-source profile H1–H9	78
29.6	1.1 Source basis and domains	78
29.7	1.2 The entire allowance profile follows from the retained monic minima	78
29.8	1.3 An exact elliptic expression for the two heat thresholds	79

Root geometry and the native late-heat receiver



Left: proved bounds OR13 and OR15, under OR12. Disk radii are schematic; no sample roots are drawn.
Right: the exact orthogonal decomposition LS6–LS9. The displayed angle is schematic; its native value is not assigned.

Figure 1: Two proved connecting constructions. OR13 and OR15 count and bound every root of the actual conductor polynomial. LS6–LS9 project the specified late-heat line through the unchanged kernel and observed minimum. The captions identify both schematics; neither assigns unknown root positions or a native projection angle.

Part I

Every original eigenclass meets the action boundary

The original polynomial source has a rank-two Hermitian action boundary. The complete root grid forces a stronger lower bound for this boundary than the largest real part of one root. We prove the bound, evaluate its effect on every original eigenclass, and carry the resulting residual through the fixed conductor projection. The source, its full relation ideal, and the physical coordinate remain unchanged.

The attained-source boundary is the earlier result NAB1–19 [1]; SD1–3 [2] identifies its norm with the rank-one quantity used in the heat calculation. The new root-sum estimate is NE6. Its eigenclass consequences are NE8–11. The supplied eigenvector continuation [4, (38)–(41)] supplies the general phase identity and the test over all projections; we prove them here with their exact receiver in the original kernel. No sign of that fixed kernel is inferred from a sign obtained by choosing a different projection.

1 Original source and the complete boundary

Fix $0 < \delta < 1/2$, $\gamma > 2$, odd $k \geq 9$, and a positive integer e . The simple-packet kernel uses $e = 1$ and $k \equiv 1 \pmod{4}$. The boundary calculation also retains the original repeated-root case $e = 1 + k(m_0 - 1)$ when it occurs. Set

$$\begin{aligned} c &= k/2, & q &= e(k+1)^2, & \lambda_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \\ \chi(S) &= \prod_{a,b=0}^k (S - \lambda_{ab})^e, & E &= \mathbb{C}[S]/(\chi), & A[P] &= [SP]. \end{aligned} \tag{NE1}$$

Use the full original positive measure $d\mu(y) = w_h^{*k}(y) dy$ on $S = c + iy$, with its mass and all original units retained. All the following finite results hold for any fixed positive source measure on this same line whose polynomial Grams through degree $N + 1$ are finite and positive definite. This includes the full arithmetic source and the separately identified Gamma source. It does not identify their metrics. Fix $N \geq q - 1$ and let

$$T_N : E \longrightarrow \mathcal{P}_N, \quad G_N = T_N^* H_N T_N$$

be the attained minimum section and its Gram matrix. Here H_N is the Gram matrix of the literal powers $1, S, \dots, S^N$ and the kernel of the source map is the whole ideal $\chi \mathcal{P}_{N-q}$. For $N = q - 1$ this ideal is zero. If W_N contains the coefficient columns of $\chi, S\chi, \dots, S^{N-q}\chi$, the section is exactly

$$T_N = E_N - W_N(W_N^* H_N W_N)^{-1} W_N^* H_N E_N,$$

where E_N inserts polynomials of degree below q . The omitted-column case means $T_N = E_N$. The formula is proved by reducing modulo χ and checking $W_N^* H_N T_N = 0$; every other lift has the additional nonnegative squared norm of its orthogonal ideal component.

Let $\ell_N = [S^N]T_N$, let b_N be the coefficient vector of χS^{N+1-q} , and set $\xi_N = T_N^* H_{N+1} b_N$, with a zero top coordinate appended to T_N in this product. Monic division gives

$$ST_N - T_N A = b_N \ell_N + Z_N, \quad \text{im } Z_N \subset \chi \mathcal{P}_{N-q}.$$

The latter term is orthogonal to T_N . On the physical line, $\bar{S} + S = k$, hence

$$D_N = A^* G_N + G_N A - k G_N = -\xi_N \ell_N - \ell_N^* \xi_N^*. \tag{NE2}$$

This proves the complete boundary without dropping a lower relation. Use $\dagger$ for the G_N adjoint and put

$$B = A + A^\dagger - kI = G_N^{-1} D_N. \quad (\text{NE3})$$

It is selfadjoint in G_N , has rank at most two, and has trace zero: $\text{Tr } B = 2\Re \text{Tr } A - kq = 0$ by the complete grid in NE1. An eigenvector at $a = k$ has Rayleigh quotient $2k\delta$ for B ; one at $a = 0$ has quotient $-2k\delta$. Such vectors exist also on each repeated-root primary block. Thus B has exactly the two nonzero eigenvalues $+\kappa_N, -\kappa_N$, with $\kappa_N > 0$. If P_∂ is its orthogonal support projection, then

$$B^2 = \kappa_N^2 P_\partial. \quad (\text{NE4})$$

All diagonalizations used here concern selfadjoint operators in the specified metric and are proved by maximizing the Rayleigh quotient on a finite unit sphere and inducting on its invariant orthogonal complement.

For the even original measure, write $y = (S - c)/i$ and $M = (A - cI)/i$. The attained compression C_N of multiplication by real y is selfadjoint. Subtracting the leading multiple of the real monic source orthogonal polynomial p_{N+1} before compression gives $M = C_N + r_N \ell_N^y$, where $r_N = [p_{N+1}]$ and $\ell_N^y = [y^N]T_N$. Reflection in y makes these two factors have opposite parity. In particular $\ell_N^y r_N = 0$. Therefore

$$B = i(r_N \ell_N^y - (r_N \ell_N^y)^\dagger), \quad \kappa_N = \|r_N \ell_N^y\|_{G_N} = \epsilon_N. \quad (\text{NE5})$$

Indeed the two normalized factor vectors are orthogonal, and the matrix of B on their span has off-diagonal entries $i\epsilon_N$ and $-i\epsilon_N$. This also extends this finite equality to $N = 2q$; the earlier SD cutoff range does not restrict the finite proof above. The real-zero fact for positive-source orthogonal polynomials, used to ensure the remainder is nonzero, is the classical result recorded by Koornwinder and coauthors [6, §18.2(vi)]. A direct proof is to multiply its real sign-changing factors: if fewer than its degree occur, this lower-degree polynomial has nonzero inner product with the orthogonal polynomial, contradicting orthogonality.

2 The whole root grid strengthens the boundary bound

Theorem 1. *At every original cutoff, including all four cutoffs $q-1, q, 2q-1, 2q$,*

$$\boxed{\kappa_N \geq \frac{e\delta(k+1)^3}{2}}. \quad (\text{NE6})$$

Proof. Let E_+ be the direct sum of the full primary spaces at roots with $a > (k/2)$. It is invariant under A and has dimension $q/2$. Existence and the direct sum follow from the coprime-factor polynomial identity: Bezout inverses give the Chinese-remainder isomorphism $E \cong \bigoplus_{a,b} \mathbb{C}[z]/(z^e)$; multiplication is $\lambda_{ab} + z$ on each summand. Choose a G_N -orthonormal basis of E_+ and let P_+ be its orthogonal projection. Its compressed action is the restriction of A , so

$$\text{Tr}(P_+ B) = 2\Re \text{Tr}(A|_{E_+}) - k \dim E_+ = e \sum_{a=(k+1)/2}^k \sum_{b=0}^k 2(2a - k)\delta.$$

This is a trace identity; the original eigenvectors have not been assumed orthogonal. If u_+, u_- are the two unit boundary eigenvectors, then $\text{Tr}(P_+ B) = \kappa_N (\|P_+ u_+\|^2 - \|P_+ u_-\|^2) \leq \kappa_N$. The positive odd integers $2a - k$ sum to $((k+1)/2)^2$. Evaluating the displayed double sum proves NE6. $\square$

3 An evaluated residual for every original eigenclass

Let $v \neq 0$ be any actual eigenvector at λ_{ab} , including the eigenline inside a repeated-root primary space. Keep its physical scale $E_v = \|v\|_{G_N}^2$. For calculation only, put

$$x = v/\sqrt{E_v}, \quad b_\lambda = 2\Re\lambda - k = 2(2a - k)\delta, \quad r_\lambda = (A^\dagger - \bar{\lambda}I)x, \quad \eta_\lambda = \|P_\partial x\|_{G_N}^2. \quad (\text{NE7})$$

No metric, physical class, or source map is changed by this notation. Since $Ax = \lambda x$, NE3 gives $r_\lambda = (B - b_\lambda I)x$. Moreover $\langle x, Bx \rangle = b_\lambda$ and $r_\lambda \perp x$. Consequently

$$\|r_\lambda\|^2 = \kappa_N^2 \eta_\lambda - b_\lambda^2, \quad |b_\lambda|/\kappa_N \leq \eta_\lambda \leq 1, \quad (\text{NE8})$$

and hence

$$|b_\lambda|(\kappa_N - |b_\lambda|) \leq \|r_\lambda\|^2 \leq \kappa_N^2 - b_\lambda^2. \quad (\text{NE9})$$

For proof, use NE4 to evaluate $\|(B - b_\lambda)x\|^2$. In the boundary eigenbasis, $|\langle x, Bx \rangle| \leq \kappa_N \|P_\partial x\|^2$, which proves the remaining inequalities.

Since k is odd, $|2a - k| \geq 1$. NE6 is larger than $2k\delta$ for the stated domain. Inserting it into NE9 gives the finite strictly positive bound for every eigenclass

$$\|r_\lambda\|^2 \geq \delta^2 |2a - k| (e(k+1)^3 - 4|2a - k|) > 0. \quad (\text{NE10})$$

For a terminal class $a = k$ the right side is $\delta^2 k(e(k+1)^3 - 4k)$. This holds before invoking an asymptotic estimate, numerical period, or EIQ statement.

The previously proved allowance estimate used by H1–7 [5] is $\log \epsilon_N^2 = 2q\psi(s_N) + O_h(k \log^2(q+2))$ on its stated source and cutoff domain. NE5 and NE9 now give, on precisely that domain and retaining that proof dependence,

$$q\psi(s_N) - O_h(k \log^2(q+2)) \leq \log \|r_\lambda\|^2 \leq 2q\psi(s_N) + O_h(k \log^2(q+2)). \quad (\text{NE11})$$

The lower bound uses $|b_\lambda| \geq 2\delta$ and eventually $\kappa_N \geq 2|b_\lambda|$; the finite form is NE9 whenever that eventual guard has not been certified. NE11 is an interval of rates, not an evaluated leading coefficient inside that interval. The new finite theorem NE10 is independent of the imported profile.

4 The receiver in the fixed conductor kernel

Return now to the simple-packet case and the actual kernel of MR1–10 [3]. Its frame is $I_K = V^{-1}\pi^T$ with $\ker \pi = \text{im}[M_A, S_k Z_k]$, and its dimension is $m = 8k - 16$. The transpose in this frame is the complex-linear dual, not an adjoint. Set

$$H_K = I_K^* G_N I_K, \quad P = I_K H_K^{-1} I_K^* G_N, \quad p = Px, \quad \theta = \|p\|^2. \quad (\text{NE12})$$

Positive definiteness of H_K proves that this is the actual orthogonal projection; multiplication verifies $P^2 = P$ and $P^* G_N = G_N P$. It retains the complete invariant image and all relations already minimized in G_N . For the specified physical eigenclass define exactly

$$z = \langle Pv, A(I - P)v \rangle_{G_N}, \quad w = \langle (I - P)v, APv \rangle_{G_N}.$$

Then the incoming identity has the following native receiver:

$$\Re(\bar{z}w) = \left| \frac{z + w}{2} \right|^2 - \frac{E_v^2}{4} |\langle Bx, p \rangle - b_\lambda \theta|^2. \quad (\text{NE13})$$

Indeed $w - z = E_v \langle r_\lambda, p \rangle$ follows by expanding $Ax = \lambda x$ and the adjoint. The elementary identity $\Re(\bar{z}w) = |(z + w)/2|^2 - |w - z|^2/4$, followed by NE7, proves NE13. In literal source coordinates the residual scalar is

$$\langle Bx, p \rangle - b_\lambda \theta = -(x^* \xi_N)(\ell_N p) - \overline{\ell_N x}(\xi_N^* p) - b_\lambda \theta. \quad (\text{NE14})$$

This follows from $G_N B = D_N$ and NE2. It evaluates the needed functional using the two actual boundary rows, while NE12 supplies their actual kernel projection. In particular no full spectral polynomial is substituted for these complex matrix elements. As $r_\lambda \perp x$ and $\|p - \theta x\|^2 = \theta(1 - \theta)$, NE8 yields

$$[-\Re(\bar{z}w)]_+ \leq \frac{E_v^2}{4}(\kappa_N^2 \eta_\lambda - b_\lambda^2)\theta(1 - \theta). \quad (\text{NE15})$$

This is a rigorous error bound for the fixed kernel. Its right side and the two terms of NE13 do not separately assign the current's sign.

For clarity there is also a fully proved relation to the test over all projections in the incoming continuation. Put $\varepsilon = \|r_\lambda\| > 0$ and $u = r_\lambda/\varepsilon$. Then $u \perp x$ and $\langle x, Au \rangle = \varepsilon$. Set $g = \lambda - \langle u, Au \rangle$, and choose the phase of u so that $\langle x, Au \rangle = -\varepsilon g/|g|$ when $g \neq 0$. For $g = 0$ choose any phase. Put

$$t = \frac{1 - |g|/\sqrt{|g|^2 + \varepsilon^2}}{2}, \quad \tilde{P}x = tx + \sqrt{t(1-t)}u.$$

This is the projection onto $\sqrt{t}x + \sqrt{1-t}u$ on their two-dimensional span; append any $m - 1$ orthogonal directions for a prescribed rank $1 \leq m \leq q - 1$. Direct expansion gives $(z + w)/(2E_v) = t(1 - t)g + \sqrt{t(1-t)}(1/2 - t)\langle x, Au \rangle = 0$. Also $|w - z|/E_v = \varepsilon\sqrt{t(1-t)}$. Thus

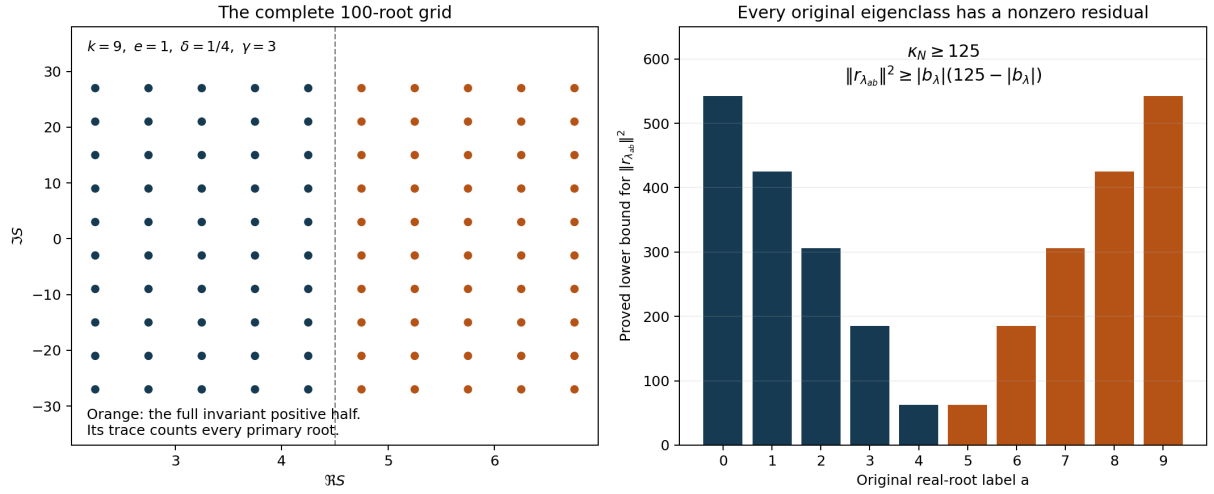
$$\Re(\bar{z}w) = -\frac{E_v^2 \varepsilon^4}{16(|g|^2 + \varepsilon^2)} < 0 \quad \text{for this explicitly constructed } \tilde{P}. \quad (\text{NE16})$$

Conversely a reducing eigenline has $r_\lambda = 0$, so NE13 is nonnegative for every projection. NE10 proves that none of the original eigenlines is reducing in the original scalar-source metric. The exact map back to the programme projection is NE12–14. Its rows must be evaluated to decide its individual sign; NE16 does not change those rows or replace that assignment.

References

- [1] Split-Zero research programme, *Native class integration*, section “The full arithmetic action boundary in the original quotient”, NAB1–19, 19 September 2026. Pinned public proof and equation location.
- [2] Split-Zero research programme, *The original spectral determinant: boundary action and small singular values*, SD1–3, 20 September 2026. Pinned public proof and equation location.
- [3] Split-Zero research programme, *The original observation kernel as an exact mixed-relation determinant*, MR1–10, 20 September 2026. Pinned public proof and equation location.
- [4] Supplied web-session continuation, *The actual arithmetic eigenvectors*, Section 7, equations (38)–(41), received 21 September 2026. Human authorship unspecified in the supplied text. The complete finite argument used here is included in this edition; its earlier source identifiers remain in the private reading ledger.
- [5] Supplied *ES/RH spectral resolution*, proof II, H1–11, 20 September 2026. The profile input is inherited at its stated written-proof status; the finite estimates are proved there.
- [6] T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, with W. P. Reinhardt, “Orthogonal Polynomials”, Chapter 18 of the *NIST Digital Library of Mathematical Functions*, F. W. J. Olver et al., editors, <https://dlmf.nist.gov/18.2#vi>.

A large action boundary is forced by the full root grid



Exact finite bounds NE6 and NE10, valid at all four source cutoffs. This displayed quartet is a declared parameter example.
Bars are proved lower bounds, not measured residuals or a claim that these parameters are zeta zeros.

Figure 2: NE6 and NE10 evaluated at the displayed parameter example. The grid uses the original $S = k/2 + iy$ coordinate. All ten imaginary labels occur at each real label. The projection used in NE6 is the orthogonal projection onto the invariant positive half of the spectrum; its trace is evaluated without assuming the root eigenvectors orthogonal. The plot does not replace the original conductor projection NE12. The scalar-source boundary is NAB1–19; its orthogonal-polynomial convention is the classical one cited in NE5.

Part II

The roots of the complete conductor polynomial

The complete conductor relation produces a polynomial $d_A = \mathcal{U}_A 1$. Its roots control the multiplication quotient connected to the actual relation quotient by KF52–55. We prove quantitative root bounds in the original physical coordinate, retaining every conductor coefficient. Only $O_A(q^{1/8})$ roots can lie beyond $q^{7/8}/8$; every root is $O_A(q)$; and their reciprocal q -scale limit is the actual exponential symbol. These statements supply root geometry for that connecting map. Its complete transformed metric remains part of the map.

5 The original polynomial and its reciprocal expression

Keep the simple-quartet domain $k \geq 9$, $k \equiv 1 \pmod{4}$, $0 < \delta < 1/2$, $\gamma > 2$, and the fixed actual conductor coefficients a_{rs} , $0 \leq r, s \leq 8$. Put

$$\begin{aligned} q &= (k+1)^2, \quad q' = (k-7)^2, \quad c = k/2, \quad c' = c - 4, \\ \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \\ \beta'_{ab} &= c' + (2a - k + 8)\delta + i(2b - k + 8)\gamma, \\ \chi(S) &= \prod_{a,b=0}^k (S - \beta_{ab}), \quad \chi'(S') = \prod_{a,b=0}^{k-8} (S' - \beta'_{ab}), \\ b_{rs} &= 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad \mathcal{T}_{Ap}(S') = \sum_{r,s} a_{rs} p(S' + b_{rs}). \end{aligned} \tag{OR1}$$

Here χ' is the named lower root polynomial, not a derivative. Write

$$E_A(w) = \sum_{r,s} a_{rs} e^{b_{rs} w}, \quad \mu_j = \sum_{r,s} a_{rs} b_{rs}^j, \quad v = \text{ord}_0 E_A, \quad g = q - q' - v. \tag{OR2}$$

The distinct shifts give $0 \leq v \leq 80$ and $\mu_v \neq 0$: otherwise the first 81 moments would annihilate the invertible Vandermonde matrix of those shifts, forcing every coefficient to be zero. The original conductor is nonzero. The derivative expansion gives $\deg \mathcal{T}_{Ap} = \deg p - v$ for $\deg p \geq v$ and leading coefficient $\mu_v \binom{\deg p}{v}$ times that of p .

For each lower root, $\beta'_{ab} + b_{rs} = \beta_{a+r, b+s}$. Thus χ' divides every shifted $\chi(S' + b_{rs})$, with full quotient

$$d_{rs}(S') = \prod_{\substack{0 \leq a, b \leq k \\ (a,b) \notin \{r, \dots, r+k-8\} \times \{s, \dots, s+k-8\}}} (S' + b_{rs} - \beta_{ab}).$$

Consequently

$$d_A(S') = \frac{\mathcal{T}_A \chi(S')}{\chi'(S')} = \sum_{r,s} a_{rs} d_{rs}(S'), \quad \deg d_A = g, \quad [S'^g] d_A = \mu_v \binom{q}{v}. \tag{OR3}$$

This is precisely KF41 and KF52, proved again here at $h = 1$.

Retain the physical coordinate $S' = c' + iy$. Set

$$y_\alpha = \frac{\beta_\alpha - c'}{i}, \quad D = \sqrt{\delta^2 + \gamma^2}, \quad B_k = 4 + kD, \quad B = \max_{a_{rs} \neq 0} |b_{rs}|, \quad A_1 = \sum_{r,s} |a_{rs}|. \tag{OR4}$$

Both upper and lower roots in this physical coordinate have modulus at most B_k . The auxiliary variable $z = q/y$ is the specified reciprocal map, with inverse $y = q/z$; it does not replace the physical coordinate in the source or metric. On $|z| < q/B_k$ put

$$G_k(z) = \frac{\mathcal{T}_A \chi(c' + iq/z)}{\chi(c' + iq/z)} = \sum_{r,s} a_{rs} \prod_{\alpha=1}^q \left(1 + \frac{-ib_{rs}z}{q - y_\alpha z}\right), \quad (\text{OR5})$$

$$F_k(z) = z^{-v} G_k(z).$$

The apparent singularity at zero is removable and

$$F_k(0) = (-i)^v \mu_v \frac{\binom{q}{v}}{q^v}. \quad (\text{OR6})$$

Indeed the polynomial degree calculation in OR3 gives $G_k(z) = \mu_v \binom{q}{v} (iq/z)^{-v} + O(z^{v+1})$. The denominators in OR5 are nonzero in its disk, proving holomorphy there. For $q \geq 2v$, with the convention $v = 0$ allowed,

$$|F_k(0)| \geq a_* := \frac{|\mu_v|}{2^v v!} > 0. \quad (\text{OR7})$$

This follows from $\binom{q}{v}/q^v = (v!)^{-1} \prod_{j=0}^{v-1} (1 - j/q)$, each factor at least $1/2$.

6 A finite count of distant roots

For $4B_k R \leq q$, the circle $|z| = 2R$ has $|q - y_\alpha z| \geq q/2$. Therefore

$$\max_{|z|=2R} |F_k(z)| \leq \frac{A_1 e^{4BR}}{(2R)^v}. \quad (\text{OR8})$$

In fact each factor in OR5 has modulus at most $1 + 2B|z|/q$, and their product is at most $e^{2B|z|}$.

We use the following finite zero count, with its proof included. For a holomorphic f on a neighborhood of $|z| \leq 2R$, $f(0) \neq 0$, the number of its zeros in $|z| \leq R$, counted with multiplicity, satisfies

$$n_f(R) \log 2 \leq \log \max_{|z|=2R} |f(z)| - \log |f(0)|. \quad (\text{OR9})$$

To prove it first choose a circle of radius t without a boundary zero. Factor all zeros ζ_j inside that circle: $f(z) = h(z) \prod_j (z - \zeta_j)$ with h zero-free there. The real part of a holomorphic logarithm of h has its mean value at zero. Also

$$\frac{1}{2\pi} \int_0^{2\pi} \log |te^{i\theta} - \zeta| d\theta = \log t \quad (|\zeta| < t),$$

by the convergent series for $\log(1 - \zeta e^{-i\theta}/t)$, whose nonconstant Fourier terms have zero mean. Subtracting the value at zero gives $\sum_j \log(t/|\zeta_j|)$, bounded by the logarithmic maximum minus $\log |f(0)|$. Let t tend to $2R$ through circles without boundary zeros. Each zero in $|z| \leq R$ contributes at least $\log 2$ in the limit. This proves OR9, including boundary zeros. It is the needed form of Jensen's classical formula [3]; the harmonic mean and Cauchy formulas are [4, 1.9.31, 1.9.33].

Let $n_A(t)$ count roots of $d_A(c' + iy)$ with $|y| \geq t$, with their full multiplicities. Under $q \geq 2v$ and $4B_k R \leq q$, the original χ and χ' have no zeros where $|y| \geq q/R$. OR3 and OR5 therefore give the exact multiplicity-preserving bijection $y \mapsto q/y$ between those roots of d_A and the zeros of F_k in $0 < |z| \leq R$. It follows that

$$n_A(q/R) \leq \frac{\log(A_1/a_*) + 4BR - v \log(2R)}{\log 2}. \quad (\text{OR10})$$

A convenient weaker bound for $R \geq 1/2$ is

$$n_A(q/R) \leq \frac{\log^+(A_1/a_*) + 4BR}{\log 2}. \quad (\text{OR11})$$

Every displayed guard is finite and part of the assertion.

In particular, impose

$$q \geq 2v, \quad 32B_k q^{1/8} \leq q. \quad (\text{OR12})$$

Using $R = 8q^{1/8}$ gives

$$\boxed{n_A(q^{7/8}/8) \leq \frac{\log^+(A_1/a_*) + 32B_k q^{1/8}}{\log 2} = O_A(q^{1/8}) = o(k).} \quad (\text{OR13})$$

All other roots lie in the disk of radius $q^{7/8}/8$. The constants depend on the unchanged actual conductor; no uniformity in a varying period or vanishing μ_v is asserted. For fixed original data the guards hold eventually because $B_k = O(k)$ and $q \asymp k^2$.

7 Every root has a finite linear bound in q

The absence of zeros near $z = 0$ can also be quantified. Define

$$R_* = \min \left\{ 1, \frac{9}{4(D+1)} \right\}, \quad M_* = \frac{A_1 e^{2BR_*}}{R_*^v}, \quad r_* = R_* \min \left\{ \frac{1}{2}, \frac{a_*}{4M_*} \right\}. \quad (\text{OR14})$$

These constants are strictly positive and independent of k . For $k \geq 9$, $B_k R_* \leq q/2$: it suffices to bound $B_k \leq k(D+4/9)$ and $q \geq k^2$ and use $R_* \leq 9/[4(D+1)]$. Thus the proof of OR8 on the radius R_* gives $\max_{|z|=R_*} |F_k(z)| \leq M_*$. Cauchy's coefficient formula then gives, for $r < R_*$,

$$|F_k(z) - F_k(0)| \leq M_* \frac{r}{R_* - r} \quad (|z| \leq r).$$

At $r = r_*$ the right side is at most $a_*/2$. For $q \geq 2v$ there is no zero in $|z| \leq r_*$. Since $q/r_* > B_k$, OR3–5 imply

$$\boxed{d_A(c' + iy) = 0 \implies |y| < q/r_*}. \quad (\text{OR15})$$

Zeros with $y = 0$ satisfy the bound directly. No root of χ' has been cancelled or lost in this argument.

A further aggregate estimate retains the entire root list. Put

$$C_0 = \frac{\log^+(A_1/a_*) + v \log^+(1/(2r_*))}{\log 2}, \quad C_1 = \frac{4B}{\log 2}, \quad T = q/r_*.$$

For $4B_k \leq t \leq T$, OR10 with $R = q/t$ gives $n_A(t) \leq C_0 + C_1 q/t$. Integrating the counting function yields

$$\boxed{\sum_{\zeta: d_A(c'+i\zeta)=0} |\zeta| \leq 4gB_k + C_0 T + C_1 q \log^+ \frac{T}{4B_k} = O_A(q \log(q+2)).} \quad (\text{OR16})$$

When $T < 4B_k$ the first term already bounds the sum; otherwise split the integral at $4B_k$ and use OR15. If $n = q/2$, the monic polynomial in the dimensionless variable $w = y/n$ has coefficient norm

$$\left\| \prod_{\zeta} (w - \zeta/n) \right\|_1 \leq \prod_{\zeta} (1 + |\zeta|/n), \quad \log \left\| \prod_{\zeta} (w - \zeta/n) \right\|_1 \leq \frac{1}{n} \sum_{\zeta} |\zeta| = O_A(\log(q+2)). \quad (\text{OR17})$$

The leading coefficient of the actual polynomial has not vanished:

$$d_A(c' + iny) = \mu_v \binom{q}{v} i^g n^g \prod_{\zeta} (y - \zeta/n).$$

OR17 concerns the explicitly displayed monic factor only.

8 The exact symbol governing the roots at q-scale

Fix $R > 0$. On $|z| \leq R$ impose $B_k R \leq q/2$ and $q \geq 4BR$. For $w_{\alpha,rs} = -ib_{rs}z/(q - y_\alpha z)$ one has $|w_{\alpha,rs}| \leq 1/2$. The convergent logarithm gives

$$\left| \sum_{\alpha} \log(1 + w_{\alpha,rs}) + ib_{rs}z \right| \leq e_k(R) := \frac{2BB_k R^2 + 4B^2 R^2}{q}. \quad (\text{OR18})$$

Indeed

$$\left| \sum_{\alpha} \frac{1}{q - y_\alpha z} - 1 \right| \leq \frac{B_k R}{q - B_k R} \leq \frac{2B_k R}{q},$$

and $|\log(1+w) - w| \leq |w|^2$ for $|w| \leq 1/2$. Summing the latter bound over q factors proves OR18. Exponentiation and $|e^u - 1| \leq e^{|u|}|u|$ give

$$\max_{|z| \leq R} |G_k(z) - E_A(-iz)| \leq A_1 e^{BR + e_k(R)} e_k(R). \quad (\text{OR19})$$

Both terms have a zero of order at least v at zero. Applying the maximum principle to their difference divided by z^v on a larger disk of radius $2R$ gives the explicit bound

$$\max_{|z| \leq R} \left| F_k(z) - \frac{E_A(-iz)}{z^v} \right| \leq \frac{A_1 e^{2BR + e_k(2R)} e_k(2R)}{(2R)^v}, \quad (\text{OR20})$$

when the guards for $2R$ hold. Thus this difference is $O_{A,R}(1/k)$ on every fixed disk, including its removable value at zero.

For any bounded region with piecewise smooth boundary, containing no zero of $E_A(-iz)/z^v$ on its boundary, OR20 preserves the full number of zeros for all sufficiently large k . Here is the counting argument. If the boundary minimum is $b > 0$ and the right side of OR20 is smaller than b , the homotopy $f + t(F_k - f)$ is nonzero on that boundary, where $f = E_A(-iz)/z^v$. Its boundary winding number is constant in t . By factoring each isolated zero and integrating the logarithmic derivative, this winding number counts all interior zeros with multiplicity. Consequently every nonzero zero z_0 of $E_A(-iz)$ of multiplicity j has exactly j corresponding roots, in arbitrarily small isolating disks for z_0 , satisfying

$$\boxed{\frac{y_\nu(k)}{q} \longrightarrow \frac{1}{z_0}}. \quad (\text{OR21})$$

Conversely each nonzero finite accumulation point of $y_\nu(k)/q$ is the reciprocal of a nonzero symbol zero: use OR20 at the reciprocal point, which must have limiting function value zero. This is a proved correspondence on the actual symbol's zero set; it does not presume a particular additional zero or identify these conductor roots with zeta zeros.

9 Return to the full conductor quotient

At $N = q - 1, q, 2q - 1, 2q$ set $L = N - q = -1, 0, q - 1, q$. The exact target is

$$Y_N = \mathcal{P}'_{L+g} \mathcal{U}_A \mathcal{P}_L, \quad \mathcal{U}_A h = \sum_{r,s} a_{rs} d_{rs}(S') h(S' + b_{rs}), \quad (\text{OR22})$$

$$\|\eta\|^2 = \int_{\mathbb{R}} |\chi'(c' + iy) \eta(c' + iy)|^2 \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy.$$

Euclidean polynomial division defines the unique maps $\mathcal{U}_A h = d_A \mathcal{C}_L h + \mathcal{R}_L h$ with $\deg \mathcal{R}_L h < g$. Their exact leading terms give

$$\det \mathcal{C}_L = \prod_{j=0}^L \frac{\binom{q+j}{v}}{\binom{q}{v}}, \quad \mathcal{F}_L = \mathcal{R}_L \mathcal{C}_L^{-1}.$$

In the decomposition $r + d_A w$, $\deg r < g$, the map

$$\mathcal{J}_L(r + d_A w) = r - \mathcal{F}_L w + d_A w \quad (\text{OR23})$$

fixes the actual low representative frame and sends $\mathcal{U}_A \mathcal{P}_L$ to $d_A \mathcal{P}_L$. Its inverse has the plus sign. It is an isometry only when the target metric is

$$H_N^{\text{target}} = (\mathcal{J}_L^{-1})^* H_N \mathcal{J}_L^{-1}. \quad (\text{OR24})$$

This follows by substituting $\mathcal{J}_L x$ in the target quadratic form; no smallness of $\mathcal{F}_L$ is used. At $L = -1$ the map is the identity.

For the nonexceptional roots in OR13, the scaled bound satisfies

$$\frac{q^{7/8}/8}{q/2} = \frac{1}{4} q^{-1/8} \leq \frac{1}{2} (q/2)^{-1/8}. \quad (\text{OR25})$$

This is exactly the root-radius guard in OOQ13 with $n = q/2$ and $\rho = n^{-1/8}$. The roots outside that disk are still present, with the explicit count OR13 and full bound OR15. Neither deleting their directions from the actual kernel nor discarding OR24 follows from their count. The metric comparison through OR24, on the unchanged frame $I_K = V^{-1} \pi^T$ for $Z = [M_A, S_k Z_k]$, remains to be evaluated. KF49 has already bounded the preceding low-degree and conductor transport costs by $o(kq)$. OR13–21 evaluate the new root geometry needed for the next part of that same calculation.

References

- [1] Split-Zero research programme, *The actual invariant kernel: degree filtration, quantitative minors, and its complete conductor graph*, 21 September 2026, KF1–55.
- [2] Split-Zero research programme, *All directions of the original outer quotient at the leading scale*, 15 September 2026, OOQ1–40, especially OOQ12–15. Pinned public proof, OOQ12–15.
- [3] J. L. W. V. Jensen, *Sur un nouvel et important théorème de la théorie des fonctions*, Acta Mathematica 22 (1899), 359–364, doi:10.1007/BF02417878. Historical attribution; OR9 includes the full proof used here.
- [4] R. Roy, F. W. J. Olver, R. A. Askey, R. Wong, with W. P. Reinhardt, *Algebraic and Analytic Methods*, NIST DLMF, chapter 1, version 1.2.8, 15 September 2026, 1.9.31 and 1.9.33.

Part III

The late heat direction through the original observation

The supplied smallest-singular-value proof identifies one late heat scale of the full original operator. We identify its surviving line as an explicit polynomial class, with a finite error, and carry it through the unchanged conductor observation. Thus the new full-source heat calculation has an exact receiver in the original kernel. The independently attained observed action retains its full coupling.

10 One specified polynomial class controls the late heat

Use the complete original polynomial Q_k and attained metric G_N in S1–S45 of the supplied proof [1]. The original Gamma comparison and reference zero law are [2, NG2–3, NG6–16]; the inherited allowance profile remains the H4–5 theorem with its stated eventual domain. All finite statements below precede that asymptotic input. Let J_N be the full remainder map, $L_N = H_N^{-1} J_N^* G_N$ its complete minimum section, and $M[P] = [yP]$ on $E = \mathbb{C}[y]/(Q_k)$. The adjoint is in G_N . Since k is odd and $\delta > 0$, $Q_k(0) \neq 0$. Define

$$u = J_N \left(\frac{Q_k - Q_k(0)}{y} \right), \quad \varphi_N(x) = (L_N x)(0), \quad \mathcal{D}p = \frac{p - p(0)}{y}, \quad B_N = J_N \mathcal{D} L_N. \quad (\text{LS1})$$

The vector u is the same algebraic quotient class at every cutoff, although its norm and φ_N depend on G_N . Multiplying by y gives $Mu = -Q_k(0)[1]$; hence $u \neq 0$. For every $x \in E$, multiplying $B_N x$ by M gives $x - \varphi_N(x)[1]$. Therefore

$$M^{-1} = B_N - \frac{u \varphi_N}{Q_k(0)}, \quad \|B_N\| \leq D_N, \quad E_N = \frac{\|u\| \|\varphi_N\|}{|Q_k(0)|}. \quad (\text{LS2})$$

The full explicit D_N is S16. Its proof from the Gamma bidiagonal inverse and the full original comparison is preserved in [1], S11–16; no covariance or source direction has been deleted.

Assume the finite, directly specified guard $E_N > 2D_N$. The triangle inequality and the kernel of φ_N give

$$\sigma_1(M^{-1}) \geq E_N - D_N > D_N \geq \sigma_2(M^{-1}). \quad (\text{LS3})$$

Thus the smallest singular value $s_{q,N}$ of M is simple. Let $P_{\min,N}$ project onto its right singular line, and let

$$P_{u,N} x = u \frac{u^* G_N x}{u^* G_N u}, \quad \eta_N = \frac{D_N}{E_N - D_N} < 1.$$

Then

$$\boxed{\|P_{\min,N} - P_{u,N}\| \leq \eta_N, \quad \|P_{\min,N} - P_{u,N}\|_1 \leq 2\eta_N.} \quad (\text{LS4})$$

To prove this, choose a unit right singular vector b for the largest singular value σ of M^{-1} , and put $a = M^{-1}b/\sigma$. Applying $I - P_u$ to LS2 kills its rank-one part and gives $\|(I - P_u)a\| \leq D_N/\sigma \leq \eta_N$. The left singular vector a of M^{-1} is the right singular vector of M for its smallest value. In the span of a, u , the difference of the two rank-one orthogonal projections has eigenvalues $\pm\sqrt{1 - |\langle a, u/\|u\| \rangle|^2}$; this follows by writing $a = \alpha u/\|u\| + \beta w$ in an orthonormal pair. Its operator and trace norms prove LS4.

For any $\tau > 0$, the entire spectrum consequently satisfies

$$\boxed{\left\| e^{-\tau M^\dagger M} - e^{-\tau s_{q,N}^2} P_{u,N} \right\|_1 \leq 2\eta_N e^{-\tau s_{q,N}^2} + (q-1)e^{-\tau/D_N^2}.} \quad (\text{LS5})$$

Indeed the other $q-1$ singular values are at least D_N^{-1} by S5, and their positive spectral projectors contribute the second term. The first term is exactly LS4 multiplied by the remaining scalar heat weight. This proof supplies the actual line; a statement about the smallest singular value alone does not supply LS5.

11 The original kernel and its complementary observation

Let $\Lambda : E \rightarrow B$ be the unchanged original observation, $K = \ker \Lambda$, and

$$I_K = V^{-1}\pi^T, \quad Z = [M_A, S_k Z_k], \quad \pi Z = 0, \quad P_K = I_K(I_K^* G_N I_K)^{-1} I_K^* G_N. \quad (\text{LS6})$$

These are exactly [3, MR1–10]. Retain

$$Q_N = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad \mathcal{L}_N = G_N^{-1} \Lambda^* Q_N, \quad P_B = \mathcal{L}_N \Lambda = I - P_K.$$

Here $\mathcal{L}_N$ is the observation's minimum section, whereas L_N in LS1 lifts the full polynomial remainder. Their domains are B and E , respectively. Direct substitution proves $\Lambda \mathcal{L}_N = I_B$, $\mathcal{L}_N^* G_N \mathcal{L}_N = Q_N$, and $\Lambda = \mathcal{L}_N^\dagger$ for these metrics.

The exact kernel fraction of the surviving class is

$$\begin{aligned} \theta_{K,N} &= \frac{u^* G_N I_K (I_K^* G_N I_K)^{-1} I_K^* G_N u}{u^* G_N u} \\ &= 1 - \frac{(\Lambda u)^* Q_N (\Lambda u)}{u^* G_N u} \in [0, 1]. \end{aligned} \quad (\text{LS7})$$

The second equality is the orthogonal decomposition $P_K + P_B = I$. It is an exact expression on the full original kernel and the actual observed class Λu .

Taking traces in LS5, first against P_K and then against P_B , gives, with its unchanged right side denoted $\epsilon_N(\tau)$,

$$\begin{aligned} \left| \text{Tr}(P_K e^{-\tau M^\dagger M}) - e^{-\tau s_{q,N}^2} \theta_{K,N} \right| &\leq \epsilon_N(\tau), \\ \left| \text{Tr}_B(\Lambda e^{-\tau M^\dagger M} \mathcal{L}_N) - e^{-\tau s_{q,N}^2} (1 - \theta_{K,N}) \right| &\leq \epsilon_N(\tau). \end{aligned} \quad (\text{LS8})$$

The second trace equality uses cyclicity: $\text{Tr}_B(\Lambda T \mathcal{L}_N) = \text{Tr}_E(P_B T)$. The estimate $|\text{Tr}(PT)| \leq \|T\|_1$ for an orthogonal projection follows by expanding T into singular vectors and using $|\langle b, Pa \rangle| \leq 1$ on each term.

The S4–S5 asymptotics show that E_N/D_N tends to infinity on the scale $\exp\{q \log(q/k) + O_h(q)\}$; this also follows from LS2 and $\|M^{-1}\| = s_{q,N}^{-1}$. Thus $\eta_N \rightarrow 0$, and the guard in LS3 holds eventually. At the common physical time S43, the smallest heat factor tends to one at $N = q-1, q$ and to zero at $N = 2q-1, 2q$. For the exact four-return with signs $+, +, -, -$, LS8 proves

$$\begin{aligned} \mathcal{R} \text{Tr}(P_K e^{-\tau_k^* M^\dagger M}) &= \theta_{K,q-1} + \theta_{K,q} + o_h(1), \\ \mathcal{R} \text{Tr}_B(\Lambda e^{-\tau_k^* M^\dagger M} \mathcal{L}_N) &= 2 - \theta_{K,q-1} - \theta_{K,q} + o_h(1). \end{aligned} \quad (\text{LS9})$$

No limit for the individual θ is assumed. Their sum cancels exactly between the two original receivers, recovering the full return $2 + o_h(1)$. LS7 is the finite next calculation for the allocation between these two receivers.

12 The exact relation to the observed action's own heat

Let $A_B = \Lambda M \mathcal{L}_N$. Use the orthogonal decomposition $E = K \oplus \mathcal{L}_N B$ with the actual metrics. Write the blocks of M as

$$M = \begin{pmatrix} A & D \\ C & A_B \end{pmatrix}, \quad D = P_K M \mathcal{L}_N, \quad C = \Lambda M|_K,$$

so that $H = M^\dagger M$ has blocks

$$H_{KK} = A^\dagger A + C^\dagger C, \quad H_{KB} = A^\dagger D + C^\dagger A_B, \quad H_{BB} = D^\dagger D + A_B^\dagger A_B. \quad (\text{LS10})$$

Every term is in the original attained metrics. For real $z > 0$, block elimination gives the complete receiver

$$\Lambda(z + H)^{-1} \mathcal{L}_N = \left[z + A_B^\dagger A_B + D^\dagger D - H_{BK}(z + H_{KK})^{-1} H_{KB} \right]^{-1}. \quad (\text{LS11})$$

The inverse exists because $z + H$ is positive definite; its Schur complement is positive definite as a complete minimum. The upper block equation solves the kernel variable as $-(z + H_{KK})^{-1} H_{KB}$ times the observed variable; substitution proves LS11.

For completeness, the time-domain connecting equation also retains these terms. Put $T_B(t) = \Lambda e^{-tH} \mathcal{L}_N$. Solving the kernel block with zero initial kernel value gives

$$T'_B(t) = -(A_B^\dagger A_B + D^\dagger D)T_B(t) + \int_0^t H_{BK} e^{-(t-s)H_{KK}} H_{KB} T_B(s) ds, \quad T_B(0) = I. \quad (\text{LS12})$$

Differentiation verifies the equation directly. In finite dimension its Laplace transform converges for $z > 0$ and yields LS11. Thus LS8–9 return the full original heat to the actual observation; LS10–12 specify the complete coupling required to compare it with $e^{-tA_B^\dagger A_B}$. Neither coupling has been set to zero, and no projected-current phase is inferred from these positive traces.

13 The surviving class is detected by the actual conductor

The full kernel in LS6 is contained in the conductor kernel $V_A = \{[p] : \chi' \mid \mathcal{T}_{Ap}\}$ of KF6 [4]. Here χ and χ' are the complete upper and lower polynomials of OR1 [5]; χ' again denotes the lower polynomial. Let $\mathcal{J}$ be the indices of nonzero actual coefficients $a_j = a_{rs}$, let $J = |\mathcal{J}|$, and retain the distinct shifts $b_j = 4 + (2r - 8)\delta + i(2s - 8)\gamma$. Define the actual rational function

$$\mathcal{R}_A(z) = \sum_{j \in \mathcal{J}} \frac{a_j}{z + b_j} = \frac{P_A(z)}{B_A(z)}, \quad B_A(z) = \prod_{j \in \mathcal{J}} (z + b_j), \quad P_A(z) = \sum_{j \in \mathcal{J}} a_j \prod_{\ell \neq j} (z + b_\ell). \quad (\text{LS13})$$

Expansion at infinity gives $\mathcal{R}_A(z) = \sum_{n \geq 0} (-1)^n \mu_n z^{-n-1}$. The original symbol order v is at most $J - 1$ by the Vandermonde argument on these J shifts. Consequently

$$\deg P_A = d := J - v - 1, \quad [z^d] P_A = (-1)^v \mu_v \neq 0. \quad (\text{LS14})$$

Every residue at $-b_j$ equals the original nonzero a_j . No denominator or zero of this rational function is inserted by changing the original conductor.

In the original S coordinate the representative of u is $\hat{u}(S) = u((S - c)/i)$. For every lower root $\eta = \beta'_{ab}$, each $\eta + b_j$ is an upper root, so LS1 gives exactly

$$(\mathcal{T}_A \hat{u})(\eta) = -iQ_k(0) \mathcal{R}_A(\eta - c). \quad (\text{LS15})$$

The denominators are nonzero: $\eta + b_j - c$ is a centered upper root, whose real part is a nonzero odd multiple of δ . The $q' = (k - 7)^2$ arguments $\eta - c$ are distinct. Thus at least $q' - d$ of these conductor values are nonzero whenever $q' > d$. In particular $u \notin V_A$, hence $u \notin K$, and $\theta_{K,N} < 1$ at every cutoff on this finite domain. This is a statement about the original conductor rows, independent of the unknown metric allocation.

A quantitative version uses the same original rows. Put $D = \sqrt{\delta^2 + \gamma^2}$, $A_1 = \sum_j |a_j|$, and impose $k \geq d + 8$. Choose the $d + 1$ lower roots $\eta_j = c' + (2j - k + 8)\delta - i(k - 8)\gamma$, $0 \leq j \leq d$. The d th finite difference of P_A on this row gives

$$\sum_{j=0}^d (-1)^{d-j} \binom{d}{j} P_A(\eta_j - c) = (2\delta)^d d! (-1)^v \mu_v.$$

The sum of the absolute binomial coefficients is 2^d . At least one index j therefore satisfies $|P_A(\eta_j - c)| \geq \delta^d d! |\mu_v|$. Each denominator factor in LS13 has modulus at most kD , because it is exactly a centered upper root. Equation LS15 gives

$$\max_{0 \leq j \leq d} |(\mathcal{T}_A \hat{u})(\eta_j)| \geq |Q_k(0)| \frac{\delta^d d! |\mu_v|}{(kD)^J}. \quad (\text{LS16})$$

Let $G_N^{\text{val}} = V^{-*} G_N V^{-1}$ be the original attained metric in the literal upper-root value coordinates, and let κ_N^{val} be its condition number in the Euclidean metric on those specified coordinates. A conductor row in LS16 is a linear functional ℓ on E that annihilates the entire K . Its coefficient vector has Euclidean norm at most A_1 . Hence $\|\ell\|_{G_N}^2 \leq A_1^2 / \lambda_{\min}(G_N^{\text{val}})$. Also $u(\beta_\alpha) = -iQ_k(0)/(\beta_\alpha - c)$, so

$$\|u\|_{G_N}^2 \leq \lambda_{\max}(G_N^{\text{val}}) q |Q_k(0)|^2 / \delta^2.$$

Since $\ell(u) = \ell(P_B u)$, the dual norm inequality and LS16 prove the explicit native observation lower bound

$$1 - \theta_{K,N} \geq \frac{\delta^{2d+2} (d! |\mu_v|)^2}{q \kappa_N^{\text{val}} A_1^2 (kD)^{2J}} > 0. \quad (\text{LS17})$$

The factor $|Q_k(0)|^2$ cancels between the two displayed physical norms, as the derivation shows. The complete invariant image is retained: a conductor row annihilates K precisely because it is a row of its full original observation, not because the other invariant rows were omitted. KF33–35 and the original full-form comparison give $\log \kappa_N^{\text{val}} = O_h(q \log(q+2))$. Thus LS17 is a quantitative positive detection bound at all four cutoffs. It does not assign a nonzero limiting fraction or the still uncomputed kq determinant coefficient.

References

- [1] *The unique small singular direction of the native arithmetic quotient*, supplied mathematical continuation, 21 September 2026, S1–45. Complete proof included in this edition; the original correspondence remains private.
- [2] Split-Zero research programme, *Gaussian spectral actions through the original arithmetic quotient*, NG1–16, public proof, NG6 and NG10 and its complete proof.
- [3] Split-Zero research programme, *The original kernel as a mixed determinant*, MR1–23, public proof, original frame and metric.
- [4] *The actual invariant kernel: degree filtration, quantitative minors, and its complete conductor graph*, KF1–55. Complete proof accompanies this edition in the included LaTeX source, particularly KF6–10 and KF33–35.
- [5] *The roots of the complete conductor polynomial*, OR1–25. Complete proof accompanies this edition in the included LaTeX source.

Part IV

The full invariant kernel and conductor graph

14 The actual invariant kernel: degree filtration, quantitative minors, and its complete conductor graph

21 September 2026. Independent derivation on the original simple-quartet five-orbit domain. All matrices below use the original period, amplitude, root ordering and units. In particular the invariant image is the entire matrix $Z=[M_A, S_k Z_k]$, not just its conductor columns.

The proved advancement is a quantitative degree-filtration theorem and a reduction of the actual kernel determinant, with error $\mathfrak{o}(kq)$, to its specified image in the complete weighted conductor-relation quotient. The fixed low-degree graph component is controlled by an actual angle estimate from invariant jets. A rank estimate alone would not prove this. The growing relation operator remains present in the final receiver. Its replacement by an outer polynomial ideal is not asserted.

14.1 1. Original data and the exact residual constraint

Retain $k \geq 9$, $k=1 \pmod{4}$,

$$q = (k+1)^2, \quad k' = k-8, \quad q' = (k-7)^2, \quad \Delta = q - q' = 16k - 48, \quad m = 8k - 16, \quad r_0 = 8k - 32. \quad (\text{KF1})$$

The original roots and their lower counterparts are

$$\beta_{ab} = k/2 + (2a - k)\delta + i(2b - k)\gamma, \quad \beta'_{ab} = (k-8)/2 + (2a - k + 8)\delta + i(2b - k + 8)\gamma,$$

with $0 < \delta < 1/2$, $\gamma > 2$. Let χ and χ' be their full monic root polynomials. The literal value matrix is V , and the full original kernel frame is

$$I_K = V^{-1}\pi^T, \quad VK = \ker Z^T = \text{im } \pi^T, \quad Z = [M_A, S_k Z_k]. \quad (\text{KF2})$$

Throughout, the four-return means exactly $Rf = f_{\{q-1\}} + f_{q-f_{\{2q-1\}}-f_{\{2q\}}}$. Transposes in this equality are complex-linear transposes. Metric adjoints will be written separately. The full conductor biform is $A = \sum_{r,s=0}^{\infty} a_{rs} e_{rs} e_{\{rs\}}^{\{8\}}$. Define

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad E_A(z) = \sum_{r,s} a_{rs} e^{b_{rs}z}, \quad \mu_j = E_A^{(j)}(0), \quad v = \text{ord}_0 E_A \leq 80. \quad (\text{KF3})$$

The constant v and the nonzero μ_v are fixed by the original data. Put

$$g = \Delta - v, \quad d = q' + v, \quad \mathcal{L} = \mathbb{C}[S]_{<v} \subset E = \mathbb{C}[S]/(\chi).$$

When $v=0$ this space is zero and every statement concerning its angles or jets below is empty.

The complete conductor operator is

$$\mathcal{T}_A p(S') = \sum_{r,s} a_{rs} p(S' + b_{rs}) = \frac{\mu_v}{v!} \partial^v B(\partial) p(S'), \quad B(z) = \frac{v! E_A(z)}{\mu_v z^v}, \quad B(0) = 1. \quad (\text{KF4})$$

On any fixed polynomial space this is a finite identity: only the derivatives up to the degree of p occur. It kills exactly the polynomials of degree below v and lowers every other degree by exactly v , with

$$[S'^{n-v}] \mathcal{T}_A S^n = \mu_v \binom{n}{v} \neq 0. \quad (\text{KF5})$$

Root addition proves $\chi' \text{ divides } T_A(\chi h)$ for every h . The conductor kernel is

$$V_A = \{[p] : \chi' \mid \mathcal{T}_A p\}, \quad \dim V_A = \Delta, \quad K \subset V_A. \quad (\text{KF6})$$

For a fixed representative of degree below q define

$$\eta(p) = \mathcal{T}_A p / \chi' \in \mathbb{C}[S']_{<g}, \quad p \in V_A. \quad (\text{KF7})$$

It is a polynomial by KF6. The quotient map $V_A \rightarrow P_{<g}$ is onto and has kernel L : onto follows by successively solving KF5 for a preimage of $\chi' \eta$, whose degree is below $q-v$; the kernel follows from KF4. This is the cutoff-independent low-polynomial version of the CEP43 map, with its unscaled μ_v factor retained.

Let R be the unique right inverse of T_A with zero coefficients in degrees below v . An explicit construction is

$$Rf = \frac{v!}{\mu_v} [B(\partial)^{-1} \partial_0^{-v} f - \text{low}_{<v}(B(\partial)^{-1} \partial_0^{-v} f)]. \quad (\text{KF8})$$

Here $\text{partial}_0^{-v} S'^j = j! S'^{j+v} / (j+v)!$, and $B(\text{partial})^{-1}$ is its finite nilpotent inverse on the degree in question. Subtracting the low part changes no image under T_A , so KF8 has the asserted properties. Thus every original representative in V_A is uniquely

$$p = l + R(\chi' \eta), \quad l \in \mathcal{L}, \quad \deg \eta < g. \quad (\text{KF9})$$

Let J_R be the complete matrix $S_k Z_k$ of the additional r_0 invariant columns. The actual remaining constraint is exactly

$$\mathcal{B}_k(l, \eta) := J_R^T V(l + R(\chi' \eta)) = 0. \quad (\text{KF10})$$

It has rank r_0 on the Delta-dimensional space in KF9, since its kernel is exactly K of dimension m . The conductor columns were used to reach V_A , and every other invariant column remains in KF10. There are no generic coefficients in this description.

14.2 2. Exact degree filtration and high-degree projection minors

For $v \leq n \leq d + v$, a polynomial p of degree at most n in V_A has $\deg T_A p < q'$. Divisibility by the monic polynomial χ' of degree q' therefore forces $T_A p = 0$, so p belongs to L . Hence

$$V_A \cap \mathcal{P}_n = \mathcal{L} \quad (v-1 \leq n < d), \quad K \cap \mathcal{P}_n = K \cap \mathcal{L} \quad (v-1 \leq n < d). \quad (\text{KF11})$$

For $0 \leq t < g$, the leading coefficient in KF5 gives

$$\deg R(\chi' \eta) = q' + v + \deg \eta \quad (\eta \neq 0).$$

Consequently the complete actual filtration is

$$\dim(K \cap \mathcal{P}_{d+t}) = v + t + 1 - \text{rank } \mathcal{B}_k|_{\mathcal{L} \oplus \mathcal{P}_t'.} \quad (\text{KF12})$$

The initial value is $s_k = \dim(K \cap \mathcal{P}_d) = v - \text{rank } \mathcal{B}_k|_{\mathcal{L}}$. All these ranks use the actual matrix KF10. At the last $t = g-1$ their value is m . This proves that every degree pivot outside the first v positions lies in the last g positions $d, \dots, q-1$. It does not assert that the last m positions alone give an invertible minor.

There is a quantitative coefficient comparison on this high block. For p in V_A write $h_j = [S'^{d+j}]p$, $0 \leq j < g$, and let η_j be the coefficients in KF7. Let C be the high-coefficient multiplication matrix of χ' : its entry C_{ij} for $i \leq j$ is the coefficient of $S'^{q'-(j-i)}$ in χ' , and it is zero for $i > j$. Let T have entries

$$T_{ij} = \binom{d+j}{v+j-i} \mu_{v+j-i} \quad (i \leq j), \quad T_{ij} = 0 \quad (i > j). \quad (\text{KF13})$$

Comparison of coefficients in $T_{Ap=chi'} \eta$ proves

$$C\eta = Th, \quad \eta = D_k h, \quad D_k = C^{-1}T, \quad \det D_k = \mu_v^g \prod_{j=0}^{g-1} \binom{d+j}{v}. \quad (KF14)$$

C has diagonal one. This determinant is nonzero for the actual data, including $v=0$, and has not been inferred from genericity.

The following explicit bounds make its conditioning quantitative. Put

$$A_{\text{abs}} = \sum |a_{rs}|, \quad B_0 = \max(1, \max |b_{rs}|), \quad C_v = A_{\text{abs}} B_0^v / |\mu_v|, \\ R_0 = B_0^{-1} \min\{1, (v+1)/(2eC_v)\}, \quad R' = \max(1, \max |\beta'_{ab}|). \quad (KF15)$$

The exponential Taylor series gives $|B(z) - 1| \leq C_v B_0 |z| \exp(B_0 |z|) / (v+1) \leq 1/2$ for $|z| \leq R_0$. Thus the coefficients c_{-1} of $1/B$ obey $|c_{-1}| \leq 2R_0^{-1}$ by Cauchy's coefficient integral. The complete inverse of T is

$$(T^{-1})_{ij} = \frac{v!}{\mu_v} c_{j-i} \frac{(q' + j)!}{(d + i)!} \quad (i \leq j). \quad (KF16)$$

This follows by applying the integration and inverse derivative series in KF8 to a single high output monomial; lower terms do not enter the high coefficient block. Since the factorial ratio is at most $q^{\wedge\{j-i\}}$, the following conservative Euclidean operator bounds hold:

$$\|C\| \leq g(1 + q'R')^{g-1}, \\ \|C^{-1}\| \leq g(1 + (q' + g)R')^{g-1}, \\ \|T\| \leq gA_{\text{abs}}(1 + qB_0)^{v+g-1}, \\ \|T^{-1}\| \leq \frac{2gv!}{|\mu_v|} \max(1, q/R_0)^{g-1}. \quad (KF17)$$

For C , use the elementary symmetric coefficient bound $\text{binom}(q', 1) (R')^{\wedge 1}$; for its inverse use the complete homogeneous coefficient bound $\text{binom}(q'+1-1, 1) (R')^{\wedge 1}$. The matrix norm is at most g times its largest entry. The T bound uses the finite binomial sum and $|\mu_j| \leq A_{\text{abs}} B_0^j$. These proofs retain every coefficient.

Let $B_+ = \|C^{\wedge -1}\| \|T\|$ and $B_- = \|T^{\wedge -1}\| \|C\|$, where the bounds mean exactly KF17. Then

$$\|D_k\| \leq B_+, \quad \|D_k^{-1}\| \leq B_-, \quad \log^+ B_+ + \log^+ B_- = O_{\text{actual}}(k \log(k+1)). \quad (KF18)$$

For any h -dimensional actual plane in V_A/L , let H and Y be its high-coefficient and eta-coefficient matrices in the same frame. KF14 says $Y = D_k H$. Cauchy–Binet and the operator bounds give

$$B_+^{-2h} \det(Y^* Y) \leq \sum_{|J|=h} |\det H_J|^2 \leq B_-^{2h} \det(Y^* Y). \quad (KF19)$$

In particular at least one actual high-degree projection minor satisfies

$$|\det H_J|^2 \geq \binom{g}{h}^{-1} B_+^{-2h} \det(Y^* Y) > 0. \quad (KF20)$$

The statement applies to the image of K after quotient by $K \cap L$, with $h = m - s_k$. Its logarithmic coefficient distortion is $O_{\text{actual}}(k^2 \log(k+1)) = o(kq)$. The Euclidean coefficient metrics used in KF17–KF20 have been specified explicitly. They are not identified with a Gamma minimum metric; the metric transport below is separate.

14.3 3. The actual low-degree angle from the full invariant image

The OCF data give an invertible matrix H_{mathsf} and the exact substitution $\mathbf{x} = H_{\text{mathsf}}(\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3, \mathbf{t}_4)^T$. Let $\rho_1, \dots, \rho_4$ be the original one-factor roots in their prescribed order, and put

$$x_i(z) = \sum_{j=1}^4 (H_{\text{mathsf}})_{ij} e^{\rho_j z}. \quad (\text{KF21})$$

A degree- k invariant monomial $F(\mathbf{x})$ restricts to the complete biform coefficient vector $\mathbf{z}_F$ and hence satisfies

$$F(x(z)) = \sum_{a,b} (z_F)_{ab} e^{\beta_{ab} z}, \quad z_F^T V p = p(\partial_z) F(x(z))|_{z=0}. \quad (\text{KF22})$$

The first equality is the original Segre multiplication rule; its second equality follows by differentiating each term. This is the exact link from invariant jets to the low polynomial directions.

Assume $v \geq 1$. Work in the finite jet algebra $J_v = \mathbb{C}[\mathbf{z}]/(\mathbf{z}^v)$. Let J_+ be the indices i with $x_i(0)$ nonzero, and J_0 its complement. There is at least one index in J_+ because H_{mathsf} is invertible and the vector $(1, 1, 1, 1)$ is nonzero. Choose any such p . Write $\mathbf{s} = |J_+|$, $\mathbf{z}_0 = |J_0| = 4 - \mathbf{s}$.

For a fixed residue epsilon modulo five define the finite base set B_{epsilon} as the exponent vectors $\mathbf{r}$ satisfying all of the following:

- $0 \leq r_i \leq 4$ for i in J_+ ;
- $0 \leq r_i \leq v-1$ for i in J_0 ;
- $\sum_{i \in J_+} r_i \equiv 0 \pmod{5}$ and $\sum_{i \in J_0} r_i \equiv \text{epsilon} \pmod{5}$.

The weights i are exactly the original C5 weights 1, 2, 3, 4. If this base set is empty, every degree- k invariant jet is zero for k in this residue, so L is contained in K ; no nonzero low angle is then needed. Otherwise let

$$D_{\epsilon} = \max_{\mathbf{r} \in B_{\epsilon}} |\mathbf{r}|, \quad A_v = (s-1)(v-1), \quad d_{\epsilon} = D_{\epsilon} + 5A_v. \quad (\text{KF23})$$

It obeys $d_{\text{epsilon}} \leq 4\mathbf{s} + (v-1)(4-\mathbf{s}) + 5(\mathbf{s}-1)(v-1) \leq 15v+1$. Every base degree has the same residue as d_{epsilon} .

For i in J_+ other than p define the unit jet $R_i = (x_i/x_p)^5$. Its difference from its constant term is nilpotent of order at most v . Thus $(R_i - R_i(0))^v = 0$, and all powers of R_i are linear combinations of $1, R_i, \dots, R_i^{v-1}$. Products of these powers reduce to exponents $0 \leq a_i < v$ with total at most A_v .

For every $\mathbf{r}$ in B_{epsilon} and every such exponent list $\mathbf{a}$, the following is an actual invariant monomial of degree d_{epsilon} :

$$F_{r,a}(x) = x^{\mathbf{r}} \prod_{i \in J_+ \setminus \{p\}} x_i^{5a_i} x_p^{d_{\epsilon} - |\mathbf{r}| - 5 \sum a_i}. \quad (\text{KF24})$$

Its last exponent is a nonnegative multiple of five by KF23. These give a fixed finite list of monomials, independent of growing k .

For every $k \geq d_{\text{epsilon}}$ of that residue, the complete space of degree- k invariant jets is exactly

$$\boxed{\text{span } j_{<v} \{F(x(z)) : F \in \mathcal{J}_k\} = j_{<v}(x_p^{k-d_{\epsilon}}) \text{span}_{r,a} j_{<v} F_{r,a}(x(z)).} \quad (\text{KF25})$$

To prove the nontrivial inclusion, first discard any monomial having an exponent at least v on a zero-constant x_i : its jet vanishes. For every nonzero-constant variable divide its exponent by five, retaining its residue between zero and four. The resulting base lies in B_{epsilon} . Factor the remaining powers as a common power of x_p times powers of the unit ratios R_i . Reduce each latter power by its nilpotent polynomial identity. KF24 is precisely the resulting finite list after extracting the common factor $x_p^{k-d_{\text{epsilon}}}$. Conversely each monomial in KF24

multiplied by $x_p^{k-d_\epsilon}$ is itself invariant and has degree k , so its jet belongs to the left side. This proves equality.

Write J_ϵ for the v by J matrix whose columns are the Taylor coefficients of the fixed jets in KF24. Let t_ϵ be its rank. If it is zero, the preceding empty-angle case again applies. Otherwise let

$$s_\epsilon = \sigma_{\min}^+(J_\epsilon) > 0, \quad a = x_p(0) \neq 0, \quad w_* = \sum_{j=1}^{v-1} \frac{|x_p^{(j)}(0)|}{j!|a|}, \quad n = k - d_\epsilon. \quad (\text{KF26})$$

All of these are fixed finite expressions in the original period and roots except n . The positive constant s_ϵ can equivalently be defined by the square root of the smallest positive eigenvalue of $J_\epsilon J_\epsilon^T$; no numerical evaluation or generic lower bound is claimed for it. Its positivity follows from its actual nonzero rank.

Multiplication by the unit jet x_p^n is an invertible matrix U_n . The nilpotent binomial inverse gives the explicit bound

$$\|U_n^{-1}\| \leq B_n^{\text{unit}} := |a|^{-n} \sum_{j=0}^{v-1} \binom{n+j-1}{j} w_*^j \quad (n \geq 1), \quad B_0^{\text{unit}} = 1. \quad (\text{KF27})$$

Indeed write the multiplication matrix of x_p as $a(I+W_0)$, with $W_0^v=0$ and $\|W_0\| \leq w_*$; expand $(I+W_0)^{-n}$ exactly through degree $v-1$. In particular $\log^+ B_n^{\text{unit}} = 0_{\text{actual}}(k + \log(k+1))$.

Use the actual invariant monomials $F_{\{r,a\}}(x) x_p^{k-d_\epsilon}$ as rows of a coefficient matrix B_k^T . Put

$$H_* = \max \left(1, \max_i \sum_j |(H_{\text{mathsf{sf}}})_{ij}| \right), \quad R_k^S = \max(1, \max |\beta_{ab}|), \quad L_k^{\text{val}} = \sqrt{qv} (R_k^S)^{v-1}. \quad (\text{KF28})$$

The multinomial theorem and the complete Segre substitution give $\|B_k^T\| \leq \sqrt{J} H_*^k$; there are no coefficients deleted in this estimate. The value map on coefficient vectors of polynomials of degree below v has norm at most L_k^{val} . By KF22 its low observation matrix is exactly

$$O_k = J_\epsilon^T U_n^T \text{diag}(0!, 1!, \dots, (v-1)!). \quad (\text{KF29})$$

Ordinary transpose preserves singular values, and the smallest singular value of the factorial matrix is one. Thus

$$\sigma_{\min}^+(O_k) \geq s_\epsilon / B_n^{\text{unit}}. \quad (\text{KF30})$$

For clarity, put $A = J_\epsilon^T$ and $B = U_n^T \text{diag}(0!, \dots, (v-1)!)$. Then $BB^* \geq (B_n^{\text{unit}})^{-2} I$, so $0_{k_0} \geq (B_n^{\text{unit}})^{-2} AA^*$. The two products have the same rank because B is invertible. The minimum characterization of their ordered eigenvalues proves KF30.

KF25 proves that the kernel of this selected low observation is exactly $K \cap L$, rather than the kernel of a smaller selection. For l in L , let distance be measured in Euclidean root-value coordinates. Then

$$\boxed{\text{dist}(Vl, VK) \geq \varepsilon_k \text{dist}(Vl, V(K \cap L))}, \quad \varepsilon_k = \min \left(1, \frac{s_\epsilon}{B_n^{\text{unit}} \sqrt{J} H_*^k L_k^{\text{val}}} \right) > 0. \quad (\text{KF31})$$

To prove it, every row of B_k^T vanishes on VK , so its value on Vl is at most $\sqrt{J} H_*^k \text{dist}(Vl, VK)$. On the other hand KF30 bounds it below by $(s_\epsilon / B_n^{\text{unit}})$ times coefficient distance to its kernel. That coefficient distance is at least the root-value distance to $V(K \cap L)$ divided by L_k^{val} . Combining the two estimates gives KF31. Its logarithmic cost is $\log(1/\varepsilon_k) = 0_{\text{actual}}(k + \log k)$. For the finitely many k below d_ϵ the

same statement uses the finite matrix of all degree- k invariant monomials in place of KF29; its positive singular values give an exact finite ϵ_k . The asymptotic statement needs only the proved stabilized range. When $v=0$ or the selected observation has rank zero, set $\epsilon_k=1$. In the latter case L is contained in K , and both distances in KF31 are zero. These conventions include every endpoint and residue class.

It also follows that

$$\dim(K \cap \mathcal{L}) = v - t_\epsilon \quad (k \geq d_\epsilon, k = \epsilon \bmod 5). \quad (\text{KF32})$$

This is an exact residue-class statement about the actual period.

14.4 4. Why top-degree transversality cannot be assumed from the five-orbit condition

The stronger assertion $K \cap L = 0$ is not a consequence of the algebraic five-orbit hypothesis alone. An explicit auxiliary matrix is

$$H_{\text{ex}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}.$$

It is invertible and sends $(1,1,1,1)^T$ to the first coordinate vector. Its transformed Segre equation is

$$Q = x_1x_4 - x_1x_2 - x_1x_3 - x_2x_3.$$

It is an irreducible rank-four quadric. Under the diagonal C5 action, its nonzero terms have weights 0,3,4, so its five translates are pairwise nonassociate: an associate would have scalar one from the weight-zero coefficient and would require both $\zeta^3 = \zeta^4 = 1$. At the first coordinate vector every invariant homogeneous monomial of degree k vanishes unless k is divisible by five. Therefore the constant polynomial has zero pairing with the entire invariant image for all k not divisible by five, and belongs to that example's actual kernel. Its conductor symbol vanishes at zero as well, since each translated quadric vanishes at that point, so this direction lies in its space L .

This is a counterexample within the stated invertible-matrix five-orbit algebra, not an assertion that H_{ex} is the programme's period matrix. For the original period KF25–KF32 give the exact deciding map and its quantitative nonzero constants. They do not replace it by the example.

14.5 5. Uniform Gamma condition bounds in the original value coordinates

Keep $S=c+iy$, $c=k/2$, and the actual mass $M_\sigma = \sqrt{2\pi}$ of $\sigma(y) = |\Gamma(1/4+iy/2)|^2/(2\pi)$. In root-value coordinates put $y_\alpha = (\beta_\alpha - c)/i$, so $|y_\alpha| \leq R_y = k\sqrt{\delta^2 + \gamma^2}$. At the four cutoffs $q-1, q, 2q-1, 2q$ let C_N be the full Gamma covariance with degrees zero through N and let $G_N^{\text{val}} = C_N^{-1}$.

The Cauchy estimate from the exact Meixner–Pollaczek generating function gives, for $N \geq 1$ and $j \leq N$,

$$|p_j(y_\alpha)|^2 / \gamma_j \leq \frac{e^2}{M_\sigma} (N+1)^{1/2} (2N+1)^{R_y+1}.$$

Here are the finite details. In $\sum_{j \leq N} p_j(y) t^j / j! = (1+t^2)^{-1/4} \exp(y \arctan t)$ take $r = N/(N+1)$. The convergent arctangent series gives $|\arctan t| \leq \arctan hr = (1/2) \log(2N+1)$ on that circle, and $1-r^2 \geq 1/(N+1)$. Cauchy's coefficient integral, together with $r^{-j} \leq e$, bounds $|p_j(y)| / j!$ by $e^{(N+1)^{1/4}} (2N+1)^{R_y/2}$. Finally $j! / (1/2)^j \leq 2j+1 \leq 2N+1$: its induction starts at one, and multiplication by $(j+1)/(j+1/2)$ takes $2j+1$ to $2j+2$. Squaring the

coefficient estimate and inserting $\text{gamma_j} = M_{\text{sigmaj}}!(1/2)_{\text{j}}$ proves the displayed bound. It follows by summing the norms of every value column that

$$\lambda_{\max}(C_N) \leq A_N^{\text{val}} := \frac{qe^2}{M_\sigma} (N+1)^{3/2} (2N+1)^{R_y+1}. \quad (\text{KF33})$$

For a lower bound retain the first q columns. The Lagrange polynomial at a root α has numerator $\text{prod}_{\{\text{beta} \neq \alpha\}}(y - y_{\text{beta}})$ and denominator $\text{prod}_{\{\text{beta} \neq \alpha\}}(y_{\text{alpha}} - y_{\text{beta}})$. Every root separation has modulus at least 2δ . Repeated use of the exact Gamma bound $\|yP\| \leq 2(\deg P + 1)\|P\|$ gives

$$\|\ell_\alpha\|_\sigma \leq \sqrt{M_\sigma} \left(\frac{2q + R_y}{2\delta} \right)^{q-1}.$$

To check the multiplier constant, in the orthonormal Gamma frame the two off-diagonal coefficients of y are $\text{sqrt}((j+1)(j+1/2))$ and $\text{sqrt}(j(j-1/2))$. The raising part on degrees through n has norm at most $n+1$ and the lowering part has norm at most n . Their sum is at most $2(n+1)$. Multiplication by each factor $y - y_{\text{beta}}$ therefore has norm at most $2q + R_y$ during the $q-1$ factors in a Lagrange numerator. This proves the asserted iteration starting from the constant polynomial of norm $\text{sqrt}(M_{\text{sigma}})$. Thus interpolation from a Euclidean unit root-value vector has squared norm at most

$$B_k^{\text{val}} := qM_\sigma \left(\frac{2q + R_y}{2\delta} \right)^{2(q-1)}. \quad (\text{KF34})$$

The minimum through degree N has no larger norm than this interpolation trial, so $G_N^{\text{val}} \leq B_k^{\text{val}}$. KF33 gives its lower bound. With

$$\kappa_k^* = \max\{1, A_{2q}^{\text{val}} B_k^{\text{val}}\},$$

all four complete Gamma metrics satisfy

$$(A_{2q}^{\text{val}})^{-1} I \preceq G_N^{\text{val}} \preceq B_k^{\text{val}} I, \quad \text{cond}(G_N^{\text{val}}) \leq \kappa_k^*, \quad \log \kappa_k^* = O_h(q \log(q+2)). \quad (\text{KF35})$$

The estimate is on the full covariance before restricting or quotienting. No original value direction is removed. The phases relating S and y have modulus one in each polynomial degree and root values are unchanged.

By comparison of distances under two positive metrics, KF31 implies the actual Gamma angle bound

$$\text{dist}_{G_N}(l, K) \geq \frac{\varepsilon_k}{\sqrt{\kappa_k^*}} \text{dist}_{G_N}(l, K \cap \mathcal{L}), \quad l \in \mathcal{L}. \quad (\text{KF36})$$

Indeed the first metric distance is at least the square root of its smallest eigenvalue times Euclidean distance. Use KF31 and then $\text{dist}_E(l, K_0) \geq \text{dist}_G(l, K_0) / \text{sqrt}(\lambda_{\max}(G))$. This proves the stated ratio.

14.6 6. Removing the fixed low graph from the actual kernel determinant

Put $K_0 = K \cap L$, $s_k = \dim K_0$, and let $\bar{K}$ be the image of K in V_A/L . Its rank is $h = m - s_k$. All spaces and frames in this paragraph are fixed across the four cutoffs. First give $\bar{K}$ the quotient metric from K/K_0 , denoted $\bar{H}^K$. Second give it the restriction of the complete quotient metric from V_A/L , denoted $\bar{H}^V$. Their definitions minimize over K_0 and L respectively.

There is the exact comparison

$$0 \leq e_N^{\text{ang}} := \log \det H_{\text{bar}, N}^K - \log \det H_{\text{bar}, N}^V \leq v \log \kappa_k^* + 2v \log(1/\varepsilon_k) =: B_k^{\text{ang}}. \quad (\text{KF37})$$

Here is the complete subspace argument. In the original Gamma metric remove K_0 orthogonally, put $A = K_{\text{orthogonal}} - K_0$, $B = L_{\text{orthogonal}} - K_0$, and put $\theta = \epsilon_k / \sqrt{\kappa_k}$. The map from K/K_0 to V_A/L is $J = (I - P_B)|_A$. KF36, applied to each b in B , gives

$$\|(I - P_A)b\|^2 \geq \theta^2 \|b\|^2, \quad \|P_A|_B\|^2 \leq 1 - \theta^2, \quad (P_B|_A)^* = P_A|_B.$$

Consequently, with $C = P_B|_A$,

$$J^*J = I_A - C^*C \succeq \theta^2 I_A, \quad \text{rank}(C^*C) \leq \dim B \leq v.$$

There are thus at most v eigenvalues different from one, and every eigenvalue belongs to $[\theta^2, 1]$. In a frame orthonormal for $\bar{H}_K$ the second metric is exactly J^*J . It follows that

$$0 \leq e_N^{\text{ang}} = -\log \det(J^*J) \leq -2v \log \theta = v \log \kappa_k^* + 2v \log(1/\epsilon_k).$$

This proves KF37, including nontrivial intersection K_0 . The fixed rank is used only after the actual quantitative angle bound has been proved.

Choose a fixed basis of K_0 and fixed lifts completing it to a frame of K . Schur complementation of its complete restricted Gram gives

$$\log \det H_{K,N} = \log \det H_{K_0,N} + \log \det H_{\text{bar},N}^K + C_{\text{frame}}, \quad (\text{KF38})$$

where the change from the original I_K frame has its fixed determinant recorded by C_{frame} . It cancels in the four-return only because that same frame is used at all four cutoffs.

The complete source minima decrease with N . Their restriction to K_0 therefore decreases, while KF35 bounds all four by the same two scalar factors in a Euclidean-orthonormal frame. Pairing the signs gives

$$0 \leq \mathcal{R} \log \det H_{K_0,N} \leq 2s_k \log \kappa_k^*. \quad (\text{KF39})$$

Also KF37 gives $|\text{Re}_N^{\text{ang}}| \leq 2B_k^{\text{ang}}$, because each of the four errors belongs to the same interval $[0, B_k^{\text{ang}}]$. Equations KF38–KF39 therefore prove

$$\left| \mathcal{R} \log \det H_{K,N}^\sigma - \mathcal{R} \log \det H_{\text{bar},N}^{V,\sigma} \right| \leq 2s_k \log \kappa_k^* + 2B_k^{\text{ang}} = O_{\text{actual}}(q \log(q+2)) = o(kq). \quad (\text{KF40})$$

This removes the finite low-degree component with a proved error for the actual invariant kernel. It is stronger than deleting a rank- v term without controlling its angle.

14.7 7. The complete relation graph and its source transport

For all four original cutoffs put $L = N - q$; it takes values $-1, 0, q-1, q$. Retain the exact outer products for every original shift,

$$d_{rs}(S') = \prod_{\substack{0 \leq i,j \leq k \\ (i,j) \notin \{r, \dots, r+k'\} \times \{s, \dots, s+k'\}}} (S' + b_{rs} - \beta_{ij}),$$

and the whole relation operator

$$\mathcal{U}_{A,L} h(S') = \sum_{r,s} a_{rs} d_{rs}(S') h(S' + b_{rs}), \quad \mathcal{T}_A(\chi h) = \chi' \mathcal{U}_{A,L} h. \quad (\text{KF41})$$

It is injective and has degree $\deg h + g$, with leading coefficient $\text{mu_vbinom}(q + \deg h, v)$ times the leading coefficient of h . Thus

$$Y_N = \mathcal{P}'_{L+g} / \mathcal{U}_{A,L} \mathcal{P}_L \quad (\text{KF42})$$

has dimension g at every cutoff. Negative degree means zero, so at the first cutoff it is exactly $\mathcal{P}'_{-1}$. Its source norm is the complete weighted lower norm

$$\|\eta\|_{\chi',1}^2 = \int_{\mathbb{R}} |\chi'(c' + iy) \eta(c' + iy)|^2 \sigma(y) dy, \quad c' = k/2 - 4. \quad (\text{KF43})$$

The source order is still one, and its mass factor has not been changed.

The actual map is

$$\Psi_N : V_A/\mathcal{L} \longrightarrow Y_N, \quad [p] \longmapsto [\mathcal{T}_{Ap}/\chi']. \quad (\text{KF44})$$

If a lift changes by χh , KF41 changes its image by U_{Ah} . If its image is zero, subtract that same χh and KF4 leaves a polynomial in L ; this proves injectivity. Surjectivity follows by solving the onto T_A source equation. The unique degree-below- q representative gives η of degree below g by KF7, so the coordinate matrix of Ψ_N on these low target representatives is independent of N . The exact actual plane in Y_N is

$$\overline{K}_N = \Psi_N(K/K_0). \quad (\text{KF45})$$

Its fixed low-polynomial frame is obtained from I_K by KF7 and quotient by its actual nullspace K_0 ; equivalently it is the projection of the kernel of the full residual constraint KF10. Thus KF45 retains the complete invariant image rather than an arbitrarily oriented rank- h plane in Y_N .

Here are finite constants for the source transport used in this map. Set $D=N-v$, let B_{-D}^+ be the original WCF12 order-one bound

$$B_D^+ = e^{2v} A_{\text{abs}} e^{2\pi\gamma} (2D+1)^{4\delta} \sqrt{2(D+v)+1} \quad (D \geq 1), \quad M_D = \max(1, B_D^+),$$

and put

$$g_0 = (-i)^v \mu_v / v!, \quad J_D = (D+1) \log^+(1/|g_0|), \quad E_N^* = (N+1) \log M_D + J_D. \quad (\text{KF46})$$

All present endpoints have $D \geq 1$. The exact orthonormal source operator $S_{Np} = (P_{<vp}, T_{Ap})$ has a block I_v and the WCF upper triangular block. Its forward norm is at most M_D . Its determinant is exactly

$$\det A_{D,1} = g_0^{D+1} \prod_{i=0}^D \rho_{i+v} / \rho_i, \quad \rho_i = \sqrt{i! / (1/2)_i}.$$

Since ρ_i increases, $|\det S_N|^{-1} \leq \exp J_D$. The singular-value complement identity therefore proves, for every rank h ,

$$\log \left\| \bigwedge^h S_N \right\| \leq h \log M_D, \quad \log \left\| \bigwedge^h S_N^{-1} \right\| \leq (N+1-h) \log M_D + J_D. \quad (\text{KF47})$$

These are bounds for the full original source map, not scalar bounds multiplied by a changing relation rank.

Its exact image of the conductor preimage is $C^{\sim} \text{vdirect-sum} \chi' P'_{\{L+g\}}$. Its image of the original relation χh is $(P_{<v(\chi h)}, \chi' U_{Ah})$. After quotient by the entire L , the free first component vanishes and the remaining metric is exactly the weighted complete quotient KF42–KF43. All low graph entries were retained before taking this quotient. The induced map on any actual subspace, in particular KF45, is a minimal lift followed by S_N and an orthogonal quotient projection. Projection contracts every exterior norm, and the same argument for S_N^{-1} supplies the reverse bound. Thus, in the transported actual frame,

$$\left| \log \det H_{\overline{K}_N, Y_N} - \log \det H_{\text{bar}, N}^{V, \sigma} \right| \leq 2E_N^*. \quad (\text{KF48})$$

This proves the source-graph comparison for the actual plane, using the complete original graph and its quotient order. WCF gives $E_N^* = O_{\text{actual}}(q \log(q+2)) = o(kq)$ at all four cutoffs.

Combining KF40 and KF48 with the previously proved JM25/MR21 arithmetic comparison gives the finite final receiver

$$\boxed{\left| \mathcal{R} \log \det H_{K, N}^{\text{ar}} - \mathcal{R} \log \det H_{\overline{K}_N, Y_N} \right| \leq 2mL_k^{\text{form}} + 2s_k \log \kappa_k^* + 2B_k^{\text{ang}} + 2 \sum_N E_N^*} \quad (\text{KF49})$$

$$= o_{\text{actual}}(kq).$$

The sum is over exactly $q-1, q, 2q-1, 2q$. Their source relation degrees are exactly $-1, 0, q-1, q$, and their target polynomial degrees are $g-1, g, q+g-1, q+g$. The arithmetic source retains $w_h = |2\pi i/h|^2/(2\pi)$ and its actual complete divisor. Its physical units enter I_K and the same original maps throughout; their prior isometries, rather than a replacement of the units by constants, are used by JM25.

14.8 8. What has been evaluated and the remaining connecting map

KF11–KF20 prove exact high-degree transversality modulo the actual low intersection and explicit growth bounds for the sum of the actual projection minors. KF25–KF32 evaluate that low intersection through a fixed finite jet matrix on each residue class and prove its quantitative nonzero-angle bound. KF49 proves that the entire fixed low component and the conductor source transport have subleading four-return cost.

The remaining rank $m-s_k$ metric is the restriction in KF45 of the complete quotient KF42. The relations are the actual sum of shifted outer products KF41. Its exact multiplication defect is

$$\mathcal{U}_A(Sh) - S'\mathcal{U}_Ah = \sum_{r,s} a_{rs} b_{rs} d_{rs}(S')h(S' + b_{rs}), \quad (\text{KF50})$$

with all original coefficients present. The identity alone does not prove that its right side has nonzero class modulo the complete relation image. Thus closure under multiplication is neither assumed nor inferred from this display. The following exact graph comparison decides the relation to an outer polynomial ideal without making that inference. It is not legitimate to replace U_A by one d_{rs} on the grounds that their source and target dimensions agree. Nor do KF18–KF20 convert coefficient control into a Gamma minimum without the metric steps above.

At each cutoff the residual determinant can be evaluated without dropping any cross Gram as follows. Choose the fixed low eta frame B_K for KF45; let H_N be the full weighted monomial Gram on $P'_\infty\{L+g\}$, let U_N be the full coefficient matrix of KF41, and let E_N insert B_K in that source. Then its exact restricted minimum is

$$H_{\overline{K}_N, Y_N} = E_N^* H_N E_N - E_N^* H_N U_N (U_N^* H_N U_N)^{-1} U_N^* H_N E_N. \quad (\text{KF51})$$

At $L=-1$ the second term is zero. The inverted matrix is positive definite because U_A is injective. The formula follows by minimizing the full quadratic form of $E_N b + U_N h$ in h ; the minimizing coefficient is $-(U_N^* H_N U_N)^{-1} U_N^* H_N E_N b$. It retains the complete source-graph correction at every endpoint. KF49 shows that evaluating its four-return determines the original kernel's kq coefficient, with the proved error, but the present argument does not yet give that remaining coefficient.

14.9 9. An exact graph over the actual outer polynomial

There is a canonical outer polynomial for this comparison, retaining every shift: put $d_A = U_A 1$. By KF41 its degree is exactly g and its leading coefficient is $a_A = \mu_v \text{binom}(q, v)$, which is nonzero. For each source degree $L \geq 0$ perform Euclidean polynomial division, with that actual leading coefficient, on the whole operator U_A :

$$\mathcal{U}_Ah = d_A \mathcal{C}_L h + \mathcal{R}_L h, \quad \deg \mathcal{R}_L h < g, \quad h \in \mathcal{P}_L. \quad (\text{KF52})$$

Here $\mathcal{C}_L$ and $\mathcal{R}_L$ are the unique linear maps given by division. They retain all original coefficients in KF41. If h has degree n , the leading term calculation in KF41 proves that $\mathcal{C}_L h$ has degree n and leading coefficient multiplied by $\text{binom}(q+n, v)/\text{binom}(q, v)$. Therefore $\mathcal{C}_L$ is upper triangular in the increasing monomial bases and

$$\det \mathcal{C}_L = \prod_{n=0}^L \frac{\binom{q+n}{v}}{\binom{q}{v}} > 0, \quad 0 \leq \log \det \mathcal{C}_L \leq v(L+1) \log \frac{q+L}{q-v+1}. \quad (\text{KF53})$$

For $v=0$ the determinant is one and the logarithmic upper bound is zero. For $v \geq 1$ each ratio is the product, over $0 \leq j < v$, of $(q+n-j)/(q-j)$, between one and $((q+L)/(q-v+1))^v$. This proves KF53 without estimating entries by a rank-only replacement. At $L \leq q$ the upper bound is $0_{\text{actual}}(q)$.

Define the complete graph coefficient map $F_L = R_{LC_L}^{-1} : P_L \rightarrow P'_{g-1}$. Equation KF52 proves exactly

$$\mathcal{U}_A \mathcal{P}_L = \{d_A w + F_L w : w \in \mathcal{P}_L\}. \quad (\text{KF54})$$

Every polynomial of degree at most $L+g$ has a unique decomposition $r + d_A w$ with $\deg r < g$, $\deg w \leq L$. The linear isomorphism

$$\mathcal{J}_L(r + d_A w) = r - F_L w + d_A w, \quad \mathcal{J}_L^{-1}(r + d_A w) = r + F_L w + d_A w \quad (\text{KF55})$$

fixes each low representative r and sends the complete relation image in KF54 onto the entire multiplication image $d_A P_L$. Its determinant in this displayed decomposition is one; its off-diagonal block is the full F_L . This is the exact relation between the original conductor quotient and that outer polynomial quotient. In particular the actual low frame B_K is preserved, and all of its cross terms remain.

To carry the metric as well, equip the outer polynomial source with the Gram $(J_L^{-1})^* H_{NJ_L^{-1}}$. Then J_L is an isometry from the complete weighted source H_N to that displayed source: direct substitution gives equality of squared norms. It descends to an isometry from Y_N onto the quotient by $d_A P_L$ with this exact transformed metric. Replacing this transformed Gram by H_N would discard F_L and is not a consequence of determinant one. At $L=-1$ set J_L to the identity, so this isometry includes the first cutoff.

Equations KF52–KF55 are finite evaluations of the complete source-graph correction as polynomial division and an invertible shear. In particular the exact obstruction to deleting that correction is the metric action of F_L , not the scalar determinant in KF53. The present proof establishes no subleading bound for that metric action. KF51 and KF55 give two equivalent exact next calculations, with all four cutoffs and the full actual frame, while KF49 has already bounded the low-angle and conductor transport corrections by $o(kq)$.

Exact programme proof sources

ORIGINAL_KERNEL_MIXED_DETERMINANT.tex, MR1 and the following complete proof.
 ORIGINAL_CONDUCTOR_STRIP_FRAME.tex, OCF1 and the following complete proof.
 WEIGHTED_CONDUCTOR_FORWARD.tex, WCF1 and the following complete proof.
 09_INPUT_CUMULATIVE_PROOFS.tex, CEP2 and the following complete proof.

Part V

The complete conductor graph volume and its invariant receiver

15 The complete conductor graph: a quantitative full-volume comparison

21 September 2026. This is an independent continuation of KF1–55 and OR1–25 on their original simple-quartet domain. The actual conductor, period, root grid, invariant columns and source order remain fixed.

The new evaluated assertion is CG24 below: the metric shear changes the determinant of the **entire g -dimensional conductor quotient** by $O_A(q \log(q+2)) = o(kq)$, at every original cutoff. This follows from a full polynomial-norm estimate that retains the exceptional conductor roots, a quantitative angle estimate and the complete WCF exterior bound. It does not yet bound the shear on the specified proper subspace $\bar{K}$. CG25–28 give the exact residual expression on that unchanged subspace; the full-volume estimate cannot be substituted for it.

15.1 1. Original objects and finite domain

Keep all notation and coefficient maps of KF1–55. In particular

$$q = (k+1)^2, \quad q' = (k-7)^2, \quad v = \text{ord}_0 E_A \leq 80, \quad g = q - q' - v, \quad a_A = \mu_v \left(\frac{q}{v} \right) \neq 0, \quad (\text{CG1})$$

and

$$\mathcal{U}_A h = \sum_{r,s=0}^8 a_{rs} d_{rs}(S') h(S' + b_{rs}), \quad d_A = \mathcal{U}_A 1, \quad \chi' \mathcal{U}_A h = \mathcal{T}_A(\chi h). \quad (\text{CG2})$$

Here χ' names the lower root polynomial, not a derivative. The entire relation and target metric are

$$Y_N = \mathcal{P}'_{L+g} / \mathcal{U}_A \mathcal{P}_L, \quad L = N - q \in \{-1, 0, q-1, q\}, \quad \|f\|_H^2 = \int_{\mathbb{R}} |\chi'(c' + iy) f(c' + iy)|^2 d\sigma(y), \quad (\text{CG3})$$

where $c' = k/2 - 4$ and

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad M_\sigma = \sqrt{2\pi}, \quad \frac{d\sigma}{dy} \geq C_L e^{-\pi|y|}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}. \quad (\text{CG4})$$

The mass, recurrence and Gamma lower bound are the established original ones. The lower bound also follows directly from Gamma reflection and the Euler integral bound on $\Gamma(3/4 - iy/2)$, as in OSP11.

Put $D_0 = \sqrt{\delta^2 + \gamma^2}$, $R_k = kD_0$, $B_k = 4 + R_k$, and retain r_*, a_*, A_1, B exactly from OR4, OR7 and OR14. In particular every zero ζ of $d_A(c' + iy)$ satisfies $|\zeta| < q/r_*$. Fix the numerical constant

$$\epsilon = 2^{-64}, \quad T = \epsilon q. \quad (\text{CG5})$$

The following estimates use the finite eventual domain

$$k \geq 29, \quad q \geq 2v, \quad k \geq 2v-1, \quad B_k \leq T/8, \quad q \geq 8/\epsilon, \quad u_4 := 8/T + (16 + R_k^2)/T^2 < 1/2. \quad (\text{CG6})$$

For each fixed original conductor these conditions hold eventually. No statement at an excluded small cutoff is inferred by asymptotic notation. The exact identities below also apply at those cutoffs.

15.2 2. Every Gamma linear factor has a polynomial lower bound

The original monic polynomials and norms are

$$p_0 = 1, \quad p_1 = y, \quad p_{j+1} = yp_j - j(j-1/2)p_{j-1}, \quad \gamma_j = M_\sigma j!(1/2)_j, \quad \phi_j = p_j/\sqrt{\gamma_j}. \quad (\text{CG7})$$

Their exact generating function is

$$\sum_{j \geq 0} p_j(y) t^j / j! = (1+t^2)^{-1/4} e^{y \arctan t}. \quad (\text{CG8})$$

This is the Meixner–Pollaczek formula with $p_j(y) = j! P_j^{(1/4)}(y/2; \pi/2)$; the primary source is DLMF 18.23.7, with recurrence 18.22.8, by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt. Differentiating this analytic identity at $t = 0$ gives the finite formula

$$\partial_y \phi_n = \sum_{\substack{1 \leq a \leq n \\ a \text{ odd}}} \frac{(-1)^{(a-1)/2}}{a} \frac{\rho_n}{\rho_{n-a}} \phi_{n-a}, \quad \rho_n = \sqrt{n!(1/2)_n}. \quad (\text{CG9})$$

The condition in this sum is simply that a is odd. Since $1 \leq \rho_n \leq \sqrt{2n+1}$, every row sum and every column sum of its matrix on degree at most D is bounded by

$$b_D = \sqrt{2D+1} H_D, \quad H_D = \sum_{j=1}^D 1/j, \quad b_0 = 0. \quad (\text{CG10})$$

For any matrix A , Cauchy–Schwarz in each row with weights $|A_{ij}|$, followed by summing columns, proves $\|A\|_2^2 \leq \|A\|_1 \|A\|_\infty$. Thus $\|\partial_y\|_{\mathcal{P}_D \rightarrow \mathcal{P}_{D-1}} \leq b_D$.

For every complex z and polynomial p of degree at most D ,

$$\boxed{\frac{\|p\|_\sigma}{2b_{D+1}} \leq \|(y-z)p\|_\sigma \leq [2(D+1) + |z|] \|p\|_\sigma.} \quad (\text{CG11})$$

For the lower bound use $p = \partial_y((y-z)p) - (y-z)\partial_y p$, pair with p , and move the multiplication operator in the second term to the other side. Its adjoint is multiplication by $y - \bar{z}$. On the real integration line $\|(y-z)p\|_\sigma = \|(y-\bar{z})p\|_\sigma$. The derivative bounds at degrees $D+1, D$ give $\|p\|^2 \leq (b_{D+1} + b_D) \|p\| \|(y-z)p\|$, proving the first inequality. The upper bound follows from CG7: the raising and lowering coefficients have norms at most $D+1, D$, respectively. This argument holds for every complex z , including a real zero inside the region carrying the norm. No factor is removed.

For the degrees below define the convenient common bounds

$$\lambda_q = (2b_{2q+1})^{-1}, \quad U_q = 4q + 2 + q/r_*, \quad U_q^0 = 4q + 2. \quad (\text{CG12})$$

Every multiplication needed below has input degree at most $2q$. A product of b factors with zeros of modulus at most q/r_* therefore has norm between λ_q^b and U_q^b times the input norm, provided the entire product stays within the displayed degree range. The same bounds apply to y^b . Their comparison costs at most $2b \log(U_q/\lambda_q)$ in the logarithm of squared norms.

15.3 3. All conductor roots, including those on the q-scale

Split the actual root list, with multiplicities, at $T/2$:

$$d_A(c' + iy) = a_A i^g D_{\text{in}}(y) D_{\text{out}}(y), \quad D_{\text{in}} = \prod_{|\zeta| \leq T/2} (y - \zeta), \quad D_{\text{out}} = \prod_{|\zeta| > T/2} (y - \zeta). \quad (\text{CG13})$$

Write $b = \deg D_{\text{out}}$, $a = g - b$. OR10–11 applied with reciprocal radius $2/\epsilon$ is allowed by CG6 and proves

$$b \leq B_A := \frac{\log^+(A_1/a_*) + 8B/\epsilon}{\log 2}. \quad (\text{CG14})$$

Thus b has a fixed finite bound for the original conductor. This is stronger than counting roots outside the shrinking relative disk in OR13. It has a different, explicitly fixed, radius. Retain also OR16's full root sum $S_k = \sum_{\zeta} |\zeta| = O_A(q \log(q+2))$.

For every polynomial p of degree $d \leq 2q$ and every $Q \geq q/2$, with $Q + a + d \leq 2q$, one has

$$e^{-E_{\text{in}}(Q,d)} \|y^{Q+a} p\|_{\sigma}^2 \leq \|y^Q D_{\text{in}} p\|_{\sigma}^2 \leq e^{E_{\text{in}}(Q,d)} \|y^{Q+a} p\|_{\sigma}^2, \quad (\text{CG15})$$

where the completely finite error is

$$\begin{aligned} \eta_k &= 4S_k/T, \\ K(Q, a, d) &= \frac{M_{\sigma}}{C_L} \frac{(d+1)^2}{q} e^{2\pi q} 10^{2d} \epsilon^{2Q} (3\epsilon/2)^{2a}, \\ E_{\text{in}}(Q, d) &= \eta_k + \log(1 + K(Q, a, d)). \end{aligned} \quad (\text{CG16})$$

Here $E_{\text{in}} = O_A(\log(q+2))$ uniformly in these degrees.

For completeness, outside $(-T, T)$, each root satisfies $|\zeta/y| \leq 1/2$, so

$$\left| \log \frac{|D_{\text{in}}(y)|^2}{|y|^{2a}} \right| \leq 2 \sum \frac{|\zeta/y|}{1 - |\zeta/y|} \leq 4S_k/T.$$

To retain the inner interval use the following finite polynomial bound:

$$\sup_{|y| \leq T} |p(y)|^2 \leq \frac{(d+1)^2 10^{2d}}{q} \int_q^{2q} |p(u)|^2 du. \quad (\text{CG17})$$

Indeed expand $p(qz)$ in the orthonormal polynomials $\sqrt{2j+1} L_j(2z-3)$ on $[1, 2]$. Rodrigues' formula and j integrations by parts give $\int_{-1}^1 L_j^2 = 2/(2j+1)$. The finite formula $L_j(x) = 2^{-j} \sum_{h=0}^j \binom{j}{h}^2 (x-1)^{j-h} (x+1)^h$ gives $|L_j(x)| \leq 10^j$ for $|x| \leq 4$, using $\sum_h \binom{j}{h}^2 \leq 4^j$. Since $|2z-3| \leq 4$ on the required inner interval, Cauchy-Schwarz proves CG17. On that interval $|y|^{2Q} |D_{\text{in}}(y)|^2 \leq T^{2Q} (3T/2)^{2a}$. The integral over $[q, 2q]$ of the reference norm is at least $q^{2(Q+a)} C_L e^{-2\pi q} \int_q^{2q} |p|^2$. CG17 and the full mass M_{σ} therefore bound both inner contributions by $K(Q, a, d)$ times that outside contribution. Combining this with the outside logarithmic comparison gives CG15. Finally $Q \geq q/2, d \leq 2q, 3\epsilon/2 < 1$ bound K by $M_{\sigma}(2q+1)^2 (C_L q)^{-1} \exp\{q(2\pi + 4\log 10 + \log \epsilon)\}$, which decays exponentially. This proves the stated order without losing the inner region or a root lying on the real axis.

The original lower polynomial satisfies the same complete estimate

$$e^{-E_{\chi'}} \|y^{q'} p\|_{\sigma}^2 \leq \|\chi'(c' + iy)p(y)\|_{\sigma}^2 \leq e^{E_{\chi'}} \|y^{q'} p\|_{\sigma}^2, \quad E_{\chi'} = O_A(1), \quad (\text{CG18})$$

uniformly for degree $p \leq 2q - q'$. Here and below modulus-one leading phases do not change the norms. An explicit bound, also needed for the shifted upper polynomial, is as follows. For $w \in \{0, 4\}$, $Q \in \{q', q\}$, put

$$\begin{aligned} u_w &= 2w/T + (w^2 + R_k^2)/T^2, \quad \eta_{Q,w} = Qu_w/(1 - u_w), \\ K_{Q,w,d} &= \frac{M_{\sigma}}{C_L} \frac{(d+1)^2}{q} e^{2\pi q} 10^{2d} \left(\frac{(T+w)^2 + R_k^2}{q^2} \right)^Q, \\ E_{Q,w,d} &= \eta_{Q,w} + \log(1 + K_{Q,w,d}). \end{aligned} \quad (\text{CG19})$$

Use $E_{\chi'} = E_{q',0,2q-q'}$. The roots of either centered original polynomial pair as $\omega, -\omega$. For the polynomial shifted by iw , each paired factor divided by y^2 is $1 + 2iw/y - (w^2 + \omega^2)/y^2$. Its

deviation from one is at most $u_w < 1/2$. This proves the outside logarithmic bound $\eta_{Q,w}$. Inside the interval its squared modulus is at most $((T+w)^2 + R_k^2)^Q$, so the same proof of CG17–16 supplies $K_{Q,w,d}$. The guards imply that $((T+w)^2 + R_k^2)/q^2 \leq 4\epsilon^2$, making the latter exponentially small for $Q \geq q/2, d \leq 2q$. Also $\eta_{Q,w} = O_A(1)$. This proves CG18 and the corresponding claim for the upper polynomial in the lower physical coordinate.

Apply CG18, then CG15 with $Q = q'$ and $p = D_{\text{out}}h$, and finally CG11 to every factor of D_{out} and to every factor of y^b . For $0 \leq L \leq q$, $\deg h \leq L$, all degrees remain at most $q' + g + L = q - v + L \leq 2q$. Thus

$$\left| \log \frac{\|\chi' d_A h\|_{\sigma, c'}^2}{|a_A|^2 \|y^{q-v} h(c' + iy)\|_{\sigma}^2} \right| \leq E_{\chi'} + E_{\text{in}}(q', L + b) + 2b \log(U_q/\lambda_q) =: E_d. \quad (\text{CG20})$$

The ratio is evaluated only for nonzero h , so its denominators are positive. The bound is uniform over every coefficient vector, including its worst direction, and $E_d = O_A(\log(q + 2))$.

15.4 4. Exact physical centers and complete relation norms

For polynomials of degree at most M , translation by $4i$ in y and its inverse have Gamma operator norm at most

$$T_M = e(M+1)(2M+1)^{5/2} \quad (M \geq 1). \quad (\text{CG21})$$

To prove it, CG8 multiplies under this translation by $e^{\pm 4i \arctan t} = ((1+it)/(1-it))^{\pm 2}$. On $|t| = M/(M+1)$ its modulus is at most $(2M+1)^2$. Cauchy's coefficient bound through degree M costs at most e . In the orthonormal Gamma frame the coefficient on the a th diagonal is multiplied by $\rho_n/\rho_{n-a} \leq \sqrt{2M+1}$. Sum the at most $M+1$ diagonal norms. The inverse has the same bound. This explicitly carries the original upper center $c = c' + 4$ to the lower center rather than identifying their coefficient norms.

Combining CG20, the $w = 4, Q = q$ case of CG19, CG21 for the entire polynomial χh , and the v factors of y in CG11 gives

$$\boxed{e^{-E_M} \|\chi h\|_{\sigma, c}^2 \leq \|\chi' d_A h\|_{\sigma, c'}^2 \leq e^{E_M} \|\chi h\|_{\sigma, c}^2, \quad E_M = O_A(\log(q + 2)).} \quad (\text{CG22})$$

One explicit common error for all $0 \leq L \leq q$ is

$$E_M = \max_{0 \leq L \leq q} E_d + E_{q,4,q} + 2 \log T_{2q} + 2 |\log |a_A|| + 2v \max\{\log U_q^0, \log(1/\lambda_q)\}.$$

The maximum can be replaced by the same displayed formulas at $d = q + \lceil B_A \rceil$, or bounded directly by $d \leq 2q$ on their actual range. Only the actual $b \leq g$ is used in the formula when the fixed coarse bound exceeds g .

The equality of coefficient vectors $h(S)$ and $h(S')$ is retained in CG22; CG21 is what transports their different evaluation lines. The modulus $|a_A|$, the v degrees and all exceptional roots were paid for explicitly. No conjectured absence of such roots or assumed nonvanishing on the integration line enters the estimate.

15.5 5. The full g-dimensional graph volume is subleading

Let $t = L + 1$ for $L \geq 0$, $N = q + L$, $D = N - v$, and let $E_L = \chi \mathcal{P}_L \subset \mathcal{P}_N$, with its original center- c Gamma metric. The degree- $< v$ space is denoted $\mathcal{L}$. It has zero intersection with E_L . Here is a quantitative angle bound that preserves the entire space. For $v \geq 1$, take the actual upper roots whose y -coordinates are

$$z_j = -\gamma + i(2j+1)\delta, \quad 0 \leq j < v.$$

They are present by $k \geq 2v - 1$, are distinct, and are independent of growing k . Let $\ell_j(y)$ be their degree- $< v$ Lagrange polynomials, and set

$$\begin{aligned} C_I &= \left(\sum_j \|\ell_j\|_\sigma^2 \right)^{1/2}, \quad R_I = \max_j |z_j|, \\ C_E(N) &= \left[\frac{ve^2}{M_\sigma} (N+1)^{3/2} (2N+1)^{R_I+1} \right]^{1/2}, \\ \theta_N &= \min\{1, (C_I C_E(N))^{-1}\}. \end{aligned} \tag{CG23}$$

Set $\theta_N = 1$ if $v = 0$. The coefficient estimate from CG8 on radius $N/(N+1)$, with $|\arctan t| \leq \frac{1}{2} \log(2N+1)$, proves $|\phi_j(z)|^2 \leq e^2 M_\sigma^{-1} (N+1)^{1/2} (2N+1)^{|z|+1}$ for $j \leq N$, exactly as in KF33. Summing gives the evaluation norm $C_E(N)$. Since every element of E_L vanishes at all z_j , interpolation gives, for $l \in \mathcal{L}$, $\|l\| \leq C_I \|\text{ev } l\| \leq C_I C_E(N) \text{dist}(l, E_L)$. Thus this angle has a polynomial lower bound. By taking the adjoint of the orthogonal cross projection, the projection of E_L onto $\mathcal{L}^\perp$ has all singular values at least θ_N and at most v different from one. Its squared determinant is therefore between θ_N^{2v} and one.

Apply the complete WCF4-13 map to that quotient. Write $M_D = \max\{1, B_D^+\}$ and $J_D = (D+1) \log^+(1/|(-i)^v \mu_v/v!|)$, exactly as in KF46. The forward exterior norm at rank t is at most M_D^t ; the inverse exterior norm is at most $M_D^{D+1-t} e^{J_D}$. Consequently, in the **same source coefficient frame** h ,

$$-2(D+1-t) \log M_D - 2J_D + 2v \log \theta_N \leq \log \frac{\det \text{Gram}(\chi' \mathcal{U}_A h)}{\det \text{Gram}(\chi h)} \leq 2t \log M_D.$$

Together with CG22, this gives the finite common bound

$$\left| \log \frac{\det \text{Gram}(\chi' \mathcal{U}_A h)}{\det \text{Gram}(\chi' d_A h)} \right| \leq B_N := 2(N+1) \log M_D + 2J_D + 2v \log(1/\theta_N) + tE_M.$$

It is $O_A(q \log(q+2))$. The inverse bound is an exterior bound before multiplying by a rank, so it does not incur a spurious q^2 cost. The polynomial angle was proved above before its fixed rank was used.

Let Q_N^U be the complete g -dimensional quotient Gram CG3 in the unchanged low remainder frame $1, S', \dots, S'^{g-1}$, and let Q_N^d be the quotient by $d_A \mathcal{P}_L$ in that same frame with the **untransformed** source metric H . KF52 gives the full coefficient matrix $\mathcal{C}_L$. The change of basis from $[1, \dots, S'^{g-1}, d_A, d_A S', \dots, d_A S'^L]$ to the basis with $\mathcal{U}_A 1, \dots, \mathcal{U}_A S'^L$ in its last slots has determinant $\det \mathcal{C}_L$. Block Gram determinants, with every cross block retained, therefore give

$$\log \det Q_N^U - \log \det Q_N^d = 2 \log \det \mathcal{C}_L - \log \frac{\det \text{Gram}(\chi' \mathcal{U}_A h)}{\det \text{Gram}(\chi' d_A h)},$$

$$\begin{aligned} |\log \det Q_N^U - \log \det Q_N^d| &\leq 2v(L+1) \log \frac{q+L}{q-v+1} + B_N, \\ B_N &= O_A(q \log(q+2)) = o(kq). \end{aligned} \tag{CG24}$$

At $L = -1$ both quotients are the identical whole low space. At $L = 0$, $\mathcal{U}_A 1 = d_A$, $\mathcal{C}_0 = 1$, $\mathcal{R}_0 = 0$, and the quotients are again identical. Thus the metric shear is exactly the identity at **both low cutoffs**, and only the two high errors occur in its four-return. CG24 evaluates their total full-target determinant cost; it does not use an assumed isometry of the untransformed metrics.

15.6 6. The exact unresolved receiver on the original kernel plane

Write the actual complete source Gram in the polynomial-division basis $r + d_A w$ as

$$H = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \quad F = \mathcal{R}_L \mathcal{C}_L^{-1}, \quad Q_N^d = A - BD^{-1}B^*. \quad (\text{CG25})$$

Its relation graph is (Fw, w) ; its attained Gram is therefore

$$Q_N^U = A - (AF + B)(D + B^*F + F^*B + F^*AF)^{-1}(F^*A + B^*). \quad (\text{CG26})$$

This is obtained by expanding and minimizing $\langle (r + Fw, w), H(r + Fw, w) \rangle$ in w . Every inverse is of a positive definite Gram of an injective relation map. Equivalently let R_d, D_d be the remainder and quotient-coefficient maps of ordinary polynomial division by d_A . The actual quotient map is exactly $R_d - FD_d$, since it kills (Fw, w) and fixes every low representative. Consequently

$$(Q_N^U)^{-1} = (R_d - FD_d)H^{-1}(R_d - FD_d)^*, \quad (Q_N^d)^{-1} = R_d H^{-1} R_d^*. \quad (\text{CG27})$$

These formulas retain the transformed metric $(\mathcal{J}_L^{-1})^* H \mathcal{J}_L^{-1}$ from KF55: CG26–27 are its quotient minima in the original low frame, not a replacement of that transformed metric.

Let E_K be the exact low coefficient frame of $\bar{K} = \Psi_N(K/(K \cap \mathcal{L}))$, defined by KF7 and KF10 with the complete $Z = [M_A, S_k Z_k]$. It is independent of N . The still unevaluated signed graph effect in the requested allocation is

$$\mathcal{G}_K = - \sum_{N \in \{2q-1, 2q\}} \log \frac{\det(E_K^* Q_N^U E_K)}{\det(E_K^* Q_N^d E_K)}. \quad (\text{CG28})$$

The missing low terms are exactly zero by CG24, not discarded by an estimate. Formula CG24 controls the corresponding full g -volume. It does not control CG28 by its rank. For example positive matrices $\text{diag}(e^a, e^{-a})$ and I_2 have identical determinants but their restrictions to the first coordinate differ by a . This elementary example is only a counterexample to that inference, not a replacement for the original matrices. The original connecting maps and exact remaining scalar are CG25–28.

15.7 7. The remaining allocation is an explicit original invariant receiver

The residual in CG28 can now be transferred, with the proved subleading full-volume cost, to the complete remaining invariant observations. This identifies the next matrix without changing their rank or columns. Use exactly KF8–10, and write their coefficient matrices as

$$B_{\mathcal{L}} = J_R^T V|_{\mathcal{L}}, \quad B_{\eta} \eta = J_R^T V \mathcal{R}(\chi' \eta), \quad J_R = S_k Z_k. \quad (\text{CG29})$$

Here $\mathcal{R}$ is KF8's fixed right inverse, not a cutoff return. Let Π be the Euclidean orthogonal projection onto the complement of $\text{im } B_{\mathcal{L}}$ in the displayed observation coefficient space. This auxiliary Euclidean projection performs only linear constraint elimination; it does not replace any original Gamma metric. Choose once and for all a row basis of ΠB_{η} , denoted $A_R : \mathbb{C}^g \rightarrow \mathbb{C}^r$. Then

$$\ker A_R = \bar{K}, \quad r = g - (m - s_k) = 8k - 32 - v + s_k. \quad (\text{CG30})$$

Indeed η lies in the kernel precisely when $B_{\eta} \eta \in \text{im } B_{\mathcal{L}}$, which is equivalent to existence of $l \in \mathcal{L}$ solving the complete KF10 constraint. KF7 identifies exactly the quotient by $K \cap \mathcal{L}$, proving the kernel and rank assertions. The full conductor columns were used before this map, and every column of J_R remains in CG29.

For $X \in \{U, d\}$ put

$$C_{R,N}^X = A_R (Q_N^X)^{-1} A_R^*, \quad \Delta_{R,N} = \log \det C_{R,N}^U - \log \det C_{R,N}^d.$$

These are the actual attained quotient covariances of that fixed observation map. If $r = 0$, their empty determinant is one. The exact block determinant identity is

$$\log \frac{\det(E_K^* Q_N^U E_K)}{\det(E_K^* Q_N^d E_K)} = \log \frac{\det Q_N^U}{\det Q_N^d} + \Delta_{R,N}. \quad (\text{CG31})$$

To prove it choose a fixed right inverse T_R of A_R and form the invertible frame $P = [E_K, T_R]$. The Schur complement of the first block of $P^* Q_N^X P$ is the attained quotient metric $(A_R(Q_N^X)^{-1} A_R^*)^{-1}$, since $A_R E_K = 0$ and $A_R T_R = I_r$. Taking determinants gives $|\det P|^2 \det Q_N^X = \det(E_K^* Q_N^X E_K) / \det C_{R,N}^X$. The fixed frame determinant cancels between U, d , proving CG31.

In particular CG24 gives the proved finite receiver

$$|\mathcal{G}_K + \Delta_{R,2q-1} + \Delta_{R,2q}| \leq \sum_{L \in \{q-1, q\}} \left[2v(L+1) \log \frac{q+L}{q-v+1} + B_{q+L} \right] = o_A(kq). \quad (\text{CG32})$$

Thus the full shear-volume calculation removes that entire part of the allocation at its required scale. The remaining leading part is the explicit signed sum of the two original invariant covariances in CG32. Its matrix is obtained by inserting A_R on both sides of CG27; its actual coefficients, all cross terms and attained minima remain. The present proof does not yet evaluate those two logarithms.

15.8 Proof-source coverage and citations

- KF1–55, *The actual invariant kernel: degree filtration, quantitative minors, and its complete conductor graph*, read completely. Its complete source must accompany publication of this continuation until a pinned public proof-file URL has been verified.
- OR1–25, *The roots of the complete conductor polynomial*, read completely. CG14 and CG20 use OR10–16, including every finite guard. Its complete source must likewise accompany publication until its public proof-file URL is verified; no live URL is invented here.
- WCF4–13 and WCF19a–f, *A polynomial forward bound for the full weighted conductor*, pinned complete proof file. The local source was read through WCF20; the exact WCF12–13 and WCF19a–f exterior statements were read directly and used as the accepted provider. No independent full rereading of the remainder of WCF is claimed. The published path and commit are identified by its existing public-readback receipt.
- OSP10–26, in the prior complete source edition, read in full. CG17 and CG19 reproduce the finite estimates needed here, retaining the original Legendre proof, Gamma mass and both integration regions. No unsupported public OSP link is asserted.
- NIST DLMF chapter 18, version 1.2.8, released 15 September 2026, human authors named at CG8. The official 18.23.7 formula and 18.22.8 recurrence were checked, and their original NIST TeX encodings were fetched, read and preserved in `graph_primary_sources/`. This argument derives its derivative estimate from those formulas; it does not attribute CG11 or CG24 to DLMF.

No subleading bound on CG28, no individual projected-current sign and no RH conclusion is claimed by the present derivation.

The pinned WCF public source was downloaded and compared byte for byte with the local source on 21 September 2026; they agree, with SHA256 `d86dcb3daf13c9b2d5a0aaa4d72cce35a1aff48066ad69c7d9846b9d097dc354`. The independent exact checker `check_graph_metric.py` passed 81 tests of CG9–11, CG24–27 and CG31. It includes a negative control showing why an equal full determinant cannot assign the determinant on the first fixed direction. The checks concern exact identities and finite auxiliary Grams; the asymptotic estimates are proved in the text above.

Part VI

Complete finite source for the eigenclass phase criterion

16 The finite eigenclass phase criterion

This complete finite proof from the supplied continuation is the source of NE13–16. The projection in the necessity construction is specified explicitly; NE12–14 give its relation to the fixed programme kernel.

16.1 7. The supplied mixed phase has an exact eigenclass criterion

Now use precisely the mixed pair C1 in the new attachment:

$$z = \langle Pv, A(I - P)v \rangle_G, \quad w = \langle (I - P)v, APv \rangle_G,$$

where P is the orthogonal projector onto the **specified** kernel. The attachment retains both of these pairings and their complete rational numerators.

Assume $Av = \lambda v$, and put

$$E = \|v\|_G^2, \quad x = v/\sqrt{E}, \quad r = (A^{\dagger G} - \bar{\lambda}I)x, \quad \theta = \|Px\|_G^2.$$

Then $r \perp x$, and direct expansion proves

$$w - z = E\langle r, Px \rangle_G.$$

Therefore

$$\Re(\bar{z}w) = \left| \frac{z + w}{2} \right|^2 - \frac{E^2}{4} |\langle r, Px \rangle_G|^2. \quad (38)$$

Since

$$\|Px - \theta x\|^2 = \theta(1 - \theta),$$

the negative part has the sharp bound

$$[-\Re(\bar{z}w)]_+ \leq \frac{E^2}{4} \|r\|_G^2 \theta(1 - \theta) \leq \frac{E^2}{16} \|r\|_G^2. \quad (39)$$

This controls the full complex phase defect by the actual adjoint-eigenvector residual, not by a determinant derivative.

16.1.1 A projection-independent sign is equivalent to a reducing eigenline

For every prescribed rank $1 \leq m \leq q - 1$,

$$\Re(\bar{z}w) \geq 0 \text{ for every rank-}m \text{ orthogonal projection } P \iff A^{\dagger G}v = \bar{\lambda}v. \quad (40)$$

The reverse implication is immediate from $w = z$.

For necessity, suppose $r \neq 0$, write $\epsilon = \|r\|$, and take

$$u_0 = r/\epsilon, \quad g = \lambda - \langle u_0, Au_0 \rangle.$$

Rotate the phase of u_0 so that $\langle x, Au_0 \rangle$ points opposite to g , and set

$$\theta = \frac{1}{2} \left(1 - \frac{|g|}{\sqrt{|g|^2 + \epsilon^2}} \right).$$

Choose P so that

$$Px = \theta x + \sqrt{\theta(1-\theta)} u_0.$$

This is a rank-one orthogonal projection on the two-dimensional span; it can be extended to any prescribed rank m by directions orthogonal to that span. The exact value is

$$\boxed{\Re(\bar{z}w) = -\frac{E^2\epsilon^4}{16(|g|^2 + \epsilon^2)} < 0.} \quad (41)$$

Equation (40) is a statement about **all** projections of the given rank. It does not say that normality is necessary for positivity at the one fixed original observation.

Part VII

The unique small singular direction

17 The unique small singular direction of the native arithmetic quotient

17.1 The theorem

Use the original simple-quartet domain: fixed h , $0 < \delta < 1/2$, $\gamma > 2$, odd $k \equiv 1 \pmod{4}$, and $q = (k+1)^2$. All metrics below are the original least-source-norm metrics for $\mu_k = w_h^{\{k\}}$, not a constructed jet metric. Write

$$Q_k(y) = \prod_{a,b=0}^k (y - (2b - k)\gamma + i(2a - k)\delta), \quad E_k = \mathbb{C}[y]/(Q_k), \quad M_k[P] = [yP].$$

Let G_N be its original metric at degree N , and order the original-metric singular values as $s_{1,N} \geq \dots \geq s_{q,N} > 0$.

The imported analytic inputs are explicitly limited to:

- NG2–NG3: the full-polynomial Gamma comparison and multiplication bound;
- NG10/NG15–NG16: the zero law of the **full** reference relation measure $Q_k^2 \sigma$, with the specified equilibrium probability ρ_s ;
- H4–H5 of the supplied *Original-metric heat transitions and complete mixed-family control*: the simple-quartet monic-norm/outgoing profile, including $N=2q$.

No numerical ξ moment is presumed known. The finite assertions proved below do not use a spectral limit. The asymptotic theorem uses the three named analytic inputs, not a claim that a hypothetical off-critical quartet has been exhibited.

Define

$$I(\delta, \gamma) = \log(\delta^2 + \gamma^2) - 3 + \frac{\delta}{\gamma} \arctan \frac{\gamma}{\delta} + \frac{\gamma}{\delta} \arctan \frac{\delta}{\gamma}, \quad (S1)$$

$$J(s) = (1+s)(\log(1+s) - 1) - \frac{s}{2} \int \log x \rho_s(x) dx, \quad J(0) = -1, \quad (S2)$$

and retain the input's **same** profile

$$\psi(0) = \log(4/\pi), \quad \psi(s) = s \log \frac{\kappa_s}{E(\kappa_s)} + \log \frac{1 + \sqrt{1 - \kappa_s^2}}{E(\kappa_s)}, \quad 1 + s = \frac{E(\kappa_s)}{\sqrt{1 - \kappa_s^2} K(\kappa_s)}. \quad (S3)$$

For any odd cutoff sequence N in $[q-1, 2q-1]$ with $(N+1-q)/q \rightarrow s$, the same conclusions hold at both N and $N+1$:

$$\boxed{\log s_{q,N} = -q \log(q/k) + q\{I(\delta, \gamma)/2 - \psi(s) - J(s)\} + o_h(q).} \quad (S4)$$

There is an explicitly given $D_{\{k,N\}} = \exp(0_h(k + \log q))$ such that

$$\boxed{s_{q-1,N} \geq D_{k,N}^{-1}} \quad (S5)$$

at **every** cutoff $q-1 \leq N \leq 2q$. Hence only one singular direction has the scale in S4. Simultaneously,

$$\log s_{1,N} = q\psi(s) + o_h(q), \quad (S6)$$

$$\boxed{\sum_{j=2}^{q-1} \log s_{j,N} = q \log q + qJ(s) + o_h(q).} \quad (S7)$$

In S4–S7, using the exact $\mathbf{s}_N$ instead of its limit gives the equivalent sequential statement; at an adjacent even cutoff one can use the preceding odd cutoff's $\mathbf{s}_N$.

For the four original signs $+, +, -, -$ at $q-1, q, 2q-1, 2q$,

$$\boxed{\lim_{q} \frac{1}{q} \mathcal{R} \log s_{q,N}^2 = 4\{\psi(1) + J(1) - \psi(0) + 1\} =: \mathfrak{c}_{\min}.} \quad (\text{S8})$$

The reproducible interval calculation below proves

$$-\frac{1354751}{10^6} < \mathfrak{c}_{\min} < -\frac{1353893}{10^6}. \quad (\text{S9})$$

The large and middle singular contributions have the limits

$$\mathfrak{c}_{\max} = 4(\psi(0) - \psi(1)), \quad \mathfrak{c}_{\text{mid}} = -4(J(1) + 1), \quad \mathfrak{c}_{\min} + \mathfrak{c}_{\max} + \mathfrak{c}_{\text{mid}} = 0. \quad (\text{S10})$$

The last cancellation is required by the unchanged algebraic determinant, not by independently choosing favorable errors.

17.2 1. The inverse itself has one unbounded direction

Let J_N and L_N be the original onto remainder map and its attained minimum section. Thus J_N is a contraction from the original polynomial source, and L_N is an isometry onto its full minimum space. Define the actual polynomial division operator

$$\mathcal{D}P = \frac{P - P(0)}{y}, \quad u_N = J_N \left[\frac{Q_k - Q_k(0)}{y} \right], \quad \varphi_N(v) = (L_N v)(0).$$

Both displayed quotients are polynomials. Direct multiplication by y modulo the **full** Q_k gives

$$\boxed{M_k^{-1} = J_N \mathcal{D} L_N - \frac{u_N \varphi_N}{Q_k(0)}.} \quad (\text{S11})$$

The functional **varphi**_N is evaluation of the specified minimum representative; it is not a purported evaluation at zero on the quotient algebra. This distinguishes the two maps.

Here is a quantitative norm bound for the first term. For the original Gamma measure

$$d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi),$$

its orthonormal Jacobi coefficients are $\mathbf{a}_j = \mathbf{sqr t}(j(j-1/2))$. The parity block of its $2n$ -dimensional Jacobi matrix is upper bidiagonal, with diagonal $\mathbf{a}_{(2j+1)}$ and superdiagonal $\mathbf{a}_{(2j+2)}$. Every squared inverse entry is at most 2: in a fixed row the next squared entry is obtained by multiplying by

$$a_{2j+2}^2 / a_{2j+3}^2 < 1,$$

and a diagonal inverse entry is at most $\mathbf{a}_1^{-2} = 2$. The full inverse has the same operator norm as this parity block; bounding that block by its Frobenius norm gives

$$\|J_{2n}^{-1}\|^2 \leq n(n+1). \quad (\text{S12})$$

Embed any polynomial of degree at most L in the least even Jacobi section of dimension at least $L+2$. With

$$n_L = \lceil (L+2)/2 \rceil,$$

full multiplication has no omitted outgoing coefficient, and S12 gives

$$\|yP\|_{\sigma}^2 \geq \frac{\|P\|_{\sigma}^2}{n_L(n_L+1)}. \quad (\text{S13})$$

The monic Gamma recurrence gives exactly

$$p_{2j}^\sigma(0) = (-1)^j 4^j (1/2)_j (1/4)_j, \quad \omega_n^\sigma = \mathbf{m}_\sigma n! (1/2)_n, \quad \mathbf{m}_\sigma = \sqrt{2\pi}.$$

Consequently

$$\mathbf{m}_\sigma K_L^\sigma(0,0) = \sum_{j=0}^{\lfloor L/2 \rfloor} \frac{(1/2)_j (1/4)_j}{j! (3/4)_j} \leq L + 1. \quad (\text{S14})$$

Apply S13 to $\mathbf{D} P$ and S14 to $P(0)$:

$$\|\mathcal{D}P\|_\sigma \leq \sqrt{n_L(n_L + 1)}(1 + \sqrt{L + 1})\|P\|_\sigma. \quad (\text{S15})$$

Let the **original** NG2 comparison on degree at least L be

$$\ell_{k,L}\|P\|_\sigma^2 \leq \|P\|_{\mu_k}^2 \leq u_k\|P\|_\sigma^2.$$

Then the usable finite constant is

$$D_{k,N} = \sqrt{u_k/\ell_{k,N}} \sqrt{n_N(n_N + 1)}(1 + \sqrt{N + 1}). \quad (\text{S16})$$

Equations S11 and S16 show that all but the largest singular value of $\mathbf{M}_k^{-1}$ are at most $D_{k,N}$. This proves S5. Its logarithm is $\mathbf{O}_h(k + \log q)$ by NG2. The source mass was retained in S14 and canceled only in the displayed evaluation inequality.

17.3 2. Parity gives a determinant identity at zero

Write $q=2n$. The original source parity is an isometry, with two quotient subspaces of dimension n . In the ordered even/odd remainder coordinates,

$$G_N = \text{diag}(G_{e,N}, G_{o,N}), \quad M_k = \begin{pmatrix} 0 & X \\ I & 0 \end{pmatrix}, \quad (\text{S17})$$

where X is multiplication by $x=y^2$ modulo $Q_k(\sqrt{x})$.

The original selfadjoint compression C_N is odd under parity and differs from $\mathbf{M}_k$ by rank one. Its rank is even. Since $\mathbf{M}_k$ is invertible, its rank is at least $q-1$; therefore

$$C_N \text{ is invertible at every one of these cutoffs.} \quad (\text{S18})$$

This does not give it a uniform spectral gap at even cutoffs.

If N is odd, the top omitted monomial is even. The lower-left block is untouched. Selfadjointness in S17 therefore gives

$$C_N = \begin{pmatrix} 0 & G_{e,N}^{-1} G_{o,N} \\ I & 0 \end{pmatrix}, \quad |\det C_N| = \frac{\det G_{o,N}}{\det G_{e,N}}. \quad (\text{S19})$$

Let p_j^μ be the source monic polynomials and U_j the monic polynomials for the **complete relation measure** $Q_k^2 \mu_k$. Put $d=N+1-q$, an even integer. Schur complementing the entire relation spaces in each parity gives

$$|\det C_N| = \frac{|p_{N+1}^{\mu_k}(0)|}{|U_d(0)|}. \quad (\text{S20})$$

To see every factor, push the even measure forward by $x=y^2$. The ratio of the odd and even source Grams is $\det H[x\lambda]/\det H[\lambda] = |p_{(N+1)/2}^\lambda(0)|$. The same ratio for the relation Grams is

$|p_{d/2}^{Q^2\lambda}(0)|$. The monic changes of polynomial basis have determinant one; source and relation masses are not changed. Their quotient is S20.

The smallest modulus of a C_N eigenvalue at an odd cutoff has a finite lower bound. Indeed the two parity metrics minimize in the same polynomial fibres, in measures λ and $x \lambda$. Apply S13 and the original comparison on the entire lifted polynomials to obtain

$$\|C_N^{-1}\|_{G_N} \leq e^{\Lambda_k/2} \sqrt{n_*(n_*+1)}, \quad (S21)$$

where $\Lambda_k = \log(u_k / \ell_1(k, 3q+4))$ and $n_* = \lceil (3q+6)/2 \rceil$ is a safe degree guard. Any sharper original multiplication bound may also be used. The logarithm of S21 is $O_h(k + \log q)$.

17.4 3. The logarithmic reference limit is justified at zero

Weak convergence alone cannot be applied to a singular logarithm. Two finite estimates resolve this issue here.

17.4.1 3.1 A hard gap for the reference relation measure

Suppose $q \geq 16$ and every root of the monic Q_k has modulus at most $q/512$. For every polynomial P of degree at most q ,

$$\int y^2 |Q_k P|^2 d\sigma \geq \frac{q^2}{524288} \int |Q_k P|^2 d\sigma. \quad (S22)$$

Here is a proof with numerical constants. Put $c = 1/512$ and $c_{\sigma} = \Gamma(1/4)^2 / (2\pi)$. Euler's product gives

$$\frac{c_{\sigma} e^{-\pi|y|/2}}{1 + 4y^2} \leq \sigma(y) \leq c_{\sigma}. \quad (S23)$$

For the lower bound, the decreasing product-log summand is bounded by its first value plus its integral, and $\int_0^{\infty} \log(1 + a^2/t^2) dt = \pi a$.

On $|y| \leq cq$, $|Q_k(y)| \leq (2cq)^q$; on $[q, 2q]$, $|Q_k(y)| \geq (q/2)^q$. On the scaled interval $[1, 2]$, the Legendre reproducing kernel bounds every degree- q polynomial at $|t| \leq c$ by

$$|P(qt)|^2 \leq (q+1)^2 100^q \int_1^2 |P(qu)|^2 du. \quad (S24)$$

This follows from the orthogonal polynomials $\sqrt{2j+1} P_j(2t-3)$ and the elementary bound $|P_j(x)| \leq 10^j$ for $|x| \leq 5$, obtained from the Legendre integral representation.

Divide the mass on $|y| \leq cq$ by the mass on $[q, 2q]$. S23–S24 bound that ratio by

$$2c(1 + 16q^2)(q+1)^2(1600e^{\pi}c^2)^q < \frac{(1 + 16q^2)(q+1)^2}{256 \cdot 2^q} \leq 1. \quad (S25)$$

The exponential bound uses only $\pi < 4$, $e < 3$. The last inequality holds at $q = 16$ and its successive ratio is below one thereafter. At least half the mass is consequently outside $|y| \leq cq$, proving S22.

All squared nodes of every relevant reference relation compression therefore lie in $[1/524288, B_{\sigma}]$ after division by q^2 . The root-radius condition is eventually satisfied because $k/q \rightarrow 0$; its displayed constant is not suppressed. NG10 now legitimately applies to $\log x$ on a fixed compact interval separated from zero.

17.4.2 3.2 Moving the negative characteristic-value comparison to zero

The NG6 characteristic-value argument is valid for **every** $a>0$, not only the source's convenient statement $a>1$. For a fixed positive a , all negative powers of $x+a$ remain bounded at zero. Its proof gives

$$|\log |p_n^\lambda(-a)| - \log |p_n^\eta(-a)|| \leq \frac{1}{2}\Lambda + \log(1 + B_\lambda/a) + \frac{5}{2}\log(1 + B_\eta/a). \quad (\text{S26})$$

For clarity, the determinant slope $f(1)-f(0)$ is $\log |p_n(-a)|$. Convexity traps it between half the $(-2,0)$ and $(0,2)$ increments. The two polynomial spaces $P_{<n}$ and $(x+a)P_{<n}$ share a codimension-one intersection. Cancel its full Gram; the two quotient minima change by at most Λ . The rational spaces $P_{<n}$ and $P_{<n}/(x+a)$ have the same codimension-one cancellation, with comparison cost $\Lambda + 2\log((a+B_\lambda)(a+B_\eta)/a^2)$. Successive Christoffel transforms interlace, bounding the three reference slope differences by $\log(1+B_\eta/a)$. Combining the two directions yields S26. This establishes its explicit $a>0$ domain without ever evaluating a rational Gram at $a=0$.

For source and relation measures alike, S13 on the full lifted polynomials supplies the common squared-node floor

$$\delta_k = \frac{e^{-\Lambda_k}}{q^2 n_*(n_* + 1)}.$$

Choose $a=\delta_k/(n+1)$. Since

$$0 \leq \log |p_n(-a)| - \log |p_n(0)| = \sum_j \log(1 + a/x_j) \leq na/\delta_k < 1,$$

S26 gives

$$\boxed{|\log |p_n^\lambda(0)| - \log |p_n^\eta(0)|| \leq 2 + \frac{1}{2}\Lambda_k + \log(1 + B_\lambda/a) + \frac{5}{2}\log(1 + B_\eta/a).} \quad (\text{S27})$$

The right side is $0_h(k+\log q)$, because the original all-degree multiplication bounds make $B_\lambda, B_\eta=0_h(1)$ on these scaled spaces and $\log(1/a)=\Lambda_k+0(\log q)$.

For source monics, the exact Gamma value and Stirling's formula give

$$\log |p_{N+1}^{\mu_k}(0)| = (N+1)\log(N+1) - (N+1) + O_h(k + \log q). \quad (\text{S28})$$

For relation monics with $d/q \rightarrow s>0$, S22, NG10 and S27 give

$$\log |U_d(0)| = d \log q + \frac{sq}{2} \int \log x \rho_s(x) dx + o_h(q). \quad (\text{S29})$$

At $s=0$ the log of the scaled relation characteristic value is $0(d)+0_h(k+\log q)=o(q)$ by the same fixed compact node bounds. Subtracting S29 from S28 in S20 proves

$$\boxed{\log |\det C_N| = q \log q + qJ(s) + o_h(q)} \quad (\text{S30})$$

for odd cutoffs. At $N=q-1$ its error is $0_h(k+\log q)$; no rate for the reference zero-law convergence at $s>0$ is invented.

17.5 4. Deriving the unique smallest singular value

For odd N , put $C=C_N$, $r=r_N$, $v=ell_N^\dagger$. Define

$$z = 1 + \ell_N C^{-1} r = \frac{\det M_k}{\det C}, \quad a = C^{-1} r, \quad b = C^{-1} v.$$

The complete Sherman–Morrison identity is

$$M_k^{-1} = C^{-1} - \frac{ab^\dagger}{z}. \quad (\text{S31})$$

Its second term is square zero: $\mathbf{e}_{11_N} C^{-2} \mathbf{r}=0$ by opposite parity. Its norm is

$$\epsilon_N^- = \epsilon_N \frac{|\det C|}{|\det M_k|} \frac{\|C^{-1}r\|}{\|r\|} \frac{\|C^{-1}v\|}{\|v\|}. \quad (\text{S32})$$

If $b_C \leq s_{\min}(C)$ and $B_C \geq \|C\|$, then the actual finite bracket is

$$\left(\frac{\epsilon_N |\det C|}{B_C^2 |\det M_k|} - b_C^{-1} \right)_+ \leq \|M_k^{-1}\| \leq b_C^{-1} + \frac{\epsilon_N |\det C|}{b_C^2 |\det M_k|}. \quad (\text{S33})$$

All norms in S31–S33 use G_N . Equations S21 and NG3 make the logs of the two resolvent-vector ratios only $O_h(k \log q)$.

The unchanged root product is

$$\log |\det M_k| = q \log k + \frac{q}{2} I(\delta, \gamma) + o_h(q). \quad (\text{S34})$$

To evaluate I , integrate $\log(\delta^2 u^2 + \gamma^2 v^2)$ over $[0, 1]^2$ and use the divergence of (u, v) times that function. The boundary integrals give S1. The midpoint Riemann sum converges despite the integrable logarithmic origin. An explicit bound, with $n = (k+1)/2$, is

$$|I_k - I - 2 \log(1 + 1/k)| \leq \frac{c_0 + \sqrt{2}(\gamma/\delta)^2 [2(n-1) + H_{n-1}]}{n^2}, \quad (\text{S35})$$

where $c_0 = |\log((\delta^2 + \gamma^2)/4) - I|$. The origin cell is integrated exactly by homogeneity. Group the remaining cells by their largest index and use $\|\text{grad} \log(\delta^2 u^2 + \gamma^2 v^2)\| \leq 2(\gamma/\delta)^2 / \sqrt{u^2 + v^2}$.

Insert S30, S34 and the supplied $H5$, $\log \epsilon_{\text{psion}} = \text{qpsi}(s) + O_h(k \log^2(q+2))$, into S32. The correction dominates C^{-1} by a factor tending to infinity on the scale $q \log(q/k)$. Taking the log of S31 proves S4 at odd cutoffs. At $N=q-1$, S35 and the explicit source value improve the remainder to $O_h(k \log^2(q+2))$.

17.5.1 The adjacent even cutoff is not omitted

Let $\omega = \omega_{N+1}$ and

$$d = \frac{\omega}{\omega + \|r\|_{G_N}^2}.$$

The exact source update is

$$G_{N+1} = G_N - \frac{G_N r r^* G_N}{\omega + \|r\|_{G_N}^2}. \quad (\text{S36})$$

In the old orthonormal frame, set $u = r/\|r\|$ and $S = I + (\sqrt{d} - 1)uu^*$. The new metric is S^*S , and operators transform by $T \rightarrow STS^{-1}$. Parity gives $Sa = a$ in S31. The rank-one inverse correction norm changes by the **exact** factor

$$\boxed{\frac{\|ab^\dagger/z\|_{G_{N+1}}^2}{\|ab^\dagger/z\|_{G_N}^2} = 1 + \frac{|z-1|^2}{\omega \|C^{-1}v\|_{G_N}^2}}. \quad (\text{S37})$$

The factor's logarithm is $O_h(k \log q)$. Indeed $z \rightarrow 0$, $\|C^{-1}v\| \geq \|v\|/B_C$, and

$$\omega \|v\|^2 = b_{N+1}(1 - d_N)$$

(with the second factor one at the first cutoff). Original monic-norm comparison gives $\log b_{N+1} = O_h(k \log q)$, and H4 makes d_N exponentially small. The transformed reference inverse satisfies

$$\|SC^{-1}S^{-1}\| \leq d^{-1/2} \|C^{-1}\|,$$

whose log is only $0_h(q)$, still negligible compared with the rank-one inverse correction's $q \log(q/k)$ size. Hence S4 holds at $N+1$ with the same order- q coefficient.

A useful exact determinant check is

$$\boxed{|\det C_{N+1}| = \frac{|Q_k(0)|^2 d}{|\det C_N|}}. \quad (S38)$$

Thus the even compression's determinant is **not** assigned the odd compression asymptotic S30. The paired proof above handles this genuine difference.

S6 follows from $|s_1 - \epsilon_N| \leq B_h q$. Subtracting S4 and S6 from S34 proves S7. This computes the logarithmic sum of all middle directions without applying weak convergence to an unbounded logarithm.

17.6 5. Evaluating the remaining endpoint constant

Let $u=u_s, v=v_s$ be the original equilibrium endpoints, $0 < u < v$, with

$$uK(\kappa_s) = 2, \quad vE(\kappa_s) = 2(1+s), \quad \kappa_s^2 = 1 - u^2/v^2.$$

Put

$$m = (u^2 + v^2)/2, \quad d = (v^2 - u^2)/2, \quad c = uv, \quad r = (v - u)/(v + u), \quad A_+ = (u + v)^2/4,$$

$$r_t = \frac{d}{m + t^2 + \sqrt{(u^2 + t^2)(v^2 + t^2)}}.$$

Then the log moment in S2 has the explicit convergent one-dimensional expression

$$\boxed{\int \log x \rho_s(x) dx = \log A_+ - \frac{2}{s} \log(1 - r^2) + \frac{1}{s} \int_0^\infty \log(1 - rr_t) dt}. \quad (S39)$$

Here the integral is absolutely convergent and negative. No original Gamma normalization or density factor is removed.

For a derivation, use $x = m + d \cos \theta$. The Fourier expansion of $\log x$ has constant term $\log A_+$ and cosine coefficients $2(-1)^{(j+1)r}/j$. For $w \geq 0$, put $R_w = \sqrt{(u^2 + w)(v^2 + w)}$ and $r_w = d/(m + w + R_w)$. Polynomial division and the elementary Poisson expansion give

$$\frac{1}{\pi} \int_{u^2}^{v^2} \frac{\sqrt{(v^2 - x)(x - u^2)}}{x + w} \log x dx = (m + w - R_w) \log A_+ - (m - c) - 2R_w \log(1 - rr_w). \quad (S40)$$

Insert this into the original equilibrium density representation GM7, with $\alpha = 2/s$, $\beta = \pi/s$, and use the retained normalization $\alpha/c = (\beta/(2\pi)) \int_0^\infty dw/(\sqrt{w}R_w)$. The constant terms cancel exactly; then $w = t^2$ gives S39. The cancellation involves convergent integrals. It is not a subtraction of separately divergent quantities.

The script `check_constants.py` re-establishes the rational root enclosure for u/v using the original complete-elliptic-integral series, 2048 terms and their full positive remainder. It uses Machin's alternating arctangent sums for an interval for π . It then encloses S39 by monotone left/right Riemann sums over $[0, 1]$ and sixteen dyadic intervals ending at $T = 65536$, each with 4096 equal subdivisions. The function $f(t) = \log(1 - rr_t)$ is increasing. Its remaining tail satisfies

$$-\frac{rd}{2T\{1 - rd/(2T^2)\}} \leq \int_T^\infty f(t) dt \leq 0. \quad (S41)$$

Only elementary interval operations, square roots and logarithms are used; no numerical quadrature error estimator or fitted asymptotic is treated as a proof.

The resulting rational enclosures include

$$\begin{aligned} \frac{1099444}{10^6} &< \int \log x \rho_1(x) dx < \frac{1099873}{10^6}, \\ -\frac{1163643}{10^6} &< J(1) < -\frac{1163427}{10^6}, \\ \frac{700182}{10^6} &< \mathfrak{c}_{\max} < \frac{700183}{10^6}, \quad \frac{653711}{10^6} < \mathfrak{c}_{\text{mid}} < \frac{654569}{10^6}. \end{aligned}$$

S9 is obtained in the same single interval computation. As a separate **non-certified locator only**, ordinary high-precision quadrature gives $J(1) \sim -1.1635249737061065$ and $\mathfrak{c}_{\min} \sim -1.3542819878252921$.

17.7 6. An explicit late-time heat return

For fixed real b , set

$$\tau_k(b) = \exp\{2q[\log(q/k) + b]\}, \quad B(s) = \psi(s) + J(s) - I(\delta, \gamma)/2.$$

S4 and S5 imply

$$\boxed{\text{Tr } e^{-\tau_k(b) M_k^{\dagger G_N} M_k} \longrightarrow \mathbf{1}_{\{b < B(s)\}}, \quad b \neq B(s).} \quad (\text{S42})$$

There is no division by q : all middle and large directions are extinguished, and the unique smallest direction either survives or is extinguished. Specifically, their total contribution is at most $(q-1) \exp[-\tau_k(b)/D_{k,N}^2] \rightarrow 0$.

The endpoint enclosures prove

$$\psi(1) + J(1) < -1 < \psi(0) + J(0).$$

Choose the **same actual time at all four cutoffs**,

$$\boxed{\tau_k^* = \exp\{2q[\log(q/k) - 1 - I(\delta, \gamma)/2]\}.} \quad (\text{S43})$$

Then

$$\boxed{\mathcal{R} \text{Tr } e^{-\tau_k^* M_k^{\dagger G_N} M_k} \longrightarrow 2.} \quad (\text{S44})$$

This is a positive-heat statement at large physical time. It is not the earlier small-time *holomorphic-minus-positive* gap, and no enormous-time holomorphic trace is silently subtracted.

A finite late-time enclosure is available directly from S11. Set $E_N = \|u_N\| \|\varphi_N\| / |Q_k(0)|$, using their actual metrics. If $\mathbf{E}_N > \mathbf{D}_N(\mathbf{k}, \mathbf{N})$, then for every $\tau > 0$,

$$e^{-\tau/(E_N - D_{k,N})^2} \leq \text{Tr } e^{-\tau M_k^{\dagger} M_k} \leq e^{-\tau/(E_N + D_{k,N})^2} + (q-1)e^{-\tau/D_{k,N}^2}. \quad (\text{S45})$$

This retains the whole finite spectrum and specifies what native inputs an effective threshold would need.

17.8 Verification and scope

`check_exact.py` checks the original quotient, both parities, the outgoing relation, the inverse relation, the division inverse, the odd zero-value determinant, the even determinant reciprocity, and the exact adjacent inverse correction factor on twelve full finite positive-measure fixtures. They have dimensions 2, 4 and 6 and include all four cutoff types. Their masses are retained and their moments are exact rationals. They are not asserted to be xi measures. The Gamma inverse bound and the finite gap prefactor are checked independently with rational arithmetic.

`check_constants.py` implements the full monotone interval certificate S39–S41. Normal and `-0` runs are compared byte-for-byte. No `assert` statement is used as a certificate gate.

The theorem evaluates a previously uncomputed native smallest-singular-value asymptotic and the associated three-way logarithmic allocation. It does not infer a sign for a different projected current or the independent proper-source target minimum. It uses the supplied native profile theorem as an analytic premise and does not turn its qualitative eventual domain into a numerical cutoff.

17.9 Sources

Split-Zero research programme. (2026, September 20). *Gaussian spectral actions through the original arithmetic quotient*. NG1–NG16; repository KokunoYumeto/zeta-function-research-reader, read at commit 5992ad50c0f2a06921f1fcaded7b4e38097c0739. NG2–NG3 supply the full-degree norm inequalities; NG10 is the reference zero-law input. The extension S26 to every positive pole and the zero-limit argument S27 are proved here.

Original-metric heat transitions and complete mixed-family control. (2026, September 20). User-supplied COMPLETE_PROOFS(4).md, H1–H9. Its exact monic-profile dependency remains an imported theorem; the complete file is preserved under `inputs/`.

Split-Zero research programme. (2026, September 20). *The original equilibrium density, its second moment, and a signed Gaussian return*. GM1–GM10 establish the density representation used to derive the new logarithmic moment formula S39. The complete density and its normalization are unchanged.

National Institute of Standards and Technology. (n.d.). *Digital Library of Mathematical Functions*, 5.8.3 (Gamma product), 18.22.8 (Meixner–Pollaczek recurrence), 18.10 (Legendre integral), and 19.2 (elliptic conventions). Exact special-function conventions are used; no external spectral-limit theorem is substituted for NG10.

The mpmath development team. (2023). *mpmath 1.3.0*, interval context. Only elementary directed interval operations are used in the reproducible constant enclosure. The library’s experimental general interval integration is not used.

Verified public inputs and human-source citations

The native comparison and full reference zero law are NG6–16 in the pinned public proof. The density representation is GM7 in the pinned public proof. The full H1–H9 profile derivation is included later in this edition. For the classical formulas used here, credit R. A. Askey and R. Roy, Gamma product, DLMF5.8.3; T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, with W. P. Reinhardt, Meixner–Pollaczek recurrence, 18.22.8 and Legendre integral, 18.10; and B. C. Carlson, elliptic-integral conventions, 19.2. The original supplied verification reports are retained as source reports. The independent root replay here has 109 exact checks (including the OR fixtures) and reproduces all five rational endpoint enclosures.

Part VIII

General-family mixing, flag quotients and rank changes

18 General-family continuation: uniform mixing bounds and the original kernel determinant

The new GitHub calculation can now be combined with a stronger, dimension-controlled mixing theorem. For the original covariance activation

$$G(t) = (G^{-1} + t\Omega)^{-1},$$

the complete motion of the observation's minimum section satisfies

$$\boxed{\int_0^\infty t \|\dot{L}(t)\|_{\text{HS}(Q(t) \rightarrow G(t))}^2 dt \leq \text{rank}(\Lambda\Omega\Lambda^*) \leq \dim \text{im } \Lambda.} \quad (1)$$

No covariance eigenvalue separation, equality of primary lengths, or bound on the condition number is required.

On the original simple-quartet observation domain, this gives

$$\boxed{\int_0^1 t \|\dot{L}_N(t)\|_{\text{HS}}^2 dt \leq q - 8k + 16.} \quad (2)$$

This improves the preceding $O_h(kq \log(q/k))$ bound for that integral to an explicit $O(q)$ bound. It concerns the **actual minimum-section motion**, not a substitute norm or an independently chosen section.

The generalization also reaches every fixed flag quotient: its determinant change has a positive spectral-shift representation, and all its derivatives are bounded by the quotient's rank. For several noncommuting source additions, the large-parameter determinant exponents are determined by an explicit filtration of their actual value ranges.

There is a necessary boundary correction. A source family can lose rank even while the supplied direction-preserving coercivity inequality remains exact. I calculate that phenomenon below, including the precise additional minimum needed to continue the family across such a point.

I incorporated the new original-kernel proof at [5992ad50c0f2a06921f1fcaded7b4e38097c0739](#). The final recheck returned [487253b5b01851da5b9b98af4e146fecf1fab902](#); its changes are publication links and metadata, not changes to the mathematical proof used here. The new source itself expressly leaves the original kernel's large-degree kq coefficient unevaluated. I do not relabel its exact determinant formula as that coefficient.

18.1 1. Retain the original mixed source and its complete minimum

At one original cutoff, retain

$$H_Z = \Gamma - C_X^* H_X^{-1} C_X, \quad F_Z = I - J_N H_X^{-1} C_X,$$

$$\Omega = F_Z H_Z^+ F_Z^*, \quad G^J = (G^{-1} + \Omega)^{-1}.$$

The inverse H_Z^+ acts on its entire positive range. The original normal representative proves

$$F_Z^* G_{k,1} F_Z \preceq \Lambda_{h,n}^k H_Z,$$

so the value map vanishes on the kernel of the normal Gram. These are the supplied full cross-Gram identities, including their interference terms.

Let

$$\Lambda : E \longrightarrow B = \text{im } \Lambda$$

be the original observation, with

$$K = \ker \Lambda, \quad m = \dim K, \quad r = \dim B, \quad q = m + r.$$

All results through §5 use these **actual dimensions**. The special formula $m = 8k - 16$ is used only on its proved simple-quartet five-orbit domain.

Write

$$H_0 = I_K^* G I_K, \quad Q_0 = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L_0 = G^{-1} \Lambda^* Q_0.$$

The explicit isometry and its inverse are

$$F = [I_K H_0^{-1/2}, L_0 Q_0^{-1/2}], \quad F^* G F = I, \quad F^{-1} = F^* G. \quad (3)$$

In these coordinates,

$$Q_0^{1/2} \Lambda F = [0, I_r].$$

Transport the actual added covariance by that same map:

$$\mathcal{K} = F^{-1} \Omega F^{-*} = \begin{pmatrix} U & V \\ V^* & W \end{pmatrix} \succeq 0. \quad (4)$$

For $t \geq 0$, put

$$\begin{aligned} D_t &= I + tW, & X_t &= tV D_t^{-1}, \\ E_t &= I + tU - t^2 V D_t^{-1} V^*. \end{aligned} \quad (5)$$

Full block inversion gives

$$F^* G(t) F = (I + t\mathcal{K})^{-1},$$

and consequently

$$\boxed{\begin{aligned} I_K^* G(t) I_K &= H_0^{1/2} E_t^{-1} H_0^{1/2}, \\ Q(t) &= Q_0^{1/2} D_t^{-1} Q_0^{1/2}, \\ L(t) &= F \begin{pmatrix} X_t \\ I_r \end{pmatrix} Q_0^{1/2}. \end{aligned}} \quad (6)$$

Thus the derivative in (1) is exactly the derivative of the original observation minimum section. It is not the derivative of an arbitrarily transported frame.

The original arithmetic action changes by the kernel leakage already calculated in the pasted source:

$$A_B^J - A_B = \Lambda A P_K \Omega \Lambda^* Q^J.$$

The analysis below retains this term throughout.

19 2. The entire kernel determinant has a positive spectral-shift representation

Define its actual logarithmic loss

$$f_K(t) = \log \det H_0 - \log \det(I_K^* G(t) I_K).$$

Equation (6) gives

$$\boxed{f_K(t) = \log \det E_t = \log \det(I + t\mathcal{K}) - \log \det(I + tW)}. \quad (7)$$

Let

$$\xi_K(s) = \#\{\lambda(\mathcal{K}) > s\} - \#\{\lambda(W) > s\}, \quad s > 0,$$

counting multiplicities. The map $v \mapsto (0, v)$ makes W the literal compression of $\mathcal{K}$. Min-max therefore proves

$$0 \leq \xi_K(s) \leq m. \quad (8)$$

Since

$$\log(1 + t\lambda) = \int_0^\lambda \frac{t}{1 + ts} ds,$$

we obtain the exact representation

$$\boxed{f_K(t) = \int_0^\infty \frac{t}{1 + ts} \xi_K(s) ds.} \quad (9)$$

Every zero covariance eigenvalue is retained: it simply contributes an empty interval in this expression.

For every integer $n \geq 1$,

$$\boxed{(-1)^{n-1} f_K^{(n)}(t) = n! \int_0^\infty \frac{s^{n-1} \xi_K(s)}{(1 + ts)^{n+1}} ds,}$$

and hence

$$\boxed{0 \leq (-1)^{n-1} f_K^{(n)}(t) \leq \frac{m(n-1)!}{t^n}, \quad t > 0.} \quad (10)$$

This simultaneously controls monotonicity, concavity, and every higher derivative. The controlling integer is the original kernel rank, not the full packet dimension.

Representation (9) is an explicit complete-Bernstein representation. That function class has an established matrix-valued theory; the result used here is proved directly from the two actual covariance matrices and their compression.

19.1 2.1 The large-activation exponent and constant are evaluated

Positivity of (4) proves

$$\ker W \subset \ker V.$$

Define the full shorted block

$$S = U - VW^+ V^* \succeq 0. \quad (11)$$

The determinant-one block congruence

$$\begin{pmatrix} I & -VW^+ \\ 0 & I \end{pmatrix} \mathcal{K} \begin{pmatrix} I & 0 \\ -W^+V^* & I \end{pmatrix} = \begin{pmatrix} S & 0 \\ 0 & W \end{pmatrix}$$

gives

$$a := \text{rank } S = \text{rank } \mathcal{K} - \text{rank } W.$$

Using the actual nonzero eigenvalues in (7),

$$\boxed{f_K(t) = a \log t + \log \frac{\text{pdet } \mathcal{K}}{\text{pdet } W} + R(t),} \quad (12)$$

with the finite bound

$$\boxed{|R(t)| \leq \frac{\text{Tr } \mathcal{K}^+ + \text{Tr } W^+}{t}.} \quad (13)$$

The pseudodeterminant of a zero-dimensional positive range is the empty product, one.

There is an expression for the constant that displays the mixing left when S is singular. Let P_0 be the orthogonal projection onto $\ker S$, and S_+ the restriction to $\text{ran } S$. Then

$$\boxed{\frac{\text{pdet } \mathcal{K}}{\text{pdet } W} = \det S_+ \det_{\ker S} (I + P_0 V W^{+2} V^* P_0).} \quad (14)$$

To prove it, the exact identity

$$E_t = I + tS + tVW^+(I + tW)^{-1}V^*$$

gives

$$E_t = tS + B_\infty + O(t^{-1}), \quad B_\infty = I + VW^{+2}V^*.$$

Block factorization on $\text{ran } S \oplus \ker S$ produces

$$\det E_t = t^a \det S_+ \det(P_0 B_\infty P_0|_{\ker S}) [1 + O(t^{-1})].$$

Comparison with (12) proves (14). In particular, the extra determinant on $\ker S$ is not omitted by treating a pseudodeterminant as multiplicative under a nonunitary congruence.

The full metric scales are also determined. With

$$J_t = \text{diag}(\sqrt{t} I_a, I_{m-a}),$$

$$\boxed{J_t E_t^{-1} J_t \longrightarrow \text{diag} (S_+^{-1}, (P_0 B_\infty P_0|_{\ker S})^{-1}).} \quad (15)$$

Thus exactly a kernel metric directions have squared-norm scale t^{-1} ; the remaining directions approach the displayed positive limiting metric.

19.2 2.2 The slope counts actual unobserved normal values

Factor the original covariance through its original normal source:

$$F^{-1}F_Z H_Z^{+1/2} = \begin{pmatrix} B_K \\ B_B \end{pmatrix}.$$

Then

$$W = B_B B_B^*, \quad V = B_K B_B^*,$$

and

$$\boxed{S = B_K P_{\ker B_B} B_K^*} \quad (16)$$

The exponent a in (12) therefore counts the independent kernel values obtainable from normal source directions whose observed value is zero. This is an exact value-map statement. It does not follow merely from the dimension of H_Z .

20 3. The section's total motion is bounded independently of all covariance scales

Differentiating (5) gives

$$X'_t = V D_t^{-2}, \quad E''_t = -2V D_t^{-3} V^*.$$

The motion energy in the original quotient norms is

$$\boxed{\mathcal{E}(t) := \|\dot{L}(t)\|_{\text{HS}(Q(t) \rightarrow G(t))}^2 = \text{Tr}(E_t^{-1} V D_t^{-3} V^*)}. \quad (17)$$

The complete curvature identity is

$$\boxed{-f''_K(t) = \text{Tr} \left[(E_t^{-1/2} E'_t E_t^{-1/2})^2 \right] + 2\mathcal{E}(t)}. \quad (18)$$

This is the metric-drift term plus the actual minimum-section-motion term.

20.1 3.1 Resolve all positive covariance modes without assuming they commute with the kernel blocks

Let P_λ be the spectral projections of W for its distinct positive eigenvalues. Define

$$B_\lambda = \lambda^{-2} V P_\lambda V^* \succeq 0.$$

The matrices B_λ need not commute.

The exact Schur expression becomes

$$\boxed{E_t = I + tS + \sum_{\lambda>0} \frac{t\lambda}{1+t\lambda} B_\lambda}, \quad (19)$$

and

$$\mathcal{E}(t) = \sum_{\lambda>0} \frac{\lambda^2}{(1+t\lambda)^3} \text{Tr}(E_t^{-1} B_\lambda). \quad (20)$$

For the pointwise bound, use

$$\frac{(t\lambda)^2}{(1+t\lambda)^3} \leq \frac{1}{4} \frac{t\lambda}{1+t\lambda}.$$

Summing inside the trace gives

$$\boxed{t^2 \mathcal{E}(t) \leq \frac{1}{4} \operatorname{Tr} [E_t^{-1}(E_t - I - tS)] \leq \frac{m}{4}.} \quad (21)$$

For the full integral, retain one λ term at a time in (19):

$$E_t \succeq I + \frac{t\lambda}{1+t\lambda} B_\lambda.$$

Set $s = t\lambda/(1+t\lambda)$. The corresponding integral is bounded by

$$\int_0^1 \operatorname{Tr} [s B_\lambda (I + s B_\lambda)^{-1}] ds.$$

For an eigenvalue $b > 0$, that scalar integral is

$$1 - \frac{\log(1+b)}{b}.$$

Define

$$\varphi(b) = \begin{cases} 1 - \log(1+b)/b, & b > 0, \\ 0, & b = 0. \end{cases}$$

Then

$$\boxed{\int_0^\infty t \mathcal{E}(t) dt \leq \sum_{\lambda>0} \operatorname{Tr} \varphi(B_\lambda) \leq \sum_{\lambda>0} \operatorname{rank}(V P_\lambda) \leq \operatorname{rank} W.} \quad (22)$$

This proves (1). For a finite upper endpoint T , each positive eigenvalue b contributes the sharper bound

$$s_T - \frac{\log(1+s_T b)}{b}, \quad s_T = \frac{T\lambda}{1+T\lambda}. \quad (23)$$

A complementary bound retains the complete coupling determinant:

$$\boxed{\int_0^\infty t \mathcal{E}(t) dt \leq \frac{1}{2} \log \det(I + V W^{+2} V^*).} \quad (24)$$

Indeed, dropping only the nonnegative tS enlarges the inverse in the energy. For the resulting $E_t^{(0)}$, integrate (18); its limit is $I + V W^{+2} V^*$, and $t(\log \det E_t^{(0)})' \rightarrow 0$.

Equations (22) and (24) may be used together by taking their minimum.

20.2 3.2 The observed-dimension constant is sharp

The bound cannot generally be replaced by the kernel dimension.

Take a one-dimensional kernel and an r -dimensional observation. For $R > 1$, set

$$W_R = \operatorname{diag}(R^{-2}, R^{-4}, \dots, R^{-2r}),$$

$$V_{R,j} = R^{-3j/2}, \quad U_R = V_R W_R^{-1} V_R^*.$$

The full covariance is positive semidefinite and $S = 0$. Its scalar B_{λ_j} equals R^j .

At times $t = u/\lambda_j$, the j -th summand dominates E_t as $R \rightarrow \infty$, while the earlier modes have already saturated and the later modes remain smaller. On every fixed $0 < a < b < \infty$, the j -th energy integral over that interval tends to

$$\int_a^b \frac{du}{(1+u)^2}.$$

The r intervals are disjoint for large R . Letting $a \downarrow 0$, $b \uparrow \infty$, and using (22), proves

$$\boxed{\int_0^\infty t \mathcal{E}_R(t) dt \longrightarrow r.} \quad (25)$$

This is an explicit general-family example, not an arithmetic packet substituted into the programme.

20.3 3.3 The finite motion does not erase determinant growth

For a single kernel and observation mode, take

$$W = 1, \quad V = \sqrt{\rho}, \quad U = \rho, \quad \rho > 0.$$

Then

$$f_K(t) = \log \frac{1 + (1 + \rho)t}{1 + t},$$

$$\mathcal{E}(t) = \frac{\rho}{(1 + t)^2 [1 + (1 + \rho)t]}.$$

The two quantities evaluate to

$$\boxed{\int_0^\infty t \mathcal{E}(t) dt = 1 - \frac{\log(1 + \rho)}{\rho} < 1,}$$

$$\boxed{f_K(\infty) = \log(1 + \rho).} \quad (26)$$

The determinant loss can therefore be arbitrarily large while the complete weighted section motion stays below one. Equation (18) locates the remaining change in the metric-drift term.

More generally, combining (12), (18), and (22) gives

$$\int_0^T t \operatorname{Tr}[(E_t^{-1/2} E_t' E_t^{-1/2})^2] dt = a \log T + O(1). \quad (27)$$

The coefficient a is the actual rank in (16).

20.4 3.4 Return the estimate through the arithmetic action

Write

$$F^{-1} A F = \begin{pmatrix} A_{KK} & A_{KB} \\ A_{BK} & A_{BB} \end{pmatrix}.$$

The observed action in the fixed value coordinates is

$$\mathcal{A}_B(t) = A_{BB} + A_{BK} X_t.$$

For $0 \leq a < b$,

$$\boxed{\mathcal{A}_B(b) - \mathcal{A}_B(a) = (b - a) A_{BK} V (I + bW)^{-1} (I + aW)^{-1}.} \quad (28)$$

The kernel leakage A_{BK} is present.
Define its actual relative norm at t by

$$\gamma(t) = \|D_t^{-1/2} A_{BK} E_t^{1/2}\|.$$

Then

$$\boxed{\|\dot{\mathcal{A}}_B(t)\|_{\text{HS}(D_t^{-1})}^2 \leq \gamma(t)^2 \mathcal{E}(t).} \quad (29)$$

On a finite interval,

$$\int_a^b t \|\dot{\mathcal{A}}_B(t)\|_{\text{HS}}^2 dt \leq \sup_{[a,b]} \gamma(t)^2 \text{rank } W.$$

This is the complete action return of the new motion bound. The actual leakage norm is not assigned a favourable value.

21 4. Every fixed flag quotient has the same rank-controlled determinant calculus

Let

$$E_0 \subset E_1 \subset E$$

be any of the original fixed flag subspaces. Give E_1/E_0 its **complete attained metric** from $G(t)$. Its rank is

$$p = \dim E_1 - \dim E_0.$$

Use the initial G -orthogonal decomposition

$$E = E_0 \oplus L \oplus E_1^{\perp G}, \quad L = E_1 \cap E_0^{\perp G}.$$

This decomposition has the explicit inclusion isometry and inverse supplied by the initial Gram. Put $O = E_1^{\perp G}$.

In these coordinates, let

$$C(t) = I + t\mathcal{K}.$$

Taking the full inverse first, restricting to E_1 , and minimizing over E_0 gives

$$\boxed{Q_{E_1/E_0}(t)^{-1} = C_{LL}(t) - C_{LO}(t)C_{OO}(t)^{-1}C_{OL}(t).} \quad (30)$$

The blocks involving E_0 disappear by this **complete minimum**, not by being set to zero in C .

Its determinant change is exactly

$$\boxed{f_{E_1/E_0}(t) = \log \det C(t)|_{E_0^{\perp G}} - \log \det C(t)|_{E_1^{\perp G}}.} \quad (31)$$

These are two nested compressions of the same covariance. Therefore §2 applies with rank p :

$$\boxed{0 \leq (-1)^{n-1} f_{E_1/E_0}^{(n)}(t) \leq \frac{p(n-1)!}{t^n}.} \quad (32)$$

The large- t exponent, constant, and metric scales follow from the same explicit shorted block in (30). This is a general result for the actual fixed subquotient, not only for the original full observation.

At the original four cutoffs, the signed expression remains the signed combination of these complete functions. Positivity at a single cutoff does not assign a sign to that four-cutoff combination.

22 5. The new GitHub kernel formula now has an explicit $o(kq)$ -accurate extremal expression

The new repository work gives the original kernel frame

$$I_K = V^{-1}\pi^T,$$

where V is the full root-value map and

$$Z = [M_{\mathcal{A}}, S_k Z_k], \quad \pi Z = 0.$$

The ordinary transpose is required by the original complex-linear observation. On the simple-quartet domain,

$$\text{rank } \pi = m = 8k - 16, \quad \text{rank } Z = r = q - m.$$

For $d = N + 1$, retain the entire orthonormal source-column matrix

$$\Phi_d = \left[\frac{f_0}{\sqrt{\omega_0}}, \dots, \frac{f_{d-1}}{\sqrt{\omega_{d-1}}} \right], \quad f_j = (p_j(v_\alpha))_\alpha.$$

Then

$$C_d = \Phi_d \Phi_d^*, \quad H_{K,N} = \bar{\pi} C_d^{-1} \pi^T,$$

and the repository proves

$$\boxed{\det H_{K,N} = g_{\pi,Z} \frac{\det(Z^T C_d \bar{Z})}{\det C_d}, \quad g_{\pi,Z} = \frac{\det(\bar{\pi} \pi^T)}{\det(Z^T \bar{Z})}.} \quad (33)$$

Both frame constants are retained before their four-cutoff cancellation.

The source derives its mixed-root expression from the Akemann–Vernizzi finite determinant theorem and an independent complete proof, including confluence and the original measure.

22.1 5.1 Use source-index orthogonality without deleting root interference

Put

$$A_d = Z^T \Phi_d.$$

Cauchy–Binet gives

$$\boxed{\det(A_d A_d^*) = \sum_{\substack{J \subset \{0, \dots, d-1\} \\ |J|=r}} |\det(A_d)_J|^2,}$$

$$\boxed{\det C_d = \sum_{\substack{J \subset \{0, \dots, d-1\} \\ |J|=q}} |\det(\Phi_d)_J|^2.} \quad (34)$$

These are positive sums over **source-column indices**. Each entry of A_d still contains the full period-dependent root sum $Z^T f_j$. Expanding one squared minor restores its mixed root terms. Equation (34) therefore does not discard the $I \neq J$ interference in the repository's mixed-relation expression.

Define

$$M_r(A_d) = \max_{|J|=r} \log |\det(A_d)_J|^2,$$

$$M_q(\Phi_d) = \max_{|J|=q} \log |\det(\Phi_d)_J|^2.$$

A zero minor has logarithm $-\infty$; at least one full-rank minor is nonzero. The finite log-sum bounds give

$$\boxed{\log \det H_{K,N} = \log g_{\pi,Z} + M_r(A_d) - M_q(\Phi_d) + e_N,} \quad (35)$$

with

$$\boxed{-\log \binom{d}{q} \leq e_N \leq \log \binom{d}{r}.} \quad (36)$$

At the four original cutoffs,

$$\boxed{|\mathcal{R} \log \det H_{K,N} - \mathcal{R}[M_r(Z^T \Phi_{N+1}) - M_q(\Phi_{N+1})]| \leq (6q+2) \log 2.} \quad (37)$$

The mass check is exact. Under the diagnostic change $d\mu \mapsto a d\mu$, every source column scales by $a^{-1/2}$. The numerator in (34) scales by a^{-r} , the denominator by a^{-q} , and their ratio by a^m , as required for the original rank- m restricted Gram.

22.2 5.2 The original Gamma comparison gives the required scale of error

The new GitHub proof retains

$$|\mathcal{R} \log \det H_{K,N} - \mathcal{R} \log \det H_{K,N}^\sigma| \leq 2mL_k^{\text{form}},$$

where

$$L_k^{\text{form}} = O_h(k \log^2(q+2)).$$

It also supplies the exact Gamma recurrence and original mass for every column of Φ_d^σ . Define the finite centre

$$\mathfrak{C}_k^\sigma = \mathcal{R} [M_r(Z^T \Phi_{N+1}^\sigma) - M_q(\Phi_{N+1}^\sigma)].$$

Then

$$\boxed{|\mathcal{R} \log \det H_{K,N}^{\text{ar}} - \mathfrak{C}_k^\sigma| \leq 2mL_k^{\text{form}} + (6q+2) \log 2 = o_h(kq).} \quad (38)$$

This is an inverse-free finite centre at the precision of the original kernel problem. Its actual period-dependent minors remain to be analysed asymptotically. **Equation (38) is not an assigned numerical or closed-form value of the kq coefficient.**

For full multiplicities, the same construction uses the complete divided-jet rows

$$\frac{p_j^{(a)}(\rho)}{a! \sqrt{\omega_j}},$$

the actual physical-unit map, and the actual kernel inclusion. The repository's confluence proof supplies those factorials and maps. The simple-quartet dimension $8k - 16$ is not extended to a new observation without its rank calculation.

23 6. Several noncommuting source additions have exact growth exponents

Take specified orthogonal groups of the **complete normal source**, after its full Gram has been formed. Their covariance contributions are

$$\mathcal{K}_i = B_i B_i^* \succeq 0.$$

No commutativity is assumed. Set

$$C(\mathbf{t}) = I + \sum_{i=1}^p t_i \mathcal{K}_i, \quad t_i \geq 0.$$

For a fixed flag quotient, use the two exact compressions from (31). Cauchy–Binet applied to the full column matrix

$$[I, B_1, \dots, B_p]$$

shows that their determinants are polynomials in the t_i with nonnegative coefficients. Every coefficient is a sum of squared full minors, so all complex mixing is kept inside those minors.

23.1 6.1 General weighted paths

Fix positive weights a_i and set

$$t_i = T^{a_i}.$$

Let

$$\mathcal{R}_s = \text{span}_{a_i \geq s} \text{ran } B_i.$$

For the original kernel/observation decomposition, let pr_B be the actual initial-metric orthogonal projection onto the observation factor. Then

$$\boxed{f_K(T^{a_1}, \dots, T^{a_p}) = \nu_K(\mathbf{a}) \log T + \log C_{\text{lead}} + o(1),} \quad (39)$$

where

$$\boxed{\nu_K(\mathbf{a}) = \int_0^\infty [\dim \mathcal{R}_s - \dim \text{pr}_B \mathcal{R}_s] ds.} \quad (40)$$

The leading constant $C_{\text{lead}} > 0$ is the ratio of the sums of the actual maximal-weight squared minors in the two determinants.

To prove (40), order the weights from largest to smallest and extend independent columns at each new level. This produces a basis of maximal total weight. Any basis has no more than $\dim \mathcal{R}_s$ columns above level s . Integrating that counting inequality proves the maximal weight $\int \dim \mathcal{R}_s ds$. Apply the same argument to the projected columns and subtract. Positivity of the squared-minor coefficients prevents cancellation of the leading terms.

For a flag quotient E_1/E_0 , first project each B_i onto $E_0^{\perp G}$, then use its nested subspace $E_1^{\perp G}$. That is the precise extension of (40), through (31).

Each determinant has at most

$$\binom{q+p}{p}$$

distinct monomials. Thus the finite log-sum-max error is at most the logarithm of this number on each side. This finite bound—not an unexamined fixed-coefficient limit—is the usable statement when k , periods, and coefficients also vary.

23.2 6.2 An exact noncommuting example

Take

$$b_1 = (1, 1, 0)^T, \quad b_2 = (1, i, 1)^T, \quad b_3 = (1, 0, 0)^T, \quad \mathcal{K}_i = b_i b_i^*.$$

Let the first coordinate be the kernel and the last two be observed.

Direct expansion gives

$$\det C = 1 + 2x + 3y + 4xy + z + xz + 2yz + xyz,$$

$$\det C_B = 1 + x + 2y + xy.$$

For

$$x = T^3, \quad y = T, \quad z = T^2,$$

their ratio has logarithm

$$\boxed{f_K = 2 \log T + o(1)}, \tag{41}$$

with leading constant one. The rank-difference filtration in (40) has height two and gives exactly that exponent.

23.3 6.3 Complex parameter control is also explicit

For $\Im z_i > 0$, both

$$C(\mathbf{z}) = I + \sum_i z_i \mathcal{K}_i$$

and its observed block are invertible. A vector in a kernel would have zero imaginary quadratic form, hence be annihilated by every relevant $\mathcal{K}_i$; the identity term would then force it to be zero.

Write the Schur complement as $E(\mathbf{z})$, and let

$$T(\mathbf{z}) = \begin{pmatrix} I \\ -C_{BB}(\mathbf{z})^{-1} C_{BK}(\mathbf{z}) \end{pmatrix}.$$

Then

$$C(\mathbf{z})T(\mathbf{z}) = \begin{pmatrix} E(\mathbf{z}) \\ 0 \end{pmatrix},$$

so

$$\boxed{\Im E(\mathbf{z}) = T(\mathbf{z})^* \left(\sum_i \Im z_i \mathcal{K}_i \right) T(\mathbf{z}) \succeq 0.} \tag{42}$$

This is an explicit holomorphic matrix family with a proved positivity property. It does not require an unconstructed period-map identification.

24 7. Rank-changing sources require a finite correction, not a continuity assertion

The supplied estimate

$$F_Z^* G_{k,1} F_Z \preceq \Lambda H_Z$$

controls small normal energies with their actual values. It does **not** imply that the literal source minimum is continuous through every rank drop.

Here is an exact source example.

Take the physical space $\mathbb{C}^2$ with its Euclidean norm and value map

$$J(x_1, x_2) = x_1 + x_2.$$

Retain the scalar source $X = \mathbb{C}e_1$ and the analytic section

$$R_z = (1 - z)e_1 + ze_2, \quad JR_z = 1.$$

Its complete normal data are

$$H_Z = |z|^2, \quad F_Z = z.$$

Thus

$$|F_Z|^2 = H_Z$$

at every parameter, including zero.

For $z \neq 0$, the joint source is $\mathbb{C}^2$, and its quotient metric is $1/2$. At zero, the literal joint source is only $\mathbb{C}e_1$, and its metric is 1:

$$G^{\text{joint}}(z) = \begin{cases} 1/2, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (43)$$

The covariance is one off zero and zero at zero. The discontinuity is exactly the disappearance of a source direction, not a failure of the displayed coercivity inequality.

24.1 7.1 Complete analysis along an analytic parameter curve

Let the full source coefficient matrix $B(z)$ be holomorphic in one complex parameter, in a fixed finite ambient source with positive real-analytic metric. Include the original scalar source among its columns.

Over the one-variable convergent power-series ring, finite elimination gives analytic invertible matrices $P(z), Q(z)$ and nonnegative integers ν_i such that

$$P(z)B(z)Q(z) = \begin{pmatrix} \text{diag}(z^{\nu_1}, \dots, z^{\nu_r}) & 0 \\ 0 & 0 \end{pmatrix}. \quad (44)$$

The elimination is constructive: choose an entry of least vanishing order, divide its nonzero analytic unit, clear its row and column, and repeat. The orders of the minors determine the successive ν_i , so they do not depend on an arbitrary pivot label.

Define

$$Y(z) = P(z)^{-1} \begin{pmatrix} I_r \\ 0 \end{pmatrix}.$$

For $z \neq 0$, $Y(z)$ spans the original source image. At zero it retains the analytically continued directions that vanished as z^{ν_i} .

The exact module discrepancy is

$$\boxed{\bigoplus_i \mathbb{C}\{z\}/(z^{\nu_i}),} \quad (45)$$

with length $\sum_i \nu_i$. The inverse-exterior exponents are the sums of the corresponding largest ν_i , in any specified ambient metric that stays positive at zero.

This gives a controlled extension of the generic source along the curve. It is not an equality with the literal zero fibre.

24.2 7.2 Return the missing directions through the actual minimum

At $z = 0$, orthogonalize the extra columns of $Y(0)$ against the **entire literal source**. Let their full normal Gram and value map be

$$H_{\text{miss}}, \quad F_{\text{miss}}.$$

The exact extension of the generic quotient metric is

$$\boxed{G_{\text{sat}}(0) = [G_{\text{lit}}(0)^{-1} + F_{\text{miss}} H_{\text{miss}}^+ F_{\text{miss}}^*]^{-1}.} \quad (46)$$

In (43), the extra direction is e_2 , with Gram one and value one; (46) is $1/(1+1) = 1/2$, exactly.

Thus the same mixed-minimum formula used in the programme supplies the correction at a rank-changing source parameter. The new family does not obtain continuity by deleting that correction.

24.3 7.3 A moving observation has an additional derivative term

The estimates in §§2–4 control covariance activation with the original observation fixed. A varying period can also move the observation kernel.

For a varying full kernel frame $B(t)$, set

$$P = B(B^*GB)^{-1}B^*G, \quad B_G^\dagger = (B^*GB)^{-1}B^*G,$$

$$Z = (I - P)B'B_G^\dagger.$$

Direct differentiation gives

$$\boxed{P' = Z + Z^\dagger G + PG^{-1}G'(I - P).} \quad (47)$$

The first two terms are the actual frame-motion contribution.

Their size cannot be bounded from dimensions alone. In $\mathbb{C}^2$, take $G = I$, no added covariance, and

$$\Lambda_\omega(t) = (-\sin \omega t, \cos \omega t).$$

The kernel is spanned by

$$(\cos \omega t, \sin \omega t),$$

and its projection derivative has norm $|\omega|$. This exact reparameterized family has unchanged dimensions and unchanged metric.

Therefore a varying-period continuation must retain its actual B' , as (47) does. The new dimension bounds remove covariance-scale dependence; they do not erase parameter-speed data.

25 8. What is now established for the programme

The general-family control has become sharper in several concrete respects.

First, the actual observation minimum section has the universal bounds

$$t^2 \|\dot{L}(t)\|_{\text{HS}}^2 \leq \frac{\dim K}{4}, \quad \int_0^\infty t \|\dot{L}(t)\|_{\text{HS}}^2 dt \leq \text{rank}(\Lambda \Omega \Lambda^*).$$

They survive repeated covariance eigenvalues, singular covariance blocks, and every original primary multiplicity.

Second, every fixed flag quotient has the exact nested-compression formula (31), positive spectral-shift representation, all-order derivative bounds, and computable long-activation exponent and constant.

Third, several noncommuting additions have the exact rank-filtration exponent (40), with positive squared-minor coefficients and finite remainder bounds. No generic orientation is used.

Fourth, the newly published original-kernel determinant now has the finite centre (38) with an explicit error below the required kq scale. The unresolved coefficient is attached to actual Gamma-polynomial/period minors, not to an unspecified matrix inverse.

Fifth, literal source-rank degeneration is accounted for by the explicit module and metric corrections (45)–(46). The direction-preserving source inequality is retained, but not promoted to a false continuity theorem.

The actual remaining quantities are still

$$\mathcal{R} \log \det(I_K^* G_N I_K)$$

at its finer coefficient scale,

$$\mathcal{M}^p = \mathcal{R} \log \det \left[(Q_N^p)^{-1/2} H_N^p (Q_N^p)^{-1/2} \right],$$

and the native projected-current phase. The source itself identifies those as unassigned values, with the original leakage retained.

Related literature: Andrzej Hanyga, *On solutions of a class of matrix-valued convolution equations*, arXiv:1805.02471v2. This is the supplied related-theory reference. The finite integral and block identities used here are proved in Sections2–7; no theorem from that paper is imported.

Part IX

Dual pencils and singular certificates

26 I. A division-free spectral pencil, all singular directions, and the moving-observation boundary

26.1 I.1 Original coordinates and the exact dual pencil

Source input. The supplied *Original-metric heat transitions and complete mixed-family control* (20 September 2026), P1–P6, fixes an endomorphism and a covariance path. Its positive one-parameter coefficient theorem is retained as prior work. We give a smaller direct determinant, a multiple-source extension, and individual singular-value certificates. The classical relation between polynomial coefficient geometry and root scales has precedents such as Akian, Gaubert, and Sharify (2017); the finite inequalities required here are proved below and are not a historical-priority claim.

Let the specified value space have dimension $q \geq 1$, original metric $G_0 > 0$, original endomorphism M , and actual positive normal-source covariances $\Omega_i \succeq 0$. Define

$$G(\mathbf{t}) = (G_0^{-1} + \sum_{i=1}^d t_i \Omega_i)^{-1}, \quad K_i = G_0^{1/2} \Omega_i G_0^{1/2}, \quad C = I + \sum_i t_i K_i, \quad t_i \geq 0,$$

$$M_0 = G_0^{1/2} M G_0^{-1/2}. \quad (\text{D1})$$

These are precisely covariances of orthogonalized source additions. Distinct, unorthogonalized physical sources cannot be assigned independent positive summands without first retaining their cross Gram. The isometry from $(E, G(\mathbf{t}))$ to Euclidean coordinates is $S = C^{-1/2} G_0^{1/2}$. Thus $T = S M S^{-1} = C^{-1/2} M_0 C^{1/2}$ and

$$T T^* = C^{-1/2} M_0 C M_0^* C^{-1/2}.$$

For every $a > 0$,

$$\boxed{\det(aI + M^{\dagger G(\mathbf{t})} M) = \frac{P(a, \mathbf{t})}{D(\mathbf{t})}, \quad P = \det(aC + M_0 C M_0^*), \quad D = \det C.} \quad (\text{D2})$$

Indeed $T^* T$ and $T T^*$ have the same characteristic polynomial by $\det(I + UV) = \det(I + VU)$, which follows by eliminating either diagonal block of a two-by-two block matrix. Multiplying the determinant of $aI + T T^*$ by $\det C$ proves D2. This uses no covariance eigenbasis and no adjugate division. At $a = 0$, continuity gives

$$P(0, \mathbf{t}) = |\det M|^2 D(\mathbf{t}). \quad (\text{D3})$$

The source's P6 adjugate formula is equal to D2, not a different determinant. In particular its exact divisibility follows from this q -square pencil.

26.2 I.2 Several noncommuting source additions still have positive coefficients

Choose any factors $K_i = B_i B_i^*$ on their actual positive ranges. Put $B_0 = I$, $t_0 = 1$. Then

$$aC + M_0 C M_0^* = \sum_{i=0}^d t_i \{a B_i B_i^* + (M_0 B_i)(M_0 B_i)^*\}. \quad (\text{D4})$$

Expand the Gram determinant by Cauchy–Binet, using the fixed columns $B_i, M_0 B_i$ and scalar column weights at_i, t_i . Every coefficient is a sum of squared moduli of **complete** q -column minors; all phases inside those determinants are retained. Therefore

$$P(a, \mathbf{t}) = \sum_{j, \alpha} p_{j, \alpha} a^j \mathbf{t}^\alpha, \quad p_{j, \alpha} \geq 0, \quad 0 \leq j \leq q, \quad |\alpha| \leq q. \quad (\text{D5})$$

Also $\alpha_i \leq \min(q, 2 \text{rank } K_i)$, because the two groups from source i have at most $2 \text{rank } K_i$ columns. No commutativity of the K_i is used. The coefficient field is that of the original matrices and their conjugates: the square-root factors above prove positivity but need not be computed to form D2.

At most

$$N_{q,d} = (q+1) \binom{q+d}{d}$$

coefficients occur. On positive parameters, if L is the maximum logarithm of their evaluated nonzero monomials,

$$0 \leq \log \det(aI + M^\dagger M) - [L - \log D] \leq \log N_{q,d}. \quad (\text{D6})$$

At four equal-dimensional endpoints with the same parameters and signs $+, +, -, -$, the absolute signed error is at most $2 \log N_{q,d}$. For fixed d this is $O_d(\log(q+1))$. It is not dimension-independent if the number of separately penalized sources grows.

26.3 I.3 Recover each squared singular value, including the smallest

Let $\lambda_1 \geq \dots \geq \lambda_q \geq 0$ be the eigenvalues of $M^\dagger M$ in the actual metric, and write

$$P(a, \mathbf{t}) = \sum_{r=0}^q a^{q-r} A_r(\mathbf{t}), \quad e_r(\lambda) = A_r/D, \quad e_0 = 1.$$

Let $R = \text{rank } M$. Then $A_r > 0$ for $0 \leq r \leq R$ and $A_r = 0$ for $r > R$, throughout the nonnegative parameter domain. For $1 \leq r \leq R$,

$$\boxed{\frac{e_r}{(q-r+1)e_{r-1}} \leq \lambda_r \leq r \frac{e_r}{e_{r-1}}}. \quad (\text{D7})$$

Proof. For each $(r-1)$ -subset of indices, add a missing member of $\{1, \dots, r\}$. The product increases by at least λ_r , and each resulting r -subset has at most r preimages. Thus $re_r \geq \lambda_r e_{r-1}$. Conversely remove from every r -subset a member whose index is at least r . Its value is at most λ_r , and each remaining subset has at most $q-r+1$ possible extensions. This gives $e_r \leq (q-r+1)\lambda_r e_{r-1}$. Zeros cause no exception when $r \leq R$. This proves D7.

At fixed parameters let m_r be the largest evaluated nonzero monomial of A_r , and let n_r count its nonzero evaluated monomials. Then $m_r \leq A_r \leq n_r m_r$, and the denominator D cancels **exactly** between adjacent coefficients. Hence

$$\boxed{\frac{m_r}{(q-r+1)n_{r-1}m_{r-1}} \leq \lambda_r \leq \frac{rn_r m_r}{m_{r-1}}}. \quad (\text{D8})$$

Here $n_r \leq \binom{q+d}{d}$. In particular, for one activation variable,

$$|\log \lambda_r - [\log m_r - \log m_{r-1}]| \leq \log(q(q+1)). \quad (\text{D9})$$

For squared-singular logarithms divided by $2q$, the uncertainty in D9 is $O(\log q/q)$, for **every rank**, including the smallest positive one. This is a finite bound; no convergence or coefficient asymptotic has been inferred.

The constants in D7 cannot uniformly be replaced by one. For equal first r roots and vanishing later roots, its upper factor r is attained. If the first $r-1$ roots tend to infinity relative to the remaining equal roots, its lower factor $q-r+1$ is approached.

26.4 I.4 Heat, inverse exterior powers, and simultaneous scales

Let l_r, u_r be the bounds in D8. For $\tau \geq 0$,

$$\sum_{r \leq R} (1 - e^{-\tau l_r}) \leq q - \text{Tr } e^{-\tau M^\dagger M} \leq \sum_{r \leq R} (1 - e^{-\tau u_r}), \quad (\text{D10})$$

$$\sum_{r \leq R} \log(1 + \tau l_r) \leq \log \det(I + \tau M^\dagger M) \leq \sum_{r \leq R} \log(1 + \tau u_r). \quad (\text{D11})$$

Zeros contribute zero to each deficit and logarithm, and one each to the heat trace itself. The same monotonicity gives bounds for $\tau\lambda/(1 + \tau\lambda)$. If M is invertible, there is a sharper direct return to the original inverse-exterior calculation:

$$\boxed{\| \wedge^p M^{-1} \|_{\text{HS}, G}^2 = \frac{A_{q-p}}{A_q}, \quad \frac{A_{q-p}}{\binom{q}{p} A_q} \leq \| \wedge^p M^{-1} \|_G^2 \leq \frac{A_{q-p}}{A_q}.} \quad (\text{D11a})$$

The squared singular values of the inverse exterior map are the products of p reciprocals. Their sum is $e_p(\lambda^{-1}) = e_{q-p}(\lambda)/e_q(\lambda) = A_{q-p}/A_q$. Their maximum lies between their average and their sum. This proves D11a, including $p = 0, q$. It retains the exact full root determinant through $A_q = |\det M|^2 D$. No singular direction is removed. For a separately prescribed conductor between two different metrics, the matching pullback and coordinate maps must be applied before this endomorphism formula is used.

For one activation $t = e^{-2q\theta}$ define

$$z_r(\theta) = \frac{1}{2q} \left[\max_n (\log p_{q-r, n} - 2q\theta n) - \max_n (\log p_{q-r+1, n} - 2q\theta n) \right].$$

This is a finite piecewise-affine function, and D9 bounds the exact logarithmic singular scale within $\log(q(q+1))/(2q)$. At $\tau = e^{-2q\sigma}$, these intervals certify which directions lie above or below the heat threshold. A direction whose interval meets σ remains unresolved. The actual coefficients are needed; the known covariance eigenvalue law alone does not determine them.

For example, fix $M = \begin{pmatrix} 1 & A \\ 0 & 1 \end{pmatrix}$ and let $D = 1 + t\kappa$. The two covariances $C_+ = \text{diag}(D, 1)$ and $C_- = \text{diag}(1, D)$ have the same eigenvalues and determinant, but

$$P_+ = (a+1)^2 D + aA^2, \quad P_- = (a+1)^2 D + aA^2 D^2. \quad (\text{D12})$$

Their squared-singular sums are $2 + A^2/D$ and $2 + A^2 D$, respectively, while both products are one. This exhibits the orientation information retained by D2.

26.5 I.5 The moving observation needs a different pencil

Use exactly the supplied M1–M4 block frame. Write

$$C(t) = \begin{pmatrix} I + tU & tV \\ tV^* & D_t \end{pmatrix}, \quad D_t = I + tW, \quad X_t = tV D_t^{-1},$$

with the old action blocks $\begin{pmatrix} a_0 & b_0 \\ c & d_0 \end{pmatrix}$. The actual observed endomorphism and metric are

$$B_t = d_0 + cX_t = Y_t D_t^{-1}, \quad Y_t = d_0 D_t + tcV, \quad Q_t = D_t^{-1}.$$

D2 applied at each fixed t proves

$$\boxed{\det(aI + B_t^{\dagger Q_t} B_t) = \frac{\det \begin{pmatrix} D_t & Y_t^* \\ -Y_t & aD_t \end{pmatrix}}{\det(D_t)^2}.} \quad (\text{D13})$$

The block determinant follows from its Schur complement. It is polynomial of degree at most r in a and $2r$ in t , where r is the observed dimension. At a fixed real $t \geq 0$ its a -coefficients are nonnegative. Its **joint** coefficients in (a, t) need not be nonnegative, because Y_t contains the moving-section phases.

Take $K = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $M = \begin{pmatrix} 2 & 1 \\ -3 & 1 \end{pmatrix}$, and observe the second coordinate. Then $D_t = 1+t$, $Y_t = 1-2t$, and the numerator in D13 is

$$a(1+t)^2 + (1-2t)^2. \quad (\text{D14})$$

Its coefficient of $a^0 t^1$ is -4 . The full M has determinant five, yet its observed endomorphism vanishes at $t = 1/2$. Thus neither full-rank return nor positive joint coefficients transfer by fiat to the moving observation.

Even the full scalar polynomial does not determine a fixed projected current. With $K = \text{diag}(1, 2)$, the same observation, $x = (1, 1)^T$, and

$$A_+ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_- = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix},$$

the full spectral polynomials, covariance, and eigenvalues agree, while the source C1 current $\Re(\bar{z}w)$ is respectively

$$\frac{1}{(1+t)(1+2t)}, \quad -\frac{1}{(1+t)(1+2t)}. \quad (\text{D15})$$

The two actions are related by a diagonal unitary commuting with the covariance; transporting the class by that unitary would preserve the current. Here the class is fixed, exactly exposing the extra data lost by the scalar spectral invariant. These are auxiliary finite examples, not xi-packet evaluations.

Part X

Calibrated heat and the observation

27 II. The resolved, calibrated heat-transition layer and the actual observed direction

27.1 II.1 What is inherited and what is being calculated

The supplied H1–H15 retains the original full companion $M = C_N + r_N \ell_N$, its original metric G_N , the complete source minimum, and the uniform norm bound $\|C_N\|_{G_N} \leq B_h q$. It imports the profile asymptotic

$$\log \epsilon_N^2 = 2q\psi(s_N) + O_h(k \log^2(q+2)), \quad \epsilon_N = \|r_N \ell_N\|_{G_N}, \quad s_N = (N+1-q)/q.$$

Consequently it proves a heat step away from $\sigma = \psi(s)$ at time $e^{-2q\sigma}$, but it does not assign a value in that unresolved profile-centred layer. We do not claim to improve that profile error. The calculation below identifies the complete transition after centring at the **actual finite** ϵ_N , with a finite error valid before an asymptotic is applied.

27.2 II.2 Finite identities and constants

Let $M = C + r\ell$ on the given q -dimensional Hermitian space, and put

$$\epsilon = \|r\ell\|_G > 0, \quad \|C\|_G \leq B, \quad d = B/\epsilon.$$

Assume a certified spectral-radius bound $|\lambda(M)| \leq R$ for the holomorphic trace. In the native instance this is the exact root bound $R = k\sqrt{\delta^2 + \gamma^2}$, including all primary multiplicities. No bound for the operator norm of M is substituted for it or inferred from it.

If $s_1 \geq \dots \geq s_q$ are the singular values of M , the rank-one minimum–maximum argument and the two triangle inequalities give

$$|s_1 - \epsilon| \leq B, \quad s_j \leq B \quad (j \geq 2). \quad (\text{H1})$$

Set $\tau = x/\epsilon^2$ for $x \geq 0$, $y_j = s_j^2/\epsilon^2$. Then

$$|y_1 - 1| \leq 2d + d^2, \quad 0 \leq y_j \leq d^2 \quad (j \geq 2). \quad (\text{H2})$$

For the full canonical heat gap, spectral logarithm, and resolvent response define

$$\mathcal{D}(x) = \Re \operatorname{Tr} e^{-xM^2/\epsilon^2} - \operatorname{Tr} e^{-xM^\dagger M/\epsilon^2},$$

$$\mathcal{L}(x) = \log \det(I + xM^\dagger M/\epsilon^2), \quad \mathcal{A}(x) = \operatorname{Tr} \frac{xM^\dagger M/\epsilon^2}{I + xM^\dagger M/\epsilon^2}.$$

Here the notation for the last quotient means multiplication by the inverse of the positive denominator, with the same operator in both factors. The full finite estimates are

$$\boxed{|\mathcal{D}(x) - (1 - e^{-x})| \leq q \frac{xR^2}{\epsilon^2} e^{xR^2/\epsilon^2} + 2xd + qxd^2,} \quad (\text{H3})$$

$$\boxed{|\mathcal{L}(x) - \log(1+x)| \leq 2xd + qxd^2, \quad |\mathcal{A}(x) - x/(1+x)| \leq 2xd + qxd^2.} \quad (\text{H4})$$

Proof. The functions $1 - e^{-xy}$, $\log(1+xy)$, and $xy/(1+xy)$, as functions of $y \geq 0$, have derivatives in $[0, x]$. Apply H2 to the first singular value and use $f(y) \leq xy$ on the others. The holomorphic trace differs from q by at most $q(xR^2/\epsilon^2)e^{xR^2/\epsilon^2}$, since on each full primary block the trace of the

exponential is its multiplicity times the exponential at its eigenvalue. This proves H3–H4 without deleting a nilpotent block from the endomorphism.

Differentiation at the calibrated variable is also controlled:

$$\boxed{|\mathcal{D}'(x) - e^{-x}| \leq q \frac{R^2}{\epsilon^2} e^{xR^2/\epsilon^2} + 2d + qd^2,} \quad (\text{H5})$$

$$\boxed{|\mathcal{L}'(x) - (1+x)^{-1}| \leq 2d + qd^2.} \quad (\text{H6})$$

For H5, $y \mapsto ye^{-xy}$ has derivative $e^{-xy}(1-xy)$ of absolute value at most one; its value on the remaining singular directions is at most y . For H6, use the derivative $1/(1+xy)^2 \leq 1$ of $y/(1+xy)$. The root-trace derivative is bounded separately as shown.

27.3 II.3 Native consequences and what remains unknown at the profile threshold

Under the supplied exponential ϵ_N profile, $qB_h^2q^2/\epsilon_N^2$ and qR_k^2/ϵ_N^2 tend to zero uniformly throughout the stated cutoff window. Thus H3–H6 give uniformly on compact x -intervals

$$\mathcal{D}_N(x) \rightarrow 1 - e^{-x}, \quad \mathcal{L}_N(x) \rightarrow \log(1+x), \quad \mathcal{A}_N(x) \rightarrow x/(1+x). \quad (\text{H7})$$

The exact calibrated logarithmic centre is

$$\hat{\psi}_N = \frac{\log \epsilon_N^2}{2q}.$$

With $x = e^z$ the time is

$$\tau_N(z) = e^{-2q\hat{\psi}_N+z}, \quad \sigma_N(z) = \hat{\psi}_N - z/(2q).$$

The transition profile is $1 - e^{-e^z}$, the regularized log profile is $\log(1 + e^z)$, and the corresponding derivatives follow from H5–H6. Its intrinsic logarithmic width is $1/q$. The supplied EIQ approximation locates its centre only to $O_h(k \log^2 q/q)$. Therefore this calculation does **not** assign a canonical value at the uncalibrated time $e^{-2q\psi(s)}$.

For $x = 1$ the limiting full gap is $1 - e^{-1}$. Calibrating each of the four cutoffs separately gives four identical limiting constants, whose signed sum is zero; those are four **different physical heat times**. This is not a change to the source's common-time four-cutoff plateau. The latter must retain one common τ , and the exact different centres $\hat{\psi}_N$ remain in H3.

If ϵ_N^2 is known only in a positive certified interval, choose a declared reference $E_0 > 0$ and set $\tau = x/E_0$. Replace x in the monotone profiles by the interval $x\epsilon_N^2/E_0$ and bound all H3 terms outward using the same interval. A profile centre estimated from a midpoint alone is not an exact native scale.

27.4 II.4 The observed heat deficit measures a product of actual projection fractions

Let $\Lambda : E \rightarrow B$ be the unchanged onto observation and

$$Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L_B = G^{-1} \Lambda^* Q_B, \quad P = L_B \Lambda.$$

Then L_B is an isometry and P is the G -orthogonal projection onto the attained observation complement. Let $v = G^{-1}\ell^*$ and define

$$\vartheta = \frac{\|Pr\|_G^2}{\|r\|_G^2} \frac{\|Pv\|_G^2}{\|v\|_G^2} \in [0, 1]. \quad (\text{H8})$$

The exact observed operator is

$$B_{\text{obs}} = \Lambda M L_B = \Lambda C L_B + (\Lambda r)(\ell L_B).$$

Its bounded term has norm at most B , and its rank-one term has squared norm $\epsilon^2 \vartheta$. This follows from $\|\Lambda r\|_{Q_B} = \|Pr\|_G$ and the corresponding dual-row identity for v . The projected rank-one term need not be square-zero.

For $r_B = \dim B \geq 1$, put

$$\mathcal{O}(x) = r_B - \text{Tr } e^{-x B_{\text{obs}}^\dagger Q_B B_{\text{obs}} / \epsilon^2}.$$

The same singular-value calculation proves

$$|\mathcal{O}(x) - (1 - e^{-x\vartheta})| \leq 2xd\sqrt{\vartheta} + r_B x d^2 \leq 2xd + r_B x d^2. \quad (\text{H9})$$

It also proves

$$\left| \log \det(I + x B_{\text{obs}}^\dagger B_{\text{obs}} / \epsilon^2) - \log(1 + x\vartheta) \right| \leq 2xd + r_B x d^2. \quad (\text{H10})$$

The case $r_B = 0$ is empty and both observed quantities are zero. No holomorphic observed trace has been replaced by the original root trace: B_{obs} is not generally an invariant restriction of M .

Thus a limiting observed transition requires the actual product ϑ_N . If $\vartheta_N \rightarrow \vartheta$, then $\mathcal{O}_N(x) \rightarrow 1 - e^{-x\vartheta}$. Codimension alone does not evaluate this product. A one-dimensional kernel containing r or v makes $\vartheta = 0$, even if its codimension fraction vanishes; a kernel orthogonal to both leaves $\vartheta = 1$.

An interval for $\mathcal{O}(x)$ gives an inverse angle certificate. If $[o_-, o_+]$ encloses it and e is the outward H9 error, set

$$L = \max(0, o_- - e), \quad U = \min(1, o_+ + e).$$

When $U < 1$,

$$\vartheta \in \left[-\frac{\log(1-L)}{x}, -\frac{\log(1-U)}{x} \right] \cap [0, 1], \quad x > 0. \quad (\text{H11})$$

If this interval fails to separate a needed value, no angle claim is made. It determines a product of two retained fractions, not their individual values or the complex current phases in the supplied C1–C4.

27.5 II.5 An exact finite example for validation

Take $G = I$, $C = I_2$, and $M = I_2 + E e_1 e_2^T$, $E > 0$. Its eigenvalues remain $(1, 1)$ while $\epsilon = E$. For projection onto the unit vector $(3/5, 4/5)$, the observed operator is the scalar $1 + 12E/25$, and $\vartheta = 144/625$. All heat and logarithmic enclosures in the checker are computed by rational power-series bounds, separately from H3–H10. This is a declared synthetic family; its entries are not substituted for native zeta moments.

Part XI

Finite unit certificates

28 III. Finite unit certificates without a Neumann smallness hypothesis

28.1 III.1 The only algebraic inverse gate is the leading coefficient

Source input. Section 5 of the supplied proof correctly retains the full length- m physical unit, its derivatives through order $2m-1$, and the maps between polynomial and physical jets. Its U6–U10 use a sufficient Neumann condition for an uncertain inverse. This section gives a different, finite certificate that uses the triangular jet structure. It does not supply the absent native derivative values.

Let S be the lower shift in the original divided-jet coordinates of $\mathbb{C}[z]/z^m$, so $S^m = 0$. Write the actual unit and its certified centre as

$$U = \sum_{j=0}^{m-1} u_j S^j, \quad \widehat{U} = \sum_{j=0}^{m-1} \widehat{u}_j S^j, \quad |u_j - \widehat{u}_j| \leq e_j.$$

Suppose only that

$$c := |\widehat{u}_0| - e_0 > 0. \quad (\text{U1})$$

A certified positive lower bound may be used in place of the displayed c . Then U is invertible: it is triangular with diagonal u_0 , and its inverse is the **finite** expression

$$U^{-1} = u_0^{-1} \sum_{r=0}^{m-1} \left(-u_0^{-1} \sum_{j=1}^{m-1} u_j S^j \right)^r. \quad (\text{U2})$$

No restriction on the size of the strictly triangular part is needed. Conversely $u_0 = 0$ makes U singular. Thus U1 is the remaining scalar certificate gate, not a suppressed denominator.

Define $a_j = |\widehat{u}_j| + e_j$ for $j \geq 1$ and nonnegative majorant coefficients

$$b_0 = c^{-1}, \quad b_n = c^{-1} \sum_{j=1}^n a_j b_{n-j} \quad (1 \leq n < m). \quad (\text{U3})$$

If $U^{-1} = \sum v_n S^n$, comparison of coefficients of $UU^{-1} = I$ proves inductively $|v_n| \leq b_n$. Compute the centre inverse $\widehat{v}_n$ by the same exact signed convolution, not by a numerical matrix inverse. From

$$U^{-1} - \widehat{U}^{-1} = -U^{-1}(U - \widehat{U})\widehat{U}^{-1}$$

we obtain coefficient radii

$$|v_n - \widehat{v}_n| \leq f_n := \sum_{i+j+l=n} b_i e_j |\widehat{v}_l|. \quad (\text{U4})$$

All sums stop before degree m .

28.2 III.2 The retained jet metric improves rather than destroys the estimate

Let $D = \text{diag}(d_0, \dots, d_{m-1}) > 0$ be the specified polynomial-jet Gram. Define

$$\beta_j = \|S^j\|_D = \max_{0 \leq l < m-j} \sqrt{d_{l+j}/d_l}, \quad \beta_0 = 1. \quad (\text{U5})$$

The formula follows because S^j sends each coordinate to a different coordinate; its weighted squared norm is the maximum of the displayed ratios. Therefore

$$\|U^{-1}\|_D \leq \sum b_j \beta_j, \quad \|U^{-1} - \widehat{U}^{-1}\|_D \leq \sum f_j \beta_j. \quad (\text{U6})$$

These are finite bounds even when an unweighted Neumann test fails.

For the supplied binomial jet family, $d = m - 1$ and

$$d_l = a(l!)^2 \binom{m-1}{l} \eta^{2l}, \quad a > 0, \quad \eta > 0.$$

The exact ratio is

$$\beta_j = \eta^j \max_l \sqrt{\frac{(l+j)!}{l!} \frac{(m-1-l)!}{(m-1-l-j)!}} \leq (\eta m/2)^j. \quad (\text{U7})$$

Indeed every one-step factor is $\eta \sqrt{(l+1)(m-1-l)} \leq \eta m/2$, and S^j is a product of j shifts. In particular higher unit coefficients are suppressed by their actual powers of η in this metric. This statement concerns the specified constructed jet metric, not the unrelated canonical scalar-source metric.

28.3 III.3 Two-sided physical-metric bounds with no smallness restriction

Retain

$$H = U^{-*} D U^{-1}, \quad \widehat{H} = \widehat{U}^{-*} D \widehat{U}^{-1}.$$

Put

$$\zeta = \sum_{n=0}^{m-1} \beta_n \sum_{i+j=n} b_i e_j, \quad \rho = \sum_{n=0}^{m-1} \beta_n \sum_{i+j=n} |\widehat{v}_i| e_j. \quad (\text{U8})$$

In the original physical coordinates the complete bound is

$$(1 + \rho)^{-2} \widehat{H} \preceq H \preceq (1 + \zeta)^2 \widehat{H}. \quad (\text{U9})$$

Proof. Let $R = U^{-1} \widehat{U}$. Then

$$R = I - U^{-1}(U - \widehat{U}), \quad R^{-1} = I + \widehat{U}^{-1}(U - \widehat{U}).$$

Finite convolution and U5 show $\|R\|_D \leq 1 + \zeta$ and $\|R^{-1}\|_D \leq 1 + \rho$. Thus

$$(1 + \rho)^{-1} \|y\|_D \leq \|Ry\|_D \leq (1 + \zeta) \|y\|_D.$$

Take $y = \widehat{U}^{-1}x$ and square. This proves U9 without requiring $\rho < 1$ or $\zeta < 1$. If $\zeta < 1$, the lower bound may be sharpened to the larger of $(1 - \zeta)^2$ and $(1 + \rho)^{-2}$. If $\rho < 1$, the upper bound may be sharpened to the smaller of $(1 + \zeta)^2$ and $(1 - \rho)^{-2}$.

A finite example outside the old sufficient domain. Let $m = 3$, $\widehat{U} = I$, $U = I + 2S$, $e_1 = 2$, $e_0 = e_2 = 0$, and $\eta = 1/100$. The unweighted error test $\|\widehat{U}^{-1}\| \sum e_j < 1$ fails because its left side is two. The actual inverse is $I - 2S + 4S^2$. With

$$D = \text{diag}(1, 2\eta^2, 4\eta^4), \quad b = 3\eta/2,$$

U7–U8 permit the rational bounds

$$\rho = 2b = 3/100, \quad \zeta = 2b + 4b^2 = 309/10000.$$

Hence

$$(100/103)^2 D \preceq U^{-*} D U^{-1} \preceq (10309/10000)^2 D. \quad (\text{U10})$$

The inverse exists with the displayed close relative metric bounds even though the generic Neumann gate failed. This is a declared triangular diagnostic, not a computation of a native unit.

28.4 III.4 Tensoring, quotient minima, and the original labels

For factorwise bounds U9, positivity of tensor products gives

$$\left(\prod_i (1 + \rho_i)^{-2} \right) \bigotimes_i \hat{H}_i \preceq \bigotimes_i H_i \preceq \left(\prod_i (1 + \zeta_i)^2 \right) \bigotimes_i \hat{H}_i. \quad (\text{U11})$$

Restriction by the **same** cyclic injection and minimization over the **same** affine value fibres preserve both scalar factors: compare the squared norm of every admissible representative and then take its infimum. A q -dimensional attained determinant therefore has absolute logarithmic error at most

$$2q \max \left\{ \sum_i \log(1 + \rho_i), \sum_i \log(1 + \zeta_i) \right\}. \quad (\text{U12})$$

When approximate units are used as value labels, the value map, observation and class must first be transported by the actual tensor product of $\hat{U}_i^{-1} U_i$. Equation U11 is not permission to change those fibres silently.

For a fixed factor with an exact nonzero leading unit and bounded higher coefficient uncertainties, U7 implies $\rho, \zeta = O(\eta)$ as $\eta \downarrow 0$. Tensoring k such factors gives a displayed exponential factor $\exp(O(k\eta))$, not an assumed equality of the metrics. With uncertain leading coefficient there is also its nonvanishing scalar contribution; it does not acquire an η factor.

The input's derivative conversion remains indispensable: for a root of complete multiplicity m , the unit jet uses $g^{(m)}, \dots, g^{(2m-1)}$, and the full local cofactor. The present certificate changes only the inverse-error propagation after those primitive derivative enclosures have actually been supplied.

Part XII

ES heat jets and full moving blocks

29 IV. The ES trace-rigidity theorem as two exact heat jets, with the full mixed return

29.1 IV.1 What the newer ES source has proved

The ES repository at commit 5787b359a73edd62130caf76f573a40d607463c1 contains *Global trace integrality and the original ES source*, TR1–TR11. The following arithmetic theorem is **an inherited result**, not a new occupancy argument:

For $x, y, z \in \mathbb{Q}_{>0}$ satisfying $1/x + 1/y + 1/z = 4/p$, the two power sums

$$s_1 = x + y + z, \quad s_2 = x^2 + y^2 + z^2$$

are integers if and only if all three denominators are integers. Rationality is essential; the same source supplies algebraic irrational counterfamilies to its omission.

Here is the length-three argument, recording the exact assumptions used by our receiver. Let $e_1 = s_1$, $e_2 = (s_1^2 - s_2)/2$. At a prime where a root has negative valuation $-a$, if precisely one root has minimal valuation then its term uniquely minimizes e_1 , contradicting integrality. If precisely two do, their product uniquely minimizes e_2 with valuation $-2a$, contradicting $v_\ell(e_2) \geq -v_\ell(2) \geq -1$. Thus all three have the same reduced denominator at every denominator prime, giving a common denominator d . It is odd, because three reduced odd numerators over an even denominator would have nonintegral first sum. The reciprocal identity $e_2/e_3 = 4/p$, together with $2e_2 \in \mathbb{Z}$, gives $d^3 \mid 2m$, where m is the reduced reciprocal numerator and divides four. An odd positive integer whose cube divides eight is one. Conversely, integer denominators give integer power sums. This is the TR1–TR4 proof specialized to three roots.

29.2 IV.2 Two centres recover both arithmetic traces

Let T be multiplication by the literal root coordinate on the four-root algebra with roots (p, x, y, z) , counted with their literal algebraic multiplicities. No Hermitian adjoint occurs in this definition. Put $\nu = 1$ for the four-dimensional root algebra and $\nu = 2$ for its eight-dimensional signed completion. The latter is free of rank two over the root algebra, so multiplication traces are doubled even at a repeated-root fibre.

For a fixed real centre $c \neq 0$, define the genuine **holomorphic** heat trace

$$H_c(t) = \text{Tr } e^{-t(T-cI)^2}. \quad (\text{E1})$$

Its trace depends only on the roots with algebraic multiplicity, not on whether T has Jordan blocks. Expanding each triangular primary block gives

$$H_c(0) = 4\nu, \quad H'_c(0) = -\nu [(p-c)^2 + (x-c)^2 + (y-c)^2 + (z-c)^2].$$

It follows exactly that

$$\boxed{s_2 = -H'_0(0)/\nu - p^2, \quad s_1 = \frac{H'_c(0) - H'_0(0) + 4\nu c^2}{2\nu c} - p.} \quad (\text{E2})$$

The reciprocal equation itself is the separate multiplication trace

$$\text{Tr } T^{-1} = 5\nu/p. \quad (\text{E3})$$

Thus on the specified positive **rational-root** source, the two quantities in E2 are integers exactly when the denominators are integral. The newer ES theorem is the arithmetic implication; E1–E2

are the exact heat receiver developed here. We have not proved that an integer pair of heat jets occurs at every prescribed prime.

If L is an original invertible conductor onto its receiving image, $T' = LTL^{-1}$ satisfies $H_c^{T'} = H_c^T$ by the entire power series and trace cyclicity. This is a holomorphic-action transport, not an identification of the positive singular-value heat trace. The latter would require the transported metric $L^{-*}GL^{-1}$ or the exact pullback correction when the target metric is separately prescribed.

29.3 IV.3 Finite samples give rigorous jet intervals

Suppose the literal roots are real and their centred magnitudes satisfy $|t_j - c| \leq R_c$, including the distinguished root p . For $t > 0$, set

$$\hat{h}_c(t) = \frac{H_c(t) - 4\nu}{t}.$$

For $v \geq 0$, the Taylor integral remainder gives $0 \leq e^{-v} - 1 + v \leq v^2/2$. Applying this to the four roots proves

$$\boxed{0 \leq \hat{h}_c(t) - H'_c(0) \leq 2\nu t R_c^4.} \quad (\text{E4})$$

If the computed $H_c(t)$ has a certified absolute error at most ϵ_c , the jet radius is at most $2\nu t R_c^4 + \epsilon_c/t$. The error in the reconstructed first sum is consequently at most

$$\boxed{\frac{t(R_c^4 + R_0^4)}{|c|} + \frac{\epsilon_c + \epsilon_0}{2\nu|c|t}}, \quad (\text{E5})$$

and the error in the reconstructed second sum is at most

$$\boxed{2tR_0^4 + \frac{\epsilon_0}{\nu t}.} \quad (\text{E6})$$

These are outward estimates at finite t , not a numerical differentiation assertion. The root bound justifies the holomorphic trace estimate because its trace is the exact root sum; it is not substituted for a native operator norm in a nonnormal norm inequality.

If a known common reduced denominator d of x, y, z is at most D , then a noninteger s_1 has distance at least $1/D$ from every integer, and a noninteger s_2 has distance at least $1/D^2$. Therefore an interval for s_1 contained strictly within $(n - 1/D, n + 1/D)$ and an interval for s_2 contained strictly within $(m - 1/D^2, m + 1/D^2)$ certify exact integrality. Their centres need not themselves equal integers. Without a denominator bound, arbitrarily close rational nonintegers prevent this inference.

For example, $p = 13$, $(x, y, z) = (4, 156/5, 39)$ is a positive rational ES triple obtained from the raw gate below, but its literal first trace is $436/5$, not an integer. Taking smaller heat time alone cannot turn it into an integral witness.

29.4 IV.4 The complete mixed minimum retains the missing second-trace term

Use the supplied original-metric observation frame. Write the fixed operator as

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and the minimum-section change of coordinates as

$$S_t = \begin{pmatrix} I & X_t \\ 0 & I \end{pmatrix}, \quad X_t = tV(I + tW)^{-1}.$$

The input's M16 supplies the exact similarity

$$S_t^{-1} T S_t = \begin{pmatrix} A_K & B_R \\ c & A_B \end{pmatrix},$$

$$A_K = a - X_t c, \quad A_B = d + c X_t, \quad B_R = b + a X_t - X_t d - X_t c X_t. \quad (\text{E7})$$

Thus

$$\boxed{\text{Tr } T = \text{Tr } A_K + \text{Tr } A_B},$$

$$\boxed{\text{Tr } T^2 = \text{Tr } A_K^2 + \text{Tr } A_B^2 + 2 \text{Tr}(B_R c)}. \quad (\text{E8})$$

Both follow by block multiplication and cyclicity of rectangular traces. The reverse block is quadratic in X_t and cannot be omitted. The two heat jets E2 can therefore be reconstructed from the full subquotient data E8; the observed block alone is not the original root trace.

An actual ES spectrum in an explicit nontrivial frame. Take $(p; x, y, z) = (13; 4, 18, 468)$ and conjugate the diagonal root matrix by the rational orthogonal rotation with first block

$$O = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}.$$

The corresponding first block of T is

$$\begin{pmatrix} 181/25 & 108/25 \\ 108/25 & 244/25 \end{pmatrix},$$

with remaining diagonal entries 18, 468. Let the first coordinate be the kernel and take a covariance supported by the all-ones positive block on the first two coordinates. Then X_t has first entry $t/(1+t)$ and its other entries zero. At every $t \geq 0$, E8 returns the exact full trace 503 and full squared trace 219533. At $t = 0$, omitting the reverse coupling loses precisely $2(108/25)^2$ from the squared trace. This example changes coordinates and introduces a declared covariance; it is not presented as a measured native period or a computed xi packet.

There is an exact finite expression beyond the jets. On the open spectral domain where the two inverses exist,

$$\det(zI - T) = \det(zI - A_K) \det(zI - A_B) \det(I - (zI - A_B)^{-1} c (zI - A_K)^{-1} B_R). \quad (\text{E9})$$

It follows from the complete block Schur complement. Clearing denominators gives a polynomial identity, so apparent factor cancellations are retained by continuation. Taking the logarithmic derivative and applying the finite Cauchy formula transports any entire test, including E1, through these same blocks with the complete correction. The formula does not claim that only $\dim K$ eigenvalues move.

29.5 IV.5 The raw gate has an exact finite spacing certificate

The ES source's TR7–TR11 fixes $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, $R = 4a - p$, and an integer $U > 0$ with $R \mid a + U$. The original rational return is

$$x = a, \quad y = \frac{pa(U + a)}{RU}, \quad z = \frac{p(a + U)}{R}.$$

The literal trace satisfies

$$\text{den}(p + x + y + z) = \frac{U}{\gcd(U, pa^2)}. \quad (\text{E10})$$

This is the inherited exact divisor test: integral trace is equivalent to $U \mid pa^2$, and its defining-prime bit selects the original M or E channel. Our finite heat/trace interval can certify it: if a rigorous interval centred at s with radius r satisfies $|s - n| + r < 1/U$ for an integer n , the

exact trace must be integral. This uses the known denominator spacing, not an assumption that approximate integrality implies integrality.

For $p = 13, a = 4, R = 3, U = 5$ gives the nonintegral trace $436/5$, whereas $U = 104$ returns the actual witness $(4, 18, 468)$ and trace 503. The formulas recover their different statuses without changing the raw gate. They provide an exact receiver for the newer trace-rigidity result; finding an occupied raw gate at every hard prime remains a separate arithmetic problem.

Part XIII

The retained full-source profile H1–H9

29.6 1.1 Source basis and domains

The two newly supplied files are preserved unchanged under `inputs/`. The mathematical input used here is their one-factor coercivity P1–P11, joint minimum J1–J8, spectral comparison S1–S5, and, on the simple-quartet domain, V1–V6 and V13. The EIQ monic-profile theorem in V1 is an imported written source theorem; the present note does not independently prove that asymptotic. The relevant GitHub source was read at commit `ff251e3adc5fd78049170b8fc94c26f882df3d14`, file `workbenches/splitzero-tandem/continuations/20260920-original-spectral-determinant/ORIGINAL_SPECTRAL_DETERMINANT.tex`, blob `11a2154819682c01c878ffb0b886f7b704fed888`. Its SD1–SD12 specify the original companion, the selfadjoint source compression, the rank-one correction, and its singular-value bounds. SD26–SD28 keep the original metric dependence. We use those original definitions; finite identities needed below are also proved here.

The evaluated heat threshold is on the programme's stated simple quartet: fixed h , fixed δ in $(0, 1/2)$, $\gamma > 2$, $k \equiv 1 \pmod{4}$, and $q = (k+1)^2$. This is not a claim to have exhibited an off-critical ξ zero. The general covariance identities in the later parts require neither a quartet nor equal multiplicities.

Let χ be the whole original sum polynomial and let M be multiplication by y in $E = C[y]/\chi_y$, with $y = (S - k/2)/i$ and $\chi_y(y) = i^{-(q)} \chi(k/2 + iy)$. Let G_N be the original attained metric from the actual convolution $m_k = w_h \hat{*} k$. Write C_N for its selfadjoint physical y -compression. Then

$$M = C_N + R_N, \quad R_N = r_N \ell_N, \quad R_N^2 = 0, \quad C_N^{\dagger G_N} = C_N, \quad \epsilon_N = \|R_N\|_{G_N} > 0. \quad (\text{H1})$$

The sign is the SD2 convention. It agrees with an earlier $M = A - Q$ convention by $R = -Q$; the positive scalar epsilon is unchanged.

Put $r = N + 1 - q$ and $s_N = r/q$. We use $q - 1 \leq N \leq 2q$, including the fourth original determinant endpoint. The finite multiplication estimate applies at degree $2q$ as well as at $2q - 1$, so one fixed constant B_h gives

$$\|C_N\|_{G_N} \leq B_h q. \quad (\text{H2})$$

When a source moment above degree $2q$ is used, the supplied all-degree coercivity is applied through degree $2q + 2$. This retains the degree-raising column; it does not silently use a truncated multiplication form.

29.7 1.2 The entire allowance profile follows from the retained monic minima

Let ω_n be the squared norm of the original real monic source polynomial p_n . Let ν_n be the squared monic norm for the complete relation measure $\chi_y \hat{*} m_k$. Define

$$d_n = \omega_n / \nu_{n-q}, \quad b_{n+1} = \omega_{n+1} / \omega_n.$$

Direct projection onto the last monic relation proves

$$\begin{aligned} \epsilon_{q-1}^2 &= b_q (1 - d_q) / d_q, \\ \epsilon_N^2 &= b_{N+1} (1 - d_N) (1 - d_{N+1}) / d_{N+1} \quad (N \geq q). \end{aligned} \quad (\text{H3})$$

Here is the full norm calculation. At cutoff N , the minimum representative of $[p_{-(N+1)}]$ is the difference between the monic relation of degree $N + 1$ and $p_{-(N+1)}$, up to sign. Its squared norm is $\nu_{-r} - \omega_{-(N+1)}$. The leading-coefficient functional on the minimum source has squared dual norm $(1 - d_N) / \omega_{-(q-1)}$ for $N \geq q$, and $1 / \omega_{-(q-1)}$ at the first cutoff. Indeed $[p_N]$ has

squared quotient norm $\omega_N - \omega_N^2 / \nu_N(N-q)$, and the leading functional is its inner product divided by ω_N . Multiplying these two norm factors gives H3. Every preceding relation is present in these orthogonal projections.

The supplied V4 gives, through $r=q+1$,

$$\log(\nu_r / \omega_{q+r}) = 2q\psi(r/q) + O_h(k \log^2(q+2)). \quad (\text{H4})$$

The same full-source comparisons applied to two consecutive monic affine fibres give $\log b_{(N+1)} = O_h(k \log^2(q+2))$. The source profile is continuous and decreasing on the required interval, with $\psi(1) > 0$. Consequently d_n is uniformly exponentially small in q on this window; both logarithms $\log(1-d_n)$ in H3 are retained and tend to zero. We obtain

$$\log \epsilon_N^2 = 2q\psi(s_N) + O_h(k \log^2(q+2)). \quad (\text{H5})$$

This is uniform in N . It is an asymptotic with the original profile theorem's eventual domain, not an effective numerical threshold in k .

29.8 1.3 An exact elliptic expression for the two heat thresholds

Denote the complete elliptic integrals by $K(u)$, $E(u)$, with modulus u in $(0,1)$, and put $u' = \sqrt{1-u^2}$. The supplied equilibrium endpoint equations and profile derivative are

$$1 + s = E(u)/(u'K(u)), \quad \psi'(s) = \log(u/E(u)), \quad \psi(0) = \log(4/\pi). \quad (\text{H6})$$

The parameter map in H6 is invertible: its derivative is

$$\frac{(K - E)(E - u'^2 K)}{u u'^3 K^2} > 0.$$

Both factors in the numerator are positive by the defining elliptic integrals. Its range is $(0, \infty)$ after subtracting one. The two derivative formulas $E' = (E-K)/u$ and $K' = E/(u u'^2) - K/u$ follow by differentiation of their defining integrals.

Integration by parts in H6 gives the complete closed expression

$$\psi(s) = s \log \frac{u_s}{E(u_s)} + \log \frac{1 + u'_s}{E(u_s)}, \quad 1 + s = \frac{E(u_s)}{u'_s K(u_s)}. \quad (\text{H7})$$

For a direct verification, differentiate the right side: the coefficient of du/ds is $((1+s)K/E-1)/u - u/(u'(1+u')) = 0$. Its limit at zero is $\log(4/\pi)$. Thus this is the same profile, through its exact inverse parameter map.

Let $a_0 = \psi(0)$, $a_1 = \psi(1)$. Then

$$a_0 = \log(4/\pi), \quad a_1 = \log \frac{u_1(1 + u'_1)}{E(u_1)^2}, \quad E(u_1) = 2u'_1 K(u_1). \quad (\text{H8})$$

The exact rational enclosure in `checks/heat_checks.py` proves

$$0 < a_1 < 1/8 < a_0. \quad (\text{H9})$$

For completeness, the enclosure uses u_1 in $(987/1000, 988/1000)$. Write $z = u^2$ and $c_n = (\text{binom}(2n, n)/4^n)^2$. The series $K = (\pi/2) \sum c_n z^n$ and $E = (\pi/2)(1 - \sum_{n \geq 1} c_n z^n / (2n-1))$ have omitted positive tails bounded by $c_{(J+1)} z^{(J+1)} / (1-z)$ and this quantity $/ (2J+1)$. At $J=512$ the rational squared inequalities E_{series}^2 versus $4(1-z)K_{\text{series}}^2$ prove the root bracket. Using $333/106 < \pi < 355/113$ and $154/1000 < u_1' < 161/1000$ then proves $1 < u_1(1+u_1')/E(u_1)^2 < 9/8$. Finally $\log(9/8) < 1/8$ and $\exp(1/8) < 8/7 < 4/(355/113)$. The checker separately derives the stated loose π bounds from Machin's alternating arctangent series.

Non-interval high-precision evaluations, not native moment evaluations, are

$$u_1 \simeq 0.9871005293061829185, \quad a_1 \simeq 0.06651895202027391967, \quad a_0 \simeq 0.24156447527049044469.$$